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CHAPTER
1

APPLICATION OF DERIVATIVE

1. INTRODUCTION

In this chapter we will study some applications of the ideas developed in previous chapter. One major objective is to see how various calculus concepts are related to graphs of functions.

We will also see how the signs and zeros of a function and its first and second derivatives can be used to study properties of the function and its graph.

Another important objectives is to illustrate how calculus can be used to solve some practical problems. This involves translating problems that arise in real-world situations into mathematical expressions that involve real variables. We will investigate some representative procedures for doing this and then see how to use the results in some problems that involve rates of change and maximum -minimum values.

2. MONOTONICITY (SIGNIFICANCE OF THE SIGN OF THE FIRST ORDER DERIVATIVE)

If the graph of a function is in upward going direction or in downward coming direction then it is called as monotonic function, and this property of the function is called monotonicity. If a function is defined in any interval, and if in some part of the interval, graph moves upwards and in the remaining part moves downward then function is not monotonic in that interval.

Increase and Decrease of a function of one argument : A function is said to be increasing on the interval $\langle a,b\rangle$ if to greater values of the argument belonging to this interval there correspond greater values of the function. In other word, $f(x)$ is an increasing function in an interval (a, b) if, whatever the values x_1 and x_2 from this interval (fig, a), from the inequality, $x_2 > x_1$ there follows the inequality $f(x_2) > f(x_1)$.

Similarly, a function is said to be decreasing on an interval (a, b) if to greater values of the argument in this interval there correspond smaller values of the function. In other words, $f(x)$ is a decreasing function (fig., b) if from the inequality $x_2 > x_1$ there follows the inequality $f(x_2) < f(x_1)$.

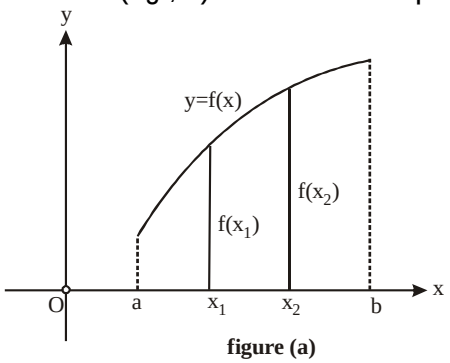


figure (a)

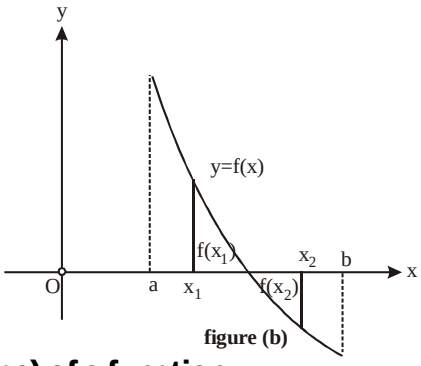


figure (b)

Theorem : Sufficiency test for the increase (decrease) of a function

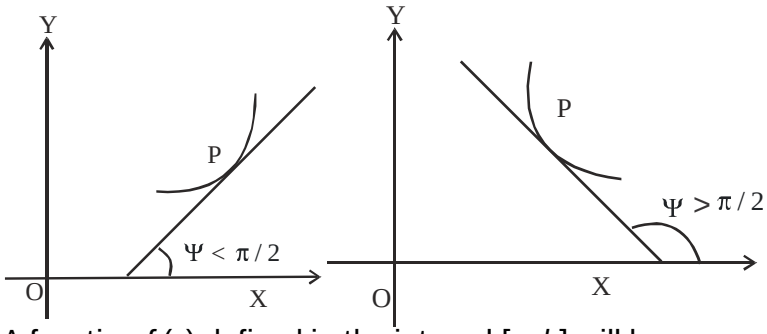
(1) If the derivative of a differentiable function is positive inside a certain interval, then the function is increasing on this interval.

(2) If the derivative of a differentiable function is negative inside a certain interval, then the function is decreasing on this interval.

A function which in a given interval is increasing or decreasing is called monotone (in that interval). An interval of variation of the independent variable on which the function is increasing is called its interval of increase, and an interval on which it is decreasing is called its interval of decrease. Both types of intervals are referred to as intervals of monotonicity of this function.

From above theorem we can rewrite the methods of testing monotonicity.

(i) At a point : A function $f(x)$ is said to be monotonic increasing (decreasing) at a point $x = a$ of its domain if it is monotonic increasing (decreasing) in the interval $(a - h, a + h)$ where h is a small positive number. Hence we may observe that $f(x)$ is monotonic increasing at $x = a$, then at this point tangent to its graph will make an acute angle with the x– axis where as if the function is monotonic decreasing these tangent will make an obtuse angle with x–axis. Consequently $f'(a)$ will be positive or negative according as $f(x)$ is monotonic increasing or decreasing at $x = a$.



(ii) In an interval : A function $f(x)$ defined in the interval $[a, b]$ will be

Monotonic increasing	$\Rightarrow f'(x) \geq 0$
Monotonic decreasing	$\Rightarrow f'(x) \leq 0$
constant	$\Rightarrow f'(x) = 0 \ \forall \ x \in (a, b)$
Strictly increasing	$\Rightarrow f'(x) > 0$
Strictly decreasing	$\Rightarrow f'(x) < 0$

Note :-

(i) In the above result $f'(x)$ should not be zero for all value of x otherwise $f(x)$ will be a constant function.

(ii) If in $[a, b]$, $f'(x) < 0$, for at least one value of x and $f'(x) > 0$ for atleast one value of x then $f(x)$ will not be monotonic in $[a, b]$.

Examples of monotonic functions : If a function is monotonic increasing (decreasing) at every point of its domain ,then it is said to be monotonic increasing (decreasing) function.
In the following table we have examples of some monotonic/ not monotonic functions.

Monotonic Increasing	Monotonic Decreasing	Not Monotonic
x^3	$1/x \quad x > 0$	x^2
$x \lvert x \rvert$	$1 - 2x$	$\lvert x \rvert$
e^x	e^{-x}	$e^x + e^{-x}$
$\log_a x, a > 1$	$\log_a x, 0 < a < 1$	$\sin x$
$\tan x$	$\cot x$	$\cos x$
$[x]$		

Properties of monotonic functions :

(i) If $f(x)$ is strictly increasing function on an interval $[a, b]$, then f^{-1} exists and it is also a strictly increasing function.

(ii) If $f(x)$ is strictly increasing function on an interval $[a, b]$ such that it is continuous, then f^{-1} is continuous on $[f(a), f(b)]$.

(iii) If $f(x)$ is continuous on $[a, b]$ such that $f'(c) \geq 0$ ($f'(c) > 0$) for each $c \in (a, b)$, then $f(x)$ is monotonically (strictly) increasing function on $[a, b]$.

(iv) If $f(x)$ is continuous on $[a, b]$ such that $f'(c) \leq 0$ ($f'(c) < 0$) for each $c \in (a, b)$, then $f(x)$ is monotonically (strictly) decreasing function on $[a, b]$

(v) If $f(x)$ and $g(x)$ are monotonically (or strictly) increasing (or decreasing) function on $[a, b]$, then $gof(x)$ is a monotonically (or strictly) increasing function on $[a, b]$

(vi) If one of the two functions $f(x)$ and $g(x)$ is strictly (or monotonically) increasing and other strictly (monotonically) decreasing, then $gof(x)$ is strictly (monotonically) decreasing on $[a, b]$

***Illustration 1*:** Function $f(x) = \sin x$ is monotonic increasing in $[0, \pi/2]$ because
$f'(x) = \cos x \ \forall \ x \in (0, \pi/2)$

***Illustration 2*:** Function $f(x) = |x|$ is not a monotonic functions in the interval [–1, 1] because

$$f(x) = \begin{cases} x; & x > 0 \\ -x; & x < 0 \end{cases} \quad ; \; f'(x) = \begin{cases} 1; & x > 0 \\ -1; & x < 0 \end{cases}$$

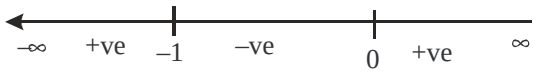
***Illustration 3*:** The function $f(x) = x - \log(1 + x)$, $x > -1$ is increasing in the interval –

(a) $(0, \infty)$ (b) $(-1, 0)$ (c) $(-\infty, 0)$ (d) None of these

Solution : **(a)**

We have, $f(x) = x - \log(1 + x)$, $x > -1$

$$\Rightarrow \quad f'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x} = \frac{x(1+x)}{(1+x)^2}. \quad \text{Sign scheme for } \frac{x(1+x)}{(1+x)^2}$$



$f'(x) > 0$ if $x < -1$ or $x > 0$ But $x > -1$.

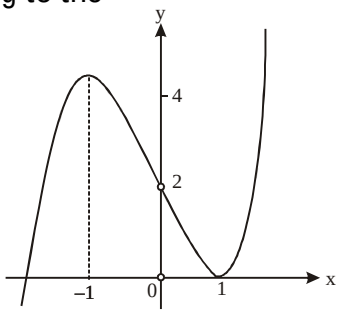
$\therefore f'(x) > 0$ if $x > 0$. Thus, $f(x)$ is increasing in the interval $(0, \infty)$

***Illustration 4*:** Test for an increase and decrease of the function $f(x) = x^3 - 3x + 2$

Solution : We find the derivative $f'(x) = 3x^2 - 3 = 3(x^2 - 1)$

The derivative vanishes for the values $x_1 = -1$ and $x_2 = 1$.

These values divide the entire infinite x-axis into three intervals : $(-\infty, -1]$, $[-1, 1]$, $[1, +\infty)$ inside each of which the derivative $f'(x)$ preserves a constant sign. Obviously, inside the first and third intervals the derivative $f'(x)$ is positive, and inside the second interval it is negative. This can be verified by taking points belonging to the



corresponding intervals. Consequently, the function $f(x)$ is increasing in the first interval, decreasing in the second, and increasing once again in the third (fig.)

***Illustration 5*:** Let $f(x) = \cot^{-1}[g(x)]$, where $g(x)$ is an increasing function for $0 < x < \pi$. Then $f(x)$ is

(a) Increasing in $(0, \pi)$

(b) Decreasing in $(0, \pi)$

(c) Increasing in $\left(0, \frac{\pi}{2}\right)$ and decreasing in $\left(\frac{\pi}{2}, \pi\right)$

(d) None of these

Solution : **(b)** We have, $f(x) = \cot^{-1}(g(x)) \Rightarrow f'(x) = \frac{-1}{1+[g(x)]^2} \times g'(x) < 0$ for $0 < x < \pi$ [$\because g(x)$ is increasing for $0 < x < \pi$, $\therefore g'(x) > 0$]

Thus, $f(x)$ is decreasing in $(0, \pi)$

Student Task

Q.1 e^{ax} is monotonic decreasing –

(a) Always (b) Never (c) When $a < 0$ (d) When $a > 0$

Q.2 If $f(x) = kx - \sin x$ is monotonically increasing, then –
(a) $k > 1$ (b) $k > -1$ (c) $k < 1$ (d) $k < -1$

Q.3 $f(x) = x + 1/x$, $x \neq 0$ is monotonic increasing when –
(a) $|x| < 1$ (b) $|x| > 1$ (c) $|x| < 2$ (d) $|x| > 2$

Q.4 Function $f(x) = 2x^3 - 9x^2 + 12x + 29$ is monotonic decreasing when –
(a) $x < 2$ (b) $x > 2$ (c) $x > 3$ (d) $1 < x < 2$

Q.5 The value of a for which the equation $x^3 - 3x + a = 0$ has two distinct roots in [0, 1] is given by -
(a) –1 (b) 1 (c) 3 (d) None of these

Q.6 Which of the following functions is not monotonic when x > 0
(a) $x + |x|$ (b) e^x (c) log x (d) sin x

ANSWER KEY

(1) (c) (2) (a) (3) (b) (4) (d) (5) (d) (6) (d)

3. MAXIMA AND MINIMA OF A FUNCTION OF ONE VARIABLE

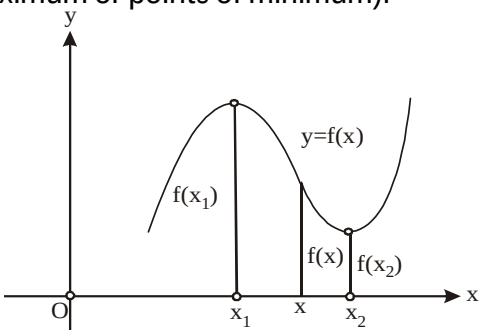
We say that a function $f(x)$ has a maximum for the value x_1 of the argument x if, in a sufficiently close neighbourhood of the point x_1 the inequality

$f(x_1) > f(x)$; $(x \neq x_1)$ is fulfilled (figure)

Analogously, we say that a function f (x) has a minimum for the value x_2 of the argument x if, within a sufficiently close neigbourhood of the point x_2 the following inequality

$f(x_2) < f(x)$; $(x \neq x_2)$ takes place (figure).

The generic term for maximum and minimum is extremum (extreme value) and those values of the argument for which extrema of a function are obtained are called the points of extremum of the function (or : points of maximum or points of minimum).



An extremum of a function, generally speaking, has a local character, since this is the greatest or the least value of the function as compared with its neighbouring values.

Therefore, the presence of an extremum of a function depends not at all on how the function behaves far from this value. From this point of view it is clear that a minimum of a function can be greater than its maximum, just as a depression in the mountains can have a greater elevation (the height above sea level) than a low summit.

Let the function $f(x)$ be defined on the closed interval $[a, b]$ and have an extremum at the point $x_0 \in [a, b]$. If x_0 is an interior point of the closed interval, then the difference $f(x) - f(x_0)$ $(x \neq x_0)$ retains a constant sign in a certain two-sided neigbourbood $x_0 - h < x < x_0 + h$ $(x \neq x_0)$ of the point x_0. Such extremum is called two-sided. For instance, the formula $f(x) = \sqrt{1 - x^2}$ has a two-sided maximum at $x_0 = 0$, since $f(x) < f(x_0) = 1$ for $-1 < x < 1$, $x \neq 0$. And if x_0 is an end-point of the closed interval $[a, b]$, say, $x_0 = a$, then $f(x) - f(x_0)$ preserves the sign only in a one-sided neighbourhood $a = x_0 < x < x_0 + h$ of the point x_0. Such an extremum is termed one-sided (or boundary). For instance, the function $f(x) = \sqrt{1 - x^2}$ has a one-sided minimum at $x_0 = -1$ and at $x_0 = 1$. Hence, under the word 'extremum' we shall understand a two-sided extremum, i.e. we shall assume that for a point of extremum of a given function $f(x)$ there is a certain neighbourhood $0 < |x - x_0| < h$ of the point x_0 in which the difference $f(x) - f(x_0)$ preserves a constant sign.

1. Necessary condition for an extremum of a function :

Theorem : If a differentiable function $f(x)$ has an extremum (two-sided) at a point x_0, then its derivative at this point is equal to zero.

Geometrical illustration : Geometrically, condition (1) means that at the point of extremum of a differentiable function $y = f(x)$ the tangent to its graph is parallel to the x-axis (figure (a)).

Corollary : A continuous may have extrema only at the points where the derivative of the function is equal to the zero or does not exist.

Indeed, if at a point x_0 of an extremum of the function $f(x)$ there exists a derivative $f''(x_0)$, then, by the above proved theorem, this derivative is equal to zero : $f'(x_0) = 0$.

The fact that at a point of extremum of a continuous function the derivative may not exist is illustrated by the function whose graph as the shape shown in figure (b).

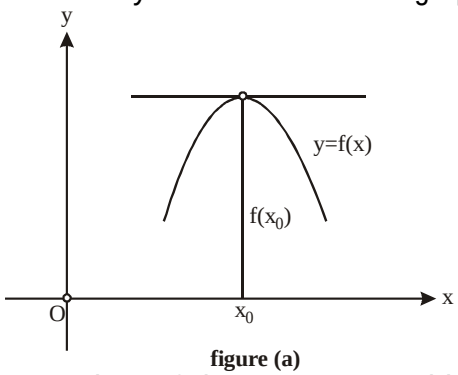


figure (a)

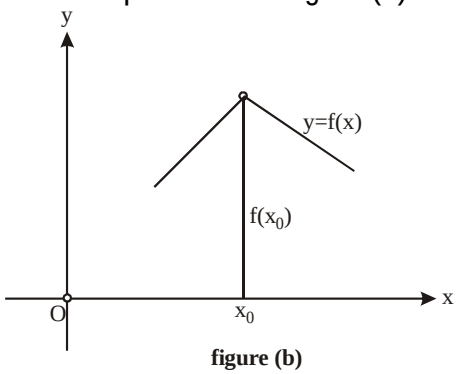


figure (b)

Those values of the argument x which for a given function $f(x)$ make its derivative $f'(x)$ vanish, or for which the derivative $f'(x)$ does not exist (for instance, turns into infinity) are called the critical values of the argument.

2. Sufficient conditions for an extremum of a function :

From the fact that $f'(x_0) = 0$ it does not follow at all that the function $f(x)$ has an extremum at $x = x_0$. Indeed, let $f(x) = x^3$. Then $f'(x) = 3x^2$, and, consequently, $f'(0) = 0$. But the value $f(0)$ is not an extremum of the given function since the difference $f(x) - f(0)$ changes its sign with the change of the sign of the argument x (fig.).

Thus, not for any critical value of the argument of a function $f(x)$ an extremum of this function takes place. Therefore, along with the necessary condition we are going to give sufficient conditions for an extremum of a function $f(x)$, where (1) the function $f(x)$ has a maximum at $x = x_0$ if the change of sign of the derivative $f'(x)$ takes place from plus to minus, (2) the function $f(x)$ has a minimum at $x = x_0$ if the derivative changes its sign from minus to plus.

Theorem 1 : If for a differentiable function $f(x)$ its derivative $f'(x)$ vanishes at $x = x_0$ but retains its sign in passing through this value, then at $x = x_0$ the function $f(x)$ has no extremum. Indeed, if, for instance, $f'(x_0) = 0$ and $f'(x) > 0$ for $x_0 - \varepsilon < x < x_0 + \varepsilon$, $x \neq x_0$ then the function $f(x)$ is increasing both on the closed interval $[x_0 - \varepsilon, x_0]$ and on the closed interval $[x_0, x_0 + \varepsilon]$. Hence at $x = x_0$ the function has neither a maximum, nor a minimum.

Using these theorems in investigating a differentiable function $f(x)$ for an extremum, we first find the critical values of the argument, i.e. those values of x_0 for which $f'(x_0) = 0$, and then, on choosing for each of such values x_0 a sufficiently small interval $(x_0 - \varepsilon, x_0 + \varepsilon)$ such that it contains no other critical values (of course, if it is possible), we check the character of this value according to the following scheme :

$f'(x_0 - h)$	$f'(x_0 + h)$	Conclusion

+	+	No extremum
+	–	Maximum
–	+	Minimum
–	–	No extremum

where the variable h runs through the interval $0 < h < \varepsilon$.

From above discussion we can rewrite the first derivative test for local maxima and minima

Let $f(x)$ be a function differentiable at $x = a$. Then

(a) $x = a$ is a point of local maximum of $f(x)$ if (i) $f'(a) = 0$ and (ii) $f'(x)$ changes sign from positive to negative as x passes $f'(x) > 0$ at every point in the left hand $(a - \varepsilon, a)$ of a and $f'(x) < 0$ at every point in the right hand $(a, a + \varepsilon)$ of a.

(b) $x = a$ is a point of local minimum of $f(x)$ if

(i) $f'(a) = 0$ and

(ii) $f'(x)$ changes sign from negative to positive as x passes through a i.e., $f'(x) < 0$ at every point in the left hand $(a - \varepsilon, a)$ of a and $f'(x) > 0$ at every point in the right hand $(a, a + \varepsilon)$ of a.

(c) If $f'(a) = 0$, but $f'(x)$ does not change sign, that is, $f'(a)$ has the same sign in the complete neighbourhood of a, then a is neither a point of local maximum nor a point of local minimum.

Algorithm determining extreme values of a function by using first derivative test :

STEP I : Put $y = f(x)$

STEP II : Find $\frac{dy}{dx}$

STEP III : Put $\frac{dy}{dx} = 0$ and solve this equation for x. Let $c_1, c_2, c_3, \ldots c_n$ be the roots of this equation. $c_1, c_2, c_3, \ldots\ldots, c_n$ are stationary values of x and these are the possible points where the function can attain a local maximum or a local minimum. So we test the function at each one of these points.

STEP IV : Consider $x = c_1$. If $\frac{dy}{dx}$ changes its sign from positive to negative as x increases through c_1, then the function attains a local maximum at $x = c_1$.

If $\frac{dy}{dx}$ changes its sign from negative to positive as x increases through c_1, then the function attains a local minimum at $x = c_1$.

If $\frac{dy}{dx}$ does not change sign as x increases through c_1, then $x = c_1$ is neither a point of local maximum nor a point of local minimum. In this case $x = c_1$ is a point of inflexion. Similarly we may deal with other values of x.

Theorem 2 : (the second rule). If for a differentiable function $f(x)$ at some point x_0 its first derivative $f'(x)$ is equal to zero and its second derivative $f''(x)$ exists and is non-zero i.e.

$f'(x_0) = 0$, $f''(x_0) \neq 0$,

then at this point the function $f(x)$ has an extremum; namely (1) if $f''(x_0) > 0$ then $f(x_0)$ is a minimum of the function $f(x)$, and (2) if $f''(x_0) < 0$, then $f(x_0)$ is a maximum of the function $f(x)$.

Higher order derivative test :

(i) The value of the function $f(x)$ at $x = a$ is maximum, if $f'(a) = 0$ and $f''(a) < 0$

(ii) The value of the function $f(x)$ at $x = a$ is minimum if $f'(a) = 0$ and $f''(a) > 0$

(iii) If $f'(a) = 0$, $f''(a) = 0$, $f'''(a) \neq 0$ then $x = a$ is not an extreme point for the function $f(x)$.

(iv) If $f'(a) = 0$, $f''(a) = 0$, $f'''(a) = 0$ then the sign of $f^{(iv)}(a)$ will determine the maximum and minimum value of function i.e. $f(x)$ is maximum, if $f^{(iv)}(a) < 0$ and minimum if $f^{(iv)}(a) > 0$.

(v) $f^n(c)$ exists and is non-zero.

Then, If n is even and $f^n(c) < 0 \Rightarrow x = c$ is a point of local maximum.

If n is even and $f^n(c) > 0 \Rightarrow x = c$ is a point of local minimum.
If n is odd $\Rightarrow x = c$ is neither a point of local maximum nor a point of local minimum.

Alogrithm for determining extreme values of a function :

From the above test criteria we obtain the following rule for determining maxima and minima of $f(x)$.

STEP I : Find $f'(x)$

STEP II : Put $f'(x) = 0$ and solve this equation for x. Let $c_1, c_2, \ldots c_n$ be the roots of this equation. $c_1, c_2, \ldots c_n$ are stationary values of x and these are the possible points where the function can attain a local maximum or a local minimum. So we test function at each one of these point.

STEP III : Find $f''(x)$. Consider $x = c_1$.
If $f''(c_1) < 0$, then $x = c_1$ is a point of local maximum.
If $f''(c_1) > 0$, then $x = c_1$ is a point of local minimum.
If $f''(c_1) = 0$, we must find $f''(x)$ and substitute in it c_1, for x.
If $f'''(c_1) \neq 0$, then $x = c_1$ is neither a point of local maximum nor a point of local minimum and is called the point of inflection.
If $f'''(c_1) = 0$, we must find $f^{IV}(x)$ and substitute in it c_1 for x.
If $f^{IV}(c_1) < 0$, then $x = c_1$ is a point of local maximum and if $f^{IV}(c_1) > 0$, then $x = c_1$ is a point of local minimum.
If $f^{IV}(c_1) = 0$, we must find $f^{V}(x)$, and, so on. Similarly the values of $c_2, c_3 \ldots$, may be tested.

Greatest and least values of a function in a given interval :

If a function $f(x)$ is defined in an interval $[a, b]$, then greatest or least values of this function occurs either at $x = a$ or $x = b$ or at those values of x where $f'(x) = 0$.

Remember that a maximum value of the function $f(x)$ in any interval $[a, b]$ is not necessarily its greatest value in that interval.
Thus greatest value of $f(x)$ in interval $[a, b]$ = Max.$[f(a), f(b), f(c)]$
Least value of $f(x)$ in interval $[a, b]$ = Min. $[f(a), f(b), f(c)]$
Where $x = c$ is a point such that $f'(c) = 0$

Properties of maxima and minima :

(i) If $f(x)$ is continuous function in its domain, then at least one maxima and one minima must lie between two equal values of x

(ii) Maxima and minima occur alternately that is between two maxima there is one minimum and vice-versa.

(iii) If $f(x)$ is a maximum (minimum) at a point $x = a$, then $1/f(x)$, $[f(x) \neq 0]$ will be minimum (maximum) at that point.

(iv) If $f(x) \to \infty$ as $x \to a$ or b and $f'(x) = 0$ only for one value of x (say c) between a and b, then $f(c)$ is necessarily the minimum and the least value.

(v) If $f(x) \to \infty$ as $x \to a$ or b, then $f(c)$ is necessarily the maximum and the greatest values of the function

Test for Local Maximum/Minimum at $x = a$ if $f(x)$ is not differentiable at $x = a$

Case 1 : When $f(x)$ is continuous at $x = a$ and $f'(a - h)$ and $f'(a + h)$ exist and are non-zero, then $f(x)$ has a local maximum or minimum at $x = a$ if $f'(a - h)$ and $f'(a + h)$ are of opposite signs.
If $f'(a - h) > 0$ and $f'(a + h) < 0$ then $x = a$ will be a point of local maximum.
If $f'(a - h) < 0$ and $f'(a + h) > 0$ then $x = a$ will be a point of local minimum.

Case 2: When $f(x)$ is continuous and $f'(a - h)$ and $f'(a + h)$ exist but one of them is zero, we should infer the information about the existence of local maxima/minima from the basic definition of local maxima/minima.

Case 3: If $f(x)$ is not continuous at $x = a$ and $f'(a - h)$ and/or $f'(a + h)$ are not finite, then compare

the values of $f(x)$ at the neighboring points of $x = a$.

Maxima and minima of function of two variable :

If a function is defined in terms of two variables and if these are associated with a given relation then by eliminating one variable, we convert function in terms of one variable and then find the maxima and minima by known methods.

Concept of Global Maximum/Minimum

Let $y = f(x)$ be a given function with domain D. Let $[a, b] \subseteq D$. Global maximum/minimum of $f(x)$ in $[a, b]$ is basically the greatest/least value of $f(x)$ in $[a, b]$.

Global maximum and minimum in $[a, b]$ would always occur at critical points of $f(x)$ within $[a, b]$ or at the end points of the interval, if f is continuous in $[a, b]$.

Global Maximum/Minimum in $[a, b]$

In order to find the global maximum and minimum of a continuous function $f(x)$ in $[a, b]$, find out all the critical points of $f(x)$ in

(a, b). Let $c_1, c_2, \ldots, c_n$ be the different critical points. Find the value of the function at these critical points. Let $f(c_1), f(c_2), \ldots, f(c_n)$ be the values of the function at critical points.

Say, $M_1 = \max \{f(a), f(c_1), f(c_2), \ldots, f(c_n), f(b)\}$ and $M_2 = \min \{f(a), f(c), f(c_2), \ldots, f(c_n), f(b)\}$

Then M_1 is the greatest value of $f(x)$ in $[a, b]$ and M_2 is the least value of $f(x)$ in $[a, b]$.

Global Maximum/Minimum in (a, b)

Method for obtaining the greatest and least values of $f(x)$ in (a, b) is almost same as the method used for obtaining the greatest and least values in $[a, b]$ however with a caution.

Let $y = f(x)$ be a continuous function and $c_1, c_2 \ldots c_3$ be the different critical points of the function in (a, b).

Let $M_1 = \max. \{f(c_1), f(c_2), f(c_3) \ldots f(c_n)\}$ and $M_2 = \min \{f(c_1), f(c_2), f(c_3) \ldots f(c_n)\}$

Now if $\lim\limits_{\substack{x \to a+0 \\ (or\ x \to b-0)}} f(x) > M_1$ or $< M_2$, $f(x)$ would not have global maximum (or global minimum) in (a, b).

This means that if the limiting values at the end points are greater than M_1 or less than M_2, then $f(x)$ would not have global maximum/minimum in (a, b). On the other hand if $M_1 > \lim\limits_{\substack{x \to a+0 \\ (and\ x \to b-0)}} f(x)$ and $M_2 < \lim\limits_{\substack{x \to a+0 \\ (and\ x \to b-0)}} f(x)$, then M_1 and M_2 would respectively be the global maximum and global minimum of $f(x)$ in (a, b).

Some standard geometrical results related to maxima and minima :

The following results can easily be established.

(i) The area of rectangle with given perimeter is greatest when it is a square.

(ii) The perimeter of a rectangle with given area is least when it is a square.

(iii) The greatest rectangle inscribed in a given circle is a square.

(iv) The greatest triangle inscribed in a given circle is equilateral.

(v) The semi vertical angle of a cone with given slant height and maximum volume is $\tan^{-1}\sqrt{2}$.

(vi) The height of a cylinder of maximum volume inscribed in a sphere of radius a is $2a/\sqrt{3}$.

Some Important Results :

(i) Equilateral triangle : Area $= (\sqrt{3}/4)\, x^2$, where x is its side

(ii) Square : Area $= a^2$, perimeter $= 4a$, where a is its side.

(iii) Rectangle : Area $= ab$, perimeter $= 2(a + b)$, where a, b are its sides

(iv) Trapezium : Area $= 1/2\,(a + b)\, h$

(v) Volume of a cuboid $= \ell bh$

(vi) Surface area of a cuboid = 2 ($\ell b + bh + h\ell$)

(vii) Volume of a prism = area of the base *x* height.

(viii) Lateral surface of a prism = perimeter of the base *x* height

(ix) Total surface of a prism = lateral surface + 2 area of the base (Note that lateral surfaces of a prism are all rectangles)

(x) Volume of a pyramid = $\frac{1}{3}$ area of the base *x* height

(xi) Curved surface of a pyramid = $\frac{1}{2}$ (perimeter of the base) × slant height

(xii) Volume of a cone = $\frac{1}{3}\pi r^2 h$

(xiii) Curved surface of a cylinder = $2\pi rh$

(xiv) Total surface of a cylinder = $2\pi rh + 2\pi r^2$

(xv) Volume of a sphere = $\frac{4}{3}\pi r^3$

(xvi) Surface area of a sphere = $4\pi r^2$.

(xvii) Area of a circular sector = $\frac{1}{2}r^2\theta$, when θ is in radius.

POINT OF INFLEXION :

If at any point *P*, the curve is concave on one side and convex on other side with respect to x–axis, then the point *P* is called the point of inflexion. Thus *P* is a point of inflexion if at *P*,

$\frac{d^2y}{dx^2} = 0$, but $\frac{d^3y}{dx^3} \neq 0$

Also point *p* is a inflexion if $f''(x) = f^{IV}(x)..... = 0$ and $f^n(x) \neq 0$ for odd *n*.

***Illustration 6*:** Find the local maxima or local minima, if any, of the function $f(x) = \sin^4 x + \cos^4 x$, $0 < x < \pi/2$ using the first derivative test.

Solution : We have $y = f(x) = \sin^4 x + \cos^4 x$

$$\therefore \frac{dy}{dx} = 4\sin^3 x \cos x - 4\cos^3 x \sin x = -4\cos x \sin x (\cos^2 x - \sin^2 x)$$

$$= -2\sin 2x \cos 2x = -\sin 4x$$

For a local maximum or a local minimum, we have

$\frac{dy}{dx} = 0 \Rightarrow -\sin 4x = 0 \Rightarrow -\sin 4x = 0 \Rightarrow \sin 4x = 0$

$\Rightarrow 4x = \pi \quad \left[\because 0 < x < \frac{\pi}{2} \therefore 0 < 4x < 2\pi\right] \quad \Rightarrow x = \pi/4.$

In the left hand of $x = \pi/4$, we have $x < \pi/4 \Rightarrow 4x < \pi \Rightarrow \sin 4x > 0 \Rightarrow -\sin 4x < 0$

$\Rightarrow \frac{dy}{dx} < 0$

In the right hand of $x = \pi/4$, we have $x > \pi/4 \Rightarrow 4x > \pi \Rightarrow \sin 4x < 0 \Rightarrow -4\sin 4x > 0$

$\Rightarrow \frac{dy}{dx} > 0$

Thus, $\frac{dy}{dx}$ changes sign from negative to positive as x increases through $\pi/4$. So $x = \pi/4$ is a point of local minimum.

The local maximum value is $f\left(\frac{\pi}{4}\right) = \left(\sin\frac{\pi}{4}\right)^4 + \left(\cos\frac{\pi}{4}\right)^4 = \frac{1}{2}$

***Illustration 7*:** If the function $f(x) = 2x^3 - 9ax^2 + 12a^2 x + 1$ attains its maximum and minimum at *p* and *q* respectively such that $p^2 = q$, then a equals

(a) 0 (b) 1 (c) 2 (d) None of these

Solution : **(c)** Since $f(x) = 2x^3 - 9ax^2 + 12a^2 x + 1$ attains max. and min. at *p* and *q*, respectively.

Therefore $f'(p) = 0, f'(q) = 0, f''(P) < 0$ and $f''(q) > 0$

Now, $f'(p) = 0, f'(q) = 0$

$\Rightarrow 6p^2 - 18ap + 12a^2 = 0$ and $6q^2 - 18aq + 12a^2 = 0$

$\Rightarrow p^2 - 3ap + 2a^2 = 0$ and $q^2 - 3aq + 2a^2 = 0$

$\Rightarrow p = a, 2a, q = a, 2a$ ..... (i)

Now, $f''(p) < 0 \Rightarrow 12p - 18a < 0 \Rightarrow p < \frac{3}{2}a$ ..... (ii)

and $f''(q) > 0 \Rightarrow 12p - 18a > 0 \Rightarrow q > \frac{3}{2}a$ ..... (iii)

From (i), (ii), (iii), we get $p = a$, $q = 2a$.

Now $p^2 = q \Rightarrow a^2 = 2a \Rightarrow a = 0, 2$

But for $a = 0$, $f(x) = 2x^3 + 1$ which does not attain a max. or min. for any value of x. Hence, a = 2.

***Illustration 8*:** Find the maximum value of $x^3 - 3x$ in the interval [0, 2].

Solution : Let $f(x) = x^3 - 3x$. Then $f'(x) = 3x^2 - 3$. For maximum or minimum $f'(x) = 0 \Rightarrow x = \pm 1$. But $x = -1 \notin [0, 2]$.

therefore $x = 1$ only. It can be easily checked that $f''(x) > 0$ for $x = 1$.

So, $f(x)$ attains a local minimum at $x = 1$

Now, $f(0) = 0$, $f(1) = -2$ and $f(2) = 6$. Hence $f(x)$ attains the maximum value at $x = 2$.

***Illustration 9*:** Divide 64 into two parts such that the sum of the cubes of two parts is minimum. The two parts are

(a) 44,20 (b) 16, 48 (c) 32, 32 (d) 50, 14

Solution : **(c)**

$x + y = 64$, $y = x^3 + y^3 = x^3 + (64 - x)^3$

$\frac{dy}{dx} = 3[x^2 - (64 - x)^2] = 0 = 3[(2x - 64)\,64] = 0 \Rightarrow x = 32$

$\frac{d^2y}{dx^2}$ = + ve, Hence min.

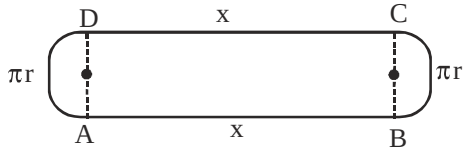
***Illustration 10*:** A running track or 440 it is to be laid out enclosing a football field, the shape of which is a rectangle with a semi -circle at each end. If the area of the rectangular portion is to be maximum then find the length of its sides.

Solution : Perimeter = 440 ft.

$2x + \pi r + \pi r = 440$ or $2x + 2\pi r = 440$ .....(1)

A = Area of rectangular portion = $x.\,2r$

$A = x\,\frac{(400 - 2x)}{\pi} = \frac{1}{\pi}(440x - 2x^2)$



$\frac{dA}{dx} = \frac{1}{\pi}(440 - 4x) \therefore x = 110$; $\frac{d^2A}{dx^2}$ = – ive

$\therefore$ A is max. when $x = 110$.

$\therefore 2r = \frac{440 - 2x}{\pi} = \frac{440 - 220}{22/7} = 70$

***Illustration 11*:** Prove that every point of the curve $y = b\sin(x/a)$ is a point of inflexion, where it meets the x–axis.

Solution : For the points of intersection of curve and x–axis, $y = 0$

$\therefore \sin x / a = 0 \Rightarrow x = a.n\pi$

Here $\frac{dy}{dx} = \frac{b}{a}\cos\frac{x}{a}$; $\frac{d^2y}{dx^2} = -\frac{b}{a^2}\sin\frac{x}{a}$; $\frac{d^3y}{dx^3} = -\frac{b}{a^3}\cos\frac{x}{a}$

Now at $x = an\pi$, we have

$\frac{d^2y}{dx^2} = -\frac{b}{a^2}\sin n\pi = 0$ and $\frac{d^3y}{dx^3} = -\frac{b}{a^3}\cos n\pi \neq 0$

Therefore $x = an\pi$ is a point of inflexion of the curve. This is the point where the curve meets the x–axis.

Student Task

Q.1 Which of the following function has maximum value ?
(a) e^x (b) $\log x$ (c) $|x|$ (d) $\sin x$

Q.2 $f(c)$ is a minimum value of $f(x)$ if -
(a) $f'(c) = 0, f''(c) > 0$ (b) $f'(c) = 0, f''(c) < 0$
(c) $f'(c) \neq 0, f''(c) = 0$ (d) $f'(c) < 0, f''(c) > 0$

Q.3 The minimum value of $\log_a x + \log_x a$, $0 < x < a$, is
(a) 1 (b) 2 (c) –2 (d) None of these

Q.4 If for a function $f(x)$, $f'(b) = 0$, $f''(b) = 0$, $f'''(b) > 0$, then $x = b$ is -
(a) A maximum point (b) A minimum point
(c) An extreme point (d) Not an extreme point

Q.5 $x^3 - 3x + 4$ is minimum at -
(a) $x = 1$ (b) $x = -1$ (c) $x = 0$ (d) No where

Q.6 If $x = p$ and $x = q$ are respectively the maximum and minimum points of the function $x^5 - 5x^4 + 5x^3 - 10$, then-
(a) $p = 0, q = 1$ (b) $p = 1, q = 0$ (c) $p = 1, q = 3$ (d) $p = 3, q = 1$

Q.7 Which of the following function has no extreme point
(a) 2^x (b) $\log_{10} x$ (c) $x - [x]$ (d) All the above

Q.8 For the curve $y = xe^x$
(a) $x = 0$ is a point of maxima (b) $x = 0$ is a point of minima
(c) $x = -1$ is a point of minima (d) $x = -1$ is a point of maxima

Q.9 The sum of two non zero number is 6. The minimum value of the sum of their reciprocals is
(a) 3 (b) 6 (c) 2/3 (d) 6/5

Q.10 The semivertical angle of a right circular cone of given slant height and maximum volume is
(a) $\tan^{-1}2$ (b) $\tan^{-1}(\sqrt{2})$ (c) $\tan^{-1}(1/2)$ (d) $\tan^{-1}(1/\sqrt{2})$

Q.11 The minimum value of $27^{\cos 2x} \cdot 81^{\sin 2x}$ is
(a) 1/243 (b) – 5 (c) 1/5 (d) None of these

Q.12 The sides of the rectangle of greatest area which can be inscribed in the ellipse $\frac{x^2}{8} + \frac{y^2}{4} = 1$ are given by
(a) $(4, 2\sqrt{2})$ (b) $(2\sqrt{2}, 2)$ (c) $(2, \sqrt{2})$ (d) $(4\sqrt{2}, 4)$

Answer Key

1.	(d)	**2.**	(a)	**3.**	(b)	**4.**	(d)	**5.**	(a)	**6.**	(c)	**7.**	(d)	**8.**	(c)
9.	(c)	**10.**	(b)	**11.**	(a)	**12.**	(a)								

4. RATE OF CHANGE OF VARIABLE

Derivative as a rate measurer

Let $y = f(x)$ be a function of x. Let Δy be the change in y corresponding to a small change Δx in x.

Then, $\frac{\Delta y}{\Delta x}$ represents the change in y due to a unit change in x. In other words, $\frac{\Delta y}{\Delta x}$ represents the average rate of change of y w.r.t. x as x changes from x to $x + \Delta x$.

As $\Delta x \to 0$, the limiting value of this average rate of change of y with respect to x in the interval $[x, x + \Delta x]$ becomes the instantaneous rate of change of y w.r.t. x.

Thus, $\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$ = Instantaneous rate of change of y w.r.t. x

$\Rightarrow$ $\frac{\Delta y}{\Delta x}$ = Rate of change of y. w.r.t. x $\left[\because \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}\right]$

The word '' instantaneous'' is often dropped.

Hence, $\frac{\Delta y}{\Delta x}$ represents the rate of change of y w.r.t. x for a definite value of x.

(a) The value of $\frac{\Delta y}{\Delta x}$ at $x = x_0$ i.e. $\left(\frac{dy}{dx}\right)_{x=x_0}$ represents the rate of change of y with respect to x at $x = x_0$

(b) If $x = \phi(t)$ and $y = \psi(t)$, then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$, provided that $\frac{dx}{dt} \neq 0$

Thus, the rate of change of y with respect to x can be calculated by using the rate of change of y and that of x each with respect to t.

(c) The term "rate of change" will mean the instantaneous rate of change unless stated otherwise.

Related rates

Generally we come across with the problems in which the rate of change of one of the quantities involved is required corresponding to the given rate of change of another quantity. For example, suppose the rate of change of volume of a spherical balloon is required when the rate of change of its radius is given. In such type of problems we must find a relation connecting such quantities and differentiate this relation w.r.t. time.

Strategy for related rated problems

(i) Carefully read the problem.
(ii) Identify and name the variables. Draw a picture, if appropriate.
(iii) Determine what you want and what you know.
(iv) Find equations that relate the variables.
(v) Differentiate the equations with respect to the time variable.
(vi) Substitute the known values of the variables and rates in the original and differentiated equations to solve for what you want.

***Illustration 12*:** Find the rate of change of the area of a circle with respect to its radius. How fast is the area changing with respect to the radius when the radius is 3cm ?

Solution : Let A be the area of the circle. Then $A = \pi r^2 \Rightarrow \frac{dA}{dr} = 2\pi r$

Thus, the rate of change of the area of the circle w.r.t. its radius r is $2\pi r$ when $r = 3$ cm, we have $\frac{dA}{dr} = (2\pi \times 3)\ \text{cm}^2/\text{cm} = 6\pi \text{cm}^2/\text{cm}$

***Illustration 13*:** A balloon, which always remains spherical, has a variable diameter $\frac{3}{2}(2x + 3)$. Determine the rate of change of volume with respect to x.

Solution : Let V be the volume of the balloon. Then

$$V = \frac{4\pi}{3}\left\{\frac{3}{4}(2x+3)\right\}^3 = \frac{9\pi}{16}.(2x + 3)^3$$

$$\Rightarrow \frac{dV}{dx} = \frac{9\pi}{16} . 3(2x + 3)^2 \frac{d}{dx}(2x + 3) \Rightarrow \frac{dV}{dx} = \frac{27\pi}{8}(2x + 3)^2.$$

***Illustration 14*:** The radius of a circle is increasing uniformly at the rate of 4 cm/sec. Find the rate at which the area of the circle is increasing when the radius is 8 cm.

Solution : Let r be the radius and A be the area of a circle at any time t. Then,

$A = \pi r^2$ and $\frac{dr}{dt} = 4$ cm/sec

Now, $A = \pi r^2$

$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$;

$\left(\frac{dA}{dt}\right)_{r=8}$ = 2π × 8 × 4 cm^2/sec = 64 π cm^2/sec.

***Illustration 15*:** The radius of a balloon is increasing at the rate of 10 cm/sec. At what rate is the surface area of the balloon increasing when the radius is 15 cm ?

Solution : Let r be the radius and S be the surface area of the balloon at any time t.

Then, $S = 4\pi r^2$ and $\frac{dr}{dt}$ = 10 cm/sec

Now, $S = 4\pi r^2$

$\frac{dS}{dt} = 8\pi r \frac{dr}{dt}$; $\frac{dS}{dt} = 80\ \pi r$ $\left[\because \frac{dr}{dt} = 10 cm/\text{sec}.\right]$

$\Rightarrow \left(\frac{dS}{dt}\right)_{r=15}$ = 80π (15) = 120 π cm^2/sec

***Illustration 16*:** The volume of a cube is increasing at a rate of 7 cm^3/sec. How fast is the surface area increasing when the length of an edge is 12 cm ?

Solution : Let x be the length of an edge of the cube, V be the volume and S be the surface area of any time t. Then $V = x^3$ and $S = 6x^2$

Also, $\frac{dV}{dt}$ = 7 cm^3/sec [Given]

$\Rightarrow \frac{d}{dt}(x^3) = 7 \Rightarrow 3x^2 \frac{dx}{dt} = 7 \Rightarrow \frac{dx}{dt} = \frac{7}{3x^2}$

Now, $S = 6x^2 \Rightarrow \frac{dS}{dt} = 12x \frac{dx}{dt}$

$\Rightarrow \frac{dS}{dt} = 12x \times \frac{7}{3x^2}$ $\left[\because \frac{dx}{dt} = \frac{7}{3x^2}\right]$

$\Rightarrow \frac{dS}{dt} = \frac{28}{x} \Rightarrow \left(\frac{dS}{dt}\right)_{x=12} = \frac{28}{12}$ cm^2/sec. = $\frac{7}{3}$ cm^2/sec.

***Illustration 17*:** The displacement x of a particle at a time t is given by $t = \sqrt{x} + 3$. Find the displacement of the particle at the instant velocity is zero.

Solution : As $t = \sqrt{x} + 3$ or $\sqrt{x} = t - 3$

Squaring both sides $x = (t - 3)^2$ ....(1)

Differentiating at x w.r.t. time we get $v = \frac{dx}{dt} = 2(t - 3)$

velocity is zero at $2(t - 3) = 0$ or $t = 3$

$\therefore$ From eq. (1) at $t = 3$ the value of x is $x = (3 - 3)^2$ or $x = 0$

***Illustration 18*:** If the motion of a particle is given by $x = at^2$ and $y = bt^2$, then find the velocity of the particle.

Solution : $x = at^2$ .....(1)

and $y = bt^2$ .....(2)

Differentiating eq. (1) and (2) w.r.t. time we get $\frac{dx}{dt} = 2at$ & $\frac{dx}{dt} = 2bt$

Let $v_1 = 2at$ & $v_2 = 2bt$

$\therefore$ Resultant velocity $v = \sqrt{v_1^2 + v_2^2}$

or $v = \sqrt{4a^2t^2 + 4b^2t^2}$ or $v = 2t\sqrt{a^2 + b^2}$

***Illustration 19*:** The relation between displacement 'x' of a particle with time 't' is given by $t = \alpha x^2 + \beta x$, where α and β are constants. Find acceleration of the particle.

Solution : To get acceleration, we should have relation between v and t

To obtain v $\left(i.e.\ \frac{dx}{dt}\right)$ differentiate given equation w.r.t. 't'

Differentiating we get

$1 = 2\alpha x \frac{dx}{dt} + \beta \frac{dx}{dt}$ or $1 = \frac{dx}{dt}(2\alpha x + \beta)$

$\Rightarrow v = \frac{dx}{dt} = \frac{1}{2\alpha x + \beta}$

$$\Rightarrow a = \frac{dv}{dt} = \frac{(2\alpha x+\beta)\frac{d(1)}{dt} - (1)\frac{d}{dt}(2\alpha x+\beta)}{(2\alpha x+\beta)^2}$$

$$\text{or } a = -\frac{2\alpha \frac{dx}{dt}}{(2\alpha x+\beta)^2} = -\frac{2\alpha}{(2\alpha x+\beta)^3} \quad \left[\because v = \frac{dx}{dt} = \frac{1}{(2\alpha x+\beta)}\right] \text{ or } a = -2\alpha v^3$$

***Illustration 20*:** A particle starts from rest with a uniform acceleration. Its displacement after t seconds is given in metres by the relation : $x = 5 + 6t + 7t^2$. Calculate the magnitude of its (i) initial velocity (ii) velocity at $t = 3$ s (iii) uniform acceleration and (iv) displacement at $t = 5$s.

Solution : The velocity v at any instant is given by

$$v = \frac{dx}{dt} = \frac{d}{dt}(5 + 6t + 7t^2) = 0 + 6 \times 1 + 7 \times 2t = 6 + 14t$$

(i) For initial velocity, $t = 0$ $\therefore$ Initial velocity $= 6 + 14 \times 0 = 6$ m/s

(ii) When $t = 3$, velocity, $v = 6 + 14 \times 3 = 6 + 42 = 48$ m/s

(iii) Acceleration, a at any time t is

$$a = \frac{dv}{dt} = \frac{d}{dt}(6 + 14t) = 0 + 14 \times 1 = 14 \text{ m/s}^2$$

(iv) At t = 5s, displacement, $x = 5 + 6 \times 5 + 7 \times 5^2 = 5 + 30 + 175 = 210$m.

Student Task

Q.1 The abscissa changes at a faster rate than the ordinate on the curve $x^3 = 12y$. Then x lies in the interval.

(a) (–2, 2) (b) (–1, 1) (c) (0, 2) (d) None of these

Q.2 The side of an equilateral triangle are increasing at the rate of 2 cm/sec. The rate at which the area increases, when the side is 10 cm is -

(a) $\sqrt{3}$ sq. units/sec. (b) 10 sq. units/sec

. (c) $10\sqrt{3}$ sq. unit/sec. (d) $10\sqrt{3}$ sq. units/sec.

Q.3 Sand falls from a tube in such a way that it forms a cone whose height is always 4/3 times the radius of its base and radius of the base increases at the rate of 1/8 cm/sec. When this radius is 1 metre, the rate at which the amount of sand increases is

(a) $\frac{1000\pi}{3}$ c.c./sec. (b) $\frac{2000}{3}$ c.c./sec.

(c) $\frac{4000}{3}$ c.c./sec. (d) $\frac{5000}{3}$ c. c./sec.

Q.4 If x and y are sides of two squares such that $y = x - x^2$.The rate of change of area of the second square w.r.t. The first square is-

(a) $2(1 - x^2)x$ (b) $2x^2 - 3x + 1$ (c) $2(2x^2 - 3x + 1)$ (d) None of these

Q.5 If the rate of change of area of a circle is equal to the rate of change of its diameter, then its radius =

(a) $2/\pi$ (b) $1/\pi$ (c) $\pi/2$ (d) π

Q.6 The side of cube is equal to the diameter of a sphere. If the side and radius increase at the same rate then the ratio of the increase of their surface is -

(a) $\pi : 6$ (b) $2\pi : 3$ (c) $3 : 2\pi$ (d) $3 : \pi$

Q.7 The side of an equilateral triangle is '*a*' units and is increasing at the rate of *k* units/sec. Rate of increases of its area is

(a) $2/\sqrt{3}\,ak$ (b) $\sqrt{3}\,ak$ (c) $\sqrt{3}/2\,ak$ (d) None of these

Q.8 If the rate of change of volume of a sphere is equal to the rate of change of its radius then its radius =

(a) 1 (b) $\sqrt{2\pi}$ (c) $\frac{1}{\sqrt{2\pi}}$ (d) $\frac{1}{2\sqrt{\pi}}$

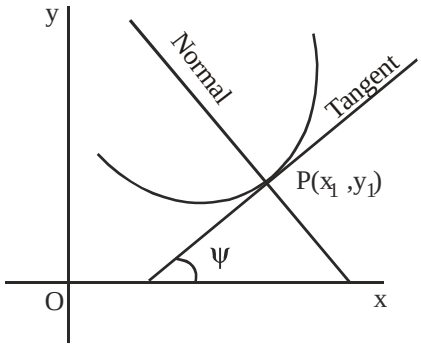
Answer Key

1. (a) **2.** (c) **3.** (d) **4.** (b) **5.** (b) **6.** (d) **7.** (b) **8.** (d)

5. SLOPES OF THE TANGENT AND THE NORMAL

Slope of the tangent :

Let $y = f(x)$ be a continuous curve, and let $P(x_1, y_1)$ be a point on it. Then, $\left(\frac{dy}{dx}\right)_p$ is the slope of the tangent to the curve $y = f(x)$ at point *P* i.e. ; $\left(\frac{dy}{dx}\right)_p = \tan\psi$ = slope of the tangent at P, where ψ is



the angle which the tangent at $P(x_1, y_1)$ makes with the positive direction of x-axis.

(i) The inclination of tangent with x–axis. = $\tan^{-1}\left(\frac{dy}{dx}\right)$

(ii) Slope of the tangent = $\frac{dy}{dx}$

(iii) If the tangent at $P(x_1,y_1)$ of the curve $y = f(x)$ is parallel to the x–axis (or perpendicular to y-axis) then $\Psi = 0$ i.e. its slope will be zero.

$m = \left(\frac{dy}{dx}\right)_{(x_1,y_1)} = 0$

The converse is also true. Hence the tangent at (x_1, y_1) is parallel to *x*–axis.

$\Rightarrow \left(\frac{dy}{dx}\right)_{(x_1,y_1)} = 0$

(iv) If the tangent at $P(x_1, y_1)$ of the curve $y = f(x)$ is parallel to y-axis (or perpendicular to x-axis) then $\Psi = \pi/2$, and its slope will be infinity i.e. $m = \left(\frac{dy}{dx}\right)_{(x_1,y_1)} = \infty$

The converser is also true. Hence the tangent at (x_1, y_1) is parallel to y–axis

$\Rightarrow \left(\frac{dy}{dx}\right)_{(x_1,y_1)} = \infty$

(v) If at any point $P(x_1, y_1)$ of the curve $y = f(x)$, the tangent makes equal angles with the axes, then at the point *P*, $\Psi = \pi/4$ or $3\pi/4$, Hence at *P*, $\tan\psi = dy/dx = \pm 1$.

The converse of the result is also true. thus at (x_1, y_1) the tangent line makes equal angles with the axes.

$$\Rightarrow \left(\frac{dy}{dx}\right)_{(x_1, y_1)} = \pm 1$$

Slope of the normal : The normal to a curve at $P(x_1, y_1)$ is a line perpendicular to the tangent at P and passing through P.

$\therefore$ Slope of the normal at $P = -\dfrac{1}{\text{slope of tangent at P}} = -\dfrac{1}{\left(\frac{dy}{dx}\right)_p} = -\left(\frac{dx}{dy}\right)_p$

Note :- If normal makes an angle of θ with positive direction of x-axis then $-\dfrac{dx}{dy} = \tan\theta$ or $\dfrac{dy}{dx} = -\cot\theta$

***Illustration 21*:** The point on the curve $y^2 = x^2 + ax + 25$ touches the axis of x are –

(a) ± 5 (b) ± 10 (c) ± 15 (d) None of these

Solution : **(b)** $\dfrac{dy}{dx} = 0$ as tangent is x - axis

$\therefore\ 2x + a = 0$ or $x = -a/2$. But point lies on the curve.

$y = x^2 + ax + 25 = \dfrac{100 - a^2}{4} = 0$ as it lies on x - axis. $\therefore\ a = \pm 10$

***Illustration 22*:** Find the slope of the tangent to the curve $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$ at the point $t = 2$.

Solution : We have, $\dfrac{dx}{dt} = 2t + 3$ and $\dfrac{dy}{dt} = 4t - 2$.

$\therefore\ \dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} = \dfrac{4t-2}{2t+3}$

Thus, slope of the tangent to the curve at the point t = 2 is

$$\left.\frac{dy}{dx}\right]_{t=2} = \frac{4(2)-2}{2(2)+3} = \frac{6}{7}$$

6. EQUATIONS OF TANGENT AND NORMAL

We know that the equation of a line passing through a point (x_1, y_1) and having slope m is

$y - y_1 = m(x - x_1)$

Slopes of the tangent and the normal to the curve $y = f(x)$ at a point $P(x_1, y_1)$ are $\left(\dfrac{dy}{dx}\right)_P$ and $-\dfrac{1}{\left(\frac{dy}{dx}\right)_P}$ respectively.

Therefore the equation of the tangent at $P(x_1, y_1)$ to the curve $y = f(x)$ is

$$y - y_1 = \left(\frac{dy}{dx}\right)_P (x - x_1) . \qquad \text{....(1)}$$

Since the normal at $P(x_1, y_1)$ passes through P and has slope $-\dfrac{1}{\left(\frac{dy}{dx}\right)_P}$, therefore the equation of the normal at $P(x_1, y_1)$ to the curve $y = f(x)$ is $y - y_1 = -\dfrac{1}{\left(\frac{dy}{dx}\right)_P}(x - x_1)$ ....(2)

(a) If $\left(\dfrac{dy}{dx}\right)_P = \infty$, then the tangent at (x_1, y_1) is parallel to y-axis and its equation is x_1.

(b) If $\left(\dfrac{dy}{dx}\right)_P = 0$, then the normal at (x_1, y_1) is parallel to y-axis and its equation is $x = x_1$.

***Illustration 23*:** Find the equation of tangent at the point 't' to the curve $x = a\cos^3 t$, $y = a\sin^3 t$.

Solution : $dx/dt = -3a\cos^2 t \sin t$;

$dy/dt = 3a\sin^2 t \cos t$

$$\Rightarrow \frac{dy}{dx} = \frac{3a\sin^2 t\cos t}{-3a\cos^2 t\sin t}$$

$$= -\frac{\sin t}{\cos t} \text{ (slope of tangent at point '}t\text{')}$$

Therefore the equation of tangent at the point 't' is written as

$$y - a\sin^3 t = -\frac{\sin t}{\cos t}(x - a\cos^3 t)$$

$$\Rightarrow \frac{y}{\sin t} - a\sin^2 t = -\frac{x}{\cos t} + a\cos^2 t \Rightarrow x\sec t + y\,\text{cosec}\, t = a$$

***Illustration 24*:** The equation of the normal to the curve $y(x-2)(x-3) - x + 7 = 0$ at the point where it cuts the x-axis is

(a) $x - 20y = 7$ (b) $20x - y = 7$ (c) $20x + y = 140$ (d) $20x - y = 140$

Solution : **(c)** $\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{y(2x-5)-1}{(x-2)(x-3)} = \frac{1}{20}$ at (7, 0)

$\therefore$ Slope of normal is -20 and its equation is $y - 0 = -20(x - 7)$

7. LENGTH OF INTERCEPTS MADE ON AXES BY THE TANGENT

Equation of tangent at any point (x_1, y_1) to the curve $y = f(x)$ is

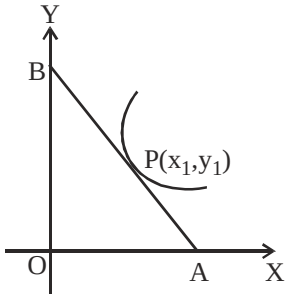
$$y - y_1 = \left(\frac{dy}{dx}\right)_{(x_1,y_1)}(x - x_1) \quad \text{....(1)}$$

Equation of x–axis $y = 0$ ....(2)

Equation of y–axis $x = 0$ .....(3)

Solving (1) and (2) , we get $x = x_1 - \left\{\frac{y_1}{(dy/dx)_{(x_1,y_1)}}\right\}$

$\therefore$ x-intercept : $OA = x_1 - \left\{\frac{y_1}{(dy/dx)_{(x_1,y_1)}}\right\}$



Similarly solving (1) and (3), we get y–intercept : $OB = y_1 - x_1\left(\frac{dy}{dx}\right)_{(x_1,y_1)}$

The length of intercept made by normal on x–axis is = $x_1 + y_1\left(\frac{dy}{dx}\right)$ and length of intercept on y-axis is $y_1 + x_1\left(\frac{dx}{dy}\right)$

***Illustration 25*:** At a point $(a/8, a/8)$ on the curve $x^{1/3} + y^{1/3} = a^{1/3}$ $(a > 0)$ tangent is drawn. If the axes be of length $\sqrt{2}$, then find the value of a.

(a) 1 (b) 2 (c) 4 (d) 8

Solution : (c) Slope of tangent $= -\frac{f_x}{f_y} = \frac{-x^{-2/3}}{y^{-2/3}} = -\left(\frac{y}{x}\right)^{2/3} = -1$ at $(a/8, a/8)$

Tangent is $y - a/8 = -1(x - a/8)$ or $x + y = a/4$

Its intercepts on axes are $A = a/4$, $B = a/4$

Portion of tangent intercepted between the axes is $\sqrt{A^2 + B^2} = \sqrt{2}$ (given)

$\therefore$ $\frac{a^2}{16} + \frac{a^2}{16} = 2$ or $a^2 = 16 \Rightarrow a = 4$

8. LENGTH OF PERPENDICULAR FROM ORIGIN TO THE TANGENT

The equation of tangent at point $P(x_1, y_1)$ of the given curve $y - y_1 = \left(\frac{dy}{dx}\right)_p (x - x_1)$

The length of perpendicular from origin (0,0) to the tangent drawn at the point (x_1, y_1) of the curve $y = f(x)$ is $p = \left|\frac{y_1 - x_1\left(\frac{dy}{dx}\right)}{\sqrt{1+\left(\frac{dy}{dx}\right)^2}}\right|$

The length of perpendicular from origin to normal is $P' = \left|\frac{x_1 + y_1\left(\frac{dy}{dx}\right)}{\sqrt{1+\left(\frac{dy}{dx}\right)^2}}\right|$

***Illustration 26*:** If p_1 and p_2 be the length of perpendiculars from the origin on the tangent and normal to the curve $x^{2/3} + y^{2/3} = a^{2/3}$ respectively, then find $4p_1^2 + p_2^2$.

Solution : Take the parametric equation as $x = a\sin^3\theta$, $y = a\cos^3\theta$

Tangent is $x\cos\theta + y\sin\theta = (a/2)\sin2\theta$. Normal is $y\cos\theta - x\sin\theta = a\cos 2\theta$

$$\therefore\ p_1 = \frac{(a/2)\sin2\theta}{\sqrt{(\cos^2\theta + \sin^2\theta)}} = \frac{a}{2}\sin 2\theta \quad ; \quad p_2 = \frac{a\cos2\theta}{\sqrt{(\cos^2\theta + \sin^2\theta)}} = a\cos2\theta$$

$$4p_1^2 + p_2^2 = a^2(\sin^2 2\theta + \cos^2 2\theta) = a^2$$

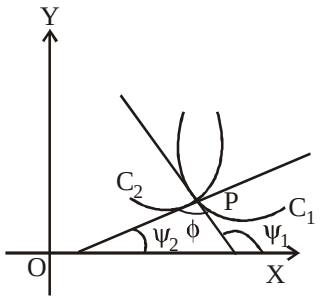
9. ANGLE OF INTERSECTION OF TWO CURVES

The angle of intersection of two curves is defined to be the angle between the tangents to the two curves at their point of intersection.

Let C_1 and C_2 be two curves having equations $y = f(x)$ and $y = g(x)$ respectively. Let PT_1 and PT_2 be tangents to the curves C_1 and C_2 respectively at their common point of intersection. Then the angle θ between PT_1 and PT_2 is the angle of intersection of C_1 and C_2 . Let ψ_1 and ψ_2 be the angles made by PT_1 and PT_2 with the positive direction of x-axis in anticlockwise sense. Then

$m_1 = \tan\psi_1$ = slope of the tangent to $y = f(x)$ at $P = \left(\frac{dy}{dx}\right)_{C_1}$.

and $m_2 = \tan\psi_2$ = slope of the tangent to $y = g(x)$ at $P = \left(\frac{dy}{dx}\right)_{C_2}$.



From figure it follow that $\phi = \psi_1 - \psi_2 \Rightarrow \tan\phi = \tan(\psi_1 - \psi_2)$

$$\Rightarrow \quad \tan\phi = \frac{\tan\psi_1 - \tan\psi_2}{1 + \tan\psi_1\tan\psi_2} \Rightarrow \tan\phi = \frac{\left(\frac{dy}{dx}\right)_{C_1} - \left(\frac{dy}{dx}\right)_{C_2}}{1 + \left(\frac{dy}{dx}\right)_{C_1}\left(\frac{dy}{dx}\right)_{C_2}}$$

The other angle between the tangents is $180° - \phi$. Generally the smaller of these two angles is taken to be the angle of intersection.

10. ORTHOGONAL CURVES

If the angle of intesection of two curves is a right angle, the two curve are said to intersect of orthogonal and the curves are called orthogonal curves.

If the curves are orthogonal, then $\phi = \frac{\pi}{2}$. $\therefore\ m_1 m_2 = -1 \Rightarrow \left(\frac{dy}{dx}\right)_{C_1} \cdot \left(\frac{dy}{dx}\right)_{C_2} = -1$

***Illustration 27*:** Find the angle of intersection of the parabola $y^2 = 4ax$ and $x^2 = 4ay$ at the origin.

Solution : $y^2 = 4ax \Rightarrow 2y\frac{dy}{dx} = 4a$ i.e. $\frac{dy}{dx} = \frac{2a}{y}$

At (0, 0), $\frac{dy}{dx} = \infty = \tan\frac{\pi}{2} = \tan\theta_1$ (say) $\quad\therefore\quad \theta_1 = \frac{\pi}{2}$

Again $x^2 = 4ay \Rightarrow 2x = 4a\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{x}{2a}$; A (0, 0), $\frac{dy}{dx} = 0 = \tan 0 = \tan\theta_2$ (say)

$\therefore\ \theta_2 = 0$

Again between the two curves = $|\theta_1 - \theta_2| = \frac{\pi}{2}$

11. LENGTH OF TANGENT, NORMAL, SUB TANGENT & SUB NORMAL

Let tangent and normal to the curve $y = f(x)$ at a point $P(x, y)$ meets the x–axis at point Q and R respectively. Then PQ and PR are called length of tangent and normal respectively, at point P. Also if PM be the perpendicular from p on x– axis, then QM and MR are called length of sub tangent and subnormal respectively at P. So from the diagram at $P(x, y)$

(i) length of tangent = $PQ = y\,\text{cosec}\,\Psi = y\sqrt{1+\left(\frac{dx}{dy}\right)^2}$

(ii) length of normal = $PR = y\sec\Psi = y\sqrt{1+\left(\frac{dy}{dx}\right)^2}$

(iii) length of sub tangent = $QM = y\cot\Psi = y\left(\frac{dx}{dy}\right)$

(iv) length of sub normal = $MR = y\tan\Psi = y\left(\frac{dy}{dx}\right)$

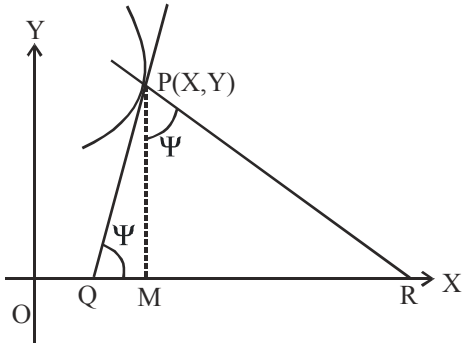


***Illustration 28*:** If $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$, then at $\theta = \pi/2$, find the length of the normal.

Solution : $\frac{dx}{d\theta} = a(1 + \cos\theta)$; $\frac{dy}{d\theta} = a\sin\theta \quad\therefore\quad \frac{dy}{dx} = \frac{\sin\theta}{1+\cos\theta} = \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}} = \tan\frac{\theta}{2}$

Length of the normal = $y\sqrt{1+\left(\frac{dy}{dx}\right)^2} = a(1-\cos\theta)\sqrt{1+\tan^2\frac{\theta}{2}} = a(1-\cos\theta).\sec\frac{\theta}{2}$

When $\theta = \frac{\theta}{2}$, Length = $a\left(1-\cos\frac{\pi}{2}\right).\sec\frac{\pi}{4} = a(1-0).\sqrt{2} = \sqrt{2}\,a$

Student Task

Q.1 If tangent at any point of the curve $y = f(x)$ makes equal intercepts with positive direction of coordinate axes, then at that point dy/dx equals –

(a) 1 (b) –1 (c) 0 (d) ∞

Q.2 If $ax + by + c = 0$ is a normal to the curve $xy = 1$, then

(a) $a > 0, b > 0$ (b) $a < 0, b < 0$

(c) $a > 0, b < 0$ or $a < 0, b > 0$ (d) $a, b \in R$

Q.3 The angle of intersection between the curves $y^2 = 4x$ and $x^2 + y^2 = 5$ at point (1, 2) is –

(a) $\pi/4$ (b) $\tan^{-1}3$ (c) $\tan^{-1}2$ (d) $\pi/2$

Q.4 If parametric equation of the curve is $x = e^t\cos t$, $y = e^t\sin t$, then the angle made by the tangent with x-axis at $t = \pi/4$ is

(a) 0 (b) $\pi/4$ (c) $\pi/3$ (d) $\pi/2$

Q.5 The distance between the origin and the normal to the curve $y = e^{2x} + x^2$ at the point $x = 0$ is

(a) $2\sqrt{5}$ (b) $2/\sqrt{5}$ (c) $\sqrt{5}$ (d) None of these

Q.6 The equation of tangent at any of the curve $x = at^2$, $y = 2at$ is

(a) $x = ty + at^2$ (b) $ty + x + at^2 = 0$ (c) $ty = x + at^2$ (d) $ty = x + at^3$

Q.7 The equation of the normal to the curve $y = x(2 - x)$ at the point (2, 0) is
(a) $x - 2y = 2$ (b) $x - 2y + 2 = 0$ (c) $2x + y = 4$ (d) $2x + y - 4 = 0$

Q.8 The length of intercepts on coordinate axes made by tangent to the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at the point $P(x_1, y_1)$ are
(a) $\sqrt{ax_1}, \sqrt{ay_1}$ (b) $\sqrt{a}, \sqrt{a}$ (c) $\sqrt{x_1}, \sqrt{y_1}$ (d) None of these

Q.9 If curve $y = 1 - ax^2$ and $y = x^2$ intersect orthogonally then the value of a is
(a) 1/2 (b) 1/3 (c) 2 (d) 3

Q.10 The equation of the normal to the curve $y^2 = 4ax$ at a point $(a, 2a)$ is -
(a) $x - y + a = 0$ (b) $x + y - 3a = 0$ (c) $x + 2y + 4a = 0$ (d) $x + y + 4a = 0$

Q.11 The length of perpendicular from (0,0) to the tangent drawn to the curve $y^2 = 4\,9\,(x + 2)$ at point (2, 4) is
(a) $1/\sqrt{2}$ (b) $3/\sqrt{5}$ (c) $6/\sqrt{5}$ (d) 1

Q.12 If curves $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $xy = C^2$ intersect orthogonally, then –
(a) $a + b = 0$ (b) $a^2 = b^2$ (c) $a + b = c$ (d) None of these

Answer Key

1.	(b)	**2.**	(c)	**3.**	(b)	**4.**	(d)	**5.**	(b)	**6.**	(c)	**7.**	(a)	**8.**	(a)
9.	(b)	**10.**	(b)	**11.**	(c)	**12.**	(b)								

12. ROLLE'S THEOREM

Definition

Let f be a real valued function defined on the closed interval $[a, b]$ such that,

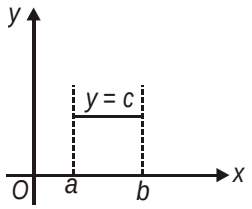
1. $f(x)$ is continuous in the closed interval $[a, b]$
2. $f(x)$ is differentiable in the open interval $]a, b[$ and
3. $f(a) = f(b)$

Analytical Interpretation

Now, Rolle's theorem is valid for a function such that

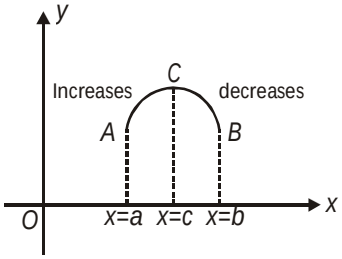
1. $f(x)$ is continuous in the closed interval $[a, b]$
2. $f(x)$ is differentiable in open interval $]a, b[$ and
3. $f(a) = f(b)$

So, generally two cases arises in such circumstances.



Case I : $f(x)$ is constant in the interval $[a, b]$ then $f'(x) = 0$ for all $x \in [a, b]$. Hence, Rolle's theorem follows, and we can say, $f'(c) = 0$, where $a < c < b$

Case II : $f(x)$ is not constant in the interval $[a, b]$ and since $f(a) = f(b)$.



The function should either increase or decrease when x takes values slightly greater than a.
Now, let $f(x)$ increases for $x > a$
Since, $f(a) = f(b)$, hence the function must size to increase at some value $x = c$ and decreasing upto $x = b$. Clearly at $x = c$ function has maximum value.
Now let h be a small positive quantity then, from definition of maximum value of the function,

$f(c+h) - f(c) < 0$ and $f(c-h) - f(c) < 0$

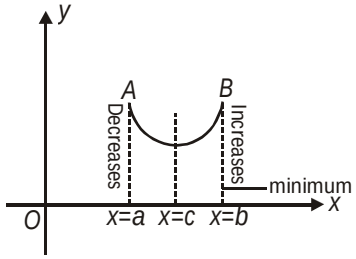
$\therefore \quad \frac{f(c+h)-f(c)}{h} < 0$ and $\frac{f(c-h)-f(c)}{-h} > 0$

So, $\lim_{h\to 0} \frac{f(c+h)-f(c)}{h} \le 0$ and $\lim_{h\to 0} \frac{f(c-h)-f(c)}{-h} \ge 0$ ..... (i)

But, if $\lim_{h\to 0} \frac{f(c+h)-f(c)}{h} \ne \frac{f(c-h)-f(c)}{-h}$,

The Rolle's theorem cannot be applicable because in such case,
RHD at $x = c \ne$ LHD at $x = c$.
Hence, $f(x)$ is not differentiable at $x = c$, which contradicts the condtion of Rolle's theorem.

$\therefore$ Only one possible solution arises, when $\lim_{h\to 0} \frac{f(c+h)-f(c)}{h} = \lim_{h\to 0} \frac{f(c-h)-f(c)}{-h} = 0$
Which implies that, $f(c) = 0$ where $a < c < b$. Hence, Rolle's theorem is proved.



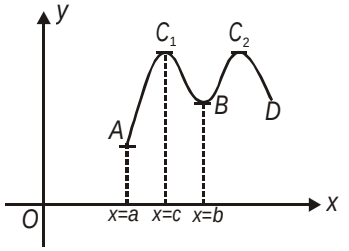
Similarly, the case where $f(x)$ decreases in the interval $a < x < c$ and then increases in the interval $c < x < b$, $f'(c) = 0$. But when $x = c$, the minimum value of $f(x)$ exists in the interval $[a, b]$.

Geometrical Interpretation

Consider the portion AB of the curve $y = f(x)$, lying between $x = a$ and $x = b$, such that

1. It goes continuously from A to B.
2. It has tangent at every point between A and B and
3. Ordinate of A = ordinate of B

From figure, it is clear that $f(x)$ increases in the interval AC_1, which implies that $f'(x) > 0$ in this region. Now, since there is unique tangent to be drawn on the curve lying in between A and B and since each of them has a unique slope i.e., unique value $f'(x)$.



$\therefore$ Due to continuity and differentiability of the function $f(x)$ in the region A to B. There is a point $x = c$ where $f'(c) = 0$. Hence, $f'(c) = 0$ where $a < c < b$
Thus Rolle's theorem is proved.

Similarly the other parts of the figure given above can be exlained, established Rolle's theorem throughout.

Note :-

1. To check the applicability of Rolle's theorem to a given function on a given interval.
2. To varify Rolle's theorem for a given function in a given interval.

In both types of problems we first check whether $f(x)$ satisfies the condition of Rolle's theorem or not.
The following results are vary helpful in doing so.

(i) A polynomial function is everywhere continuous and differentiable.

(ii) The exponential function, sine and cosine functions are everywhere continuous and differentiable.

(iii) Logarithimic functions is continuous and differentiable in its domain.

(iv) $\tan x$ is not continuous and differentiable at $x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, ...$

(v) $|x|$ is not differentiable at $x = 0$.

(vi) If $f'(x) = (2x-1)^{1/2}$, then $f'(x) = \frac{1}{\sqrt{2x-1}}$ is such that as $x \to \left(\frac{1}{2}\right)^{+} \Rightarrow f'(x) \to \infty$

So, $f(x)$ is not differentiable at $x = \frac{1}{2}$.

***Illustration 29*:** The function $f(x) = x(x+3)e^{-1/2x}$ satisfies all the condition of Rolle's theorem in $[-3, 0]$. The value of c is

(a) 0 (b) 1 (c) –2 (d) –3

Solution : **(c)** To determine 'c' in Rolle's theorem, $f'(c) = 0$.

Here $f'(c) = (x^2+3x)e^{-(1/2)x}\left(-\frac{1}{2}\right) + (2x+3)e^{-(1/2)x}$

$$\left\{-\frac{1}{2}(x^2+3x)+2x+3\right\} = -\frac{1}{2}e^{-(x/2)}\{x^2-x-6\}$$

$\therefore f'(c) = 0 \Rightarrow c^2 - c - 6 = 0 \Rightarrow c = 3, -2.$

But $c = 3 \notin [-2, 0]$, Hence $c = -2$.

***Illustration 30*:** If the function $f(x) = x^3 - 6x^2 + ax + b$ satisfies Rolle's theorem in the interval $[1, 3]$ and $f'\left(\frac{2\sqrt{3}+1}{\sqrt{3}}\right) = 0$ then

(a) $a = 11$ (b) $a = -6$ (c) $a = 6$ (d) $a = 1$

Solution : **(a)** $f(x) = x^3 - 6x^2 + ax + b \Rightarrow f'(x) = 3x^2 - 12x + a$

$$\Rightarrow f'(c) = 0 \Rightarrow f'\left(2+\frac{1}{\sqrt{3}}\right) = 0 \Rightarrow 3\left(2+\frac{1}{\sqrt{3}}\right)^2 - 12\left(2+\frac{1}{\sqrt{3}}\right) + a = 0$$

$$\Rightarrow 3\left(4+\frac{1}{3}+\frac{4}{\sqrt{3}}\right) - 12\left(2+\frac{1}{\sqrt{3}}\right) + a = 0 \Rightarrow 12+1+4\sqrt{3}-24-4\sqrt{3}+a = 0$$

Student Task

Q.1 Rolle's theorem is true for the function $f(x) = x^2 - 4$ in the interval

(a) $[-2, 0]$ (b) $[-2, 2]$ (c) $\left[0, \frac{1}{2}\right]$ (d) $[0, 2]$

Q.2 For which interval, the function $\frac{x^2-3x}{x-1}$ satisfies at the conditions of Rolle's theorem

(a) $[0, 3]$ (b) $[-3, 0]$ (c) $[1.5, 3]$ (d) For no interval

Q.3 If $f(x)$ satisfies the conditions of Rolle's theorem in $[1, 2]$ and $f(x)$ is continuous in $[1, 2]$ then $\int_1^2 f'(x)dx$ is equal to

(a) 3 (b) 0 (c) 1 (d) 2

Q.4 Condider the function $f(x) = e^{-2x} \sin 2x$ over the interval $\left(0, \frac{\pi}{2}\right)$. A real number $c \in \left(0, \frac{\pi}{2}\right)$, as guaranted by Roll's theorem, such that $f'(c) = 0$ is

(a) $\frac{\pi}{8}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$

Q.5 If the function $f(x) = ax^3 + bx^2 + 11x - 6$ satisfies the conditions of Rolle's theorem for the interval $[1, 3]$ and $f'\left(2+\frac{1}{\sqrt{3}}\right) = 0$, then the values of a and b are respectively.

(a) 1, –6 (b) –2, 1 (c) $-1, \frac{1}{2}$ (d) –1, 6

Q.6 Rolle's theorem is not applicable to the function $f(x) = |x|$ defined on [–1, 1] because
(a) f is not continuous on [–1, 1] (b) f is not differentiable on (–1, 1)
(c) $f(-1) \neq f(1)$ (d) $f(-1) = f(1) \neq 0$

Q.7 Let $f(x) = \begin{cases} x^{\alpha} \ln x, & x > 0 \\ 0, & x = 0 \end{cases}$ Rolle's theorem is applicable to f for $x \in [0, 1]$, if α =
(a) –2 (b) –1 (c) 0 (d) 1/2

Q.8 The value of a for which the equation $x^3 - 3x + a = 0$ has two distinct roots in [0, 1] is given by
(a) –1 (b) 1 (c) 3 (d) None of these

Q.9 Let a, be two distinct roots of a polynomial $f(x)$. Then there exists at least one root lying between a and b of the polynomial
(a) $f(x)$ (b) $f'(x)$ (c) $f''(x)$ (d) None of these

Q.10 If $\frac{a_0}{n+1} + \frac{a_1}{n} + \frac{a_2}{n-1} + ... + \frac{a_{n-1}}{2} + a_n = 0$. Then the function $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + ...+ a_n$ has in (0, 1)
(a) At least one zero (b) At most one zero
(c) Only 3 zeros (d) Only 2 zeros

Answer Key

1.	(b)	2.	(d)	3.	(b)	4.	(a)	5.	(a)	6.	(b)	7.	(d)	8.	(d)
9.	(b)	10.	(a)												

13. LAGRANGE'S MEAN VALUE THEOREM

If a function $f(x)$,
1. Is continuous in the closed interval $[a, b]$ and
2. Is differentiable in the open interval (a, b)

Then there is atleast one value $c \in (a, b)$, such that ; $f'(c) = \frac{f(b) - f(a)}{b - a}$

Analytical Interpretation

First form : Consider the function, $\phi(x) = f(x) - \frac{f(b) - f(a)}{b - a}x$

Since, $f(x)$ is continuous in $[a, b]$

$\therefore$ $f(x)$ is also continuous in $[a, b]$

since, $f'(x)$ exists in (a, b) hence $\phi'(x)$ also exists in (a, b) and $\phi'(x) = f'(x) - \frac{f(b) - f(b)}{b - a}$ ..... (i)

Clearly, $\phi(x)$ satisfies all the condition of Rolle's theorem

$\therefore$ There is atleast one value of c and x between a and b such that $f'(c) = 0$ substituting $x = c$ in (1) we get, $f'(x) = \frac{f(b) - f(a)}{b - a}$ which proves the theorem.

Second form : If we write $b = a + h$ then $a < c < b$, $c = a + \theta h$ where $0 < \theta < 1$

Thus, the mean value theorem can be stated as follows : If

(i) $f(x)$ is continuous in closed interval $[a, a + h]$

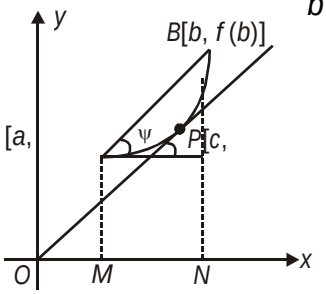
(ii) $f'(x)$ exists in the open interval $(a, a + h)$ then there exists at least one number $\theta (0 < \theta < 1)$ Such that $f(a + h) = f(a) + hf'(a + \theta h)$.

Geometrical Interpretation

Let $f(x)$ be a function defined on $[a, b]$ and let APB be the curve represented by $x = f(x)$. Then co-ordinates of A and B are $(a, f(a))$ and $(b, f(b))$ respectively. Suppose the chord AB makes an angle ψ with the axis of x. Then from the triangle ARB, we have

$$\tan\psi = \frac{BR}{AR} \Rightarrow \tan\psi = \frac{f(b) - f(a)}{b - a}$$

By Lagrange's Mean value theorem, we have, $f'(c)=\frac{f(b)-f(a)}{b-a}$ $\therefore$ $\tan\psi = f'(c)$



$\Rightarrow$ Slope of the chord AB = slope of the tangent at $(c, f(c))$

***Illustration 31*:** In the mean-value theorem $\frac{f(b)-f(a)}{b-a} = f'(c)$, if $a = 0$, $b=\frac{1}{2}$ and $f(x) = x(x-1)(x-2)$, the value of c is

(a) $1-\frac{\sqrt{15}}{6}$ (b) $1+\sqrt{15}$ (c) $1-\frac{\sqrt{21}}{6}$ (d) $1+\sqrt{21}$

Solution : **(c)** From mean value theorem $f'(c)=\frac{f(b)-f(b)}{b-a}$

$a = 0, f(a) = 0 \Rightarrow b = \frac{1}{2}, f(b) = \frac{3}{8}$

$f'(x) = (x-1)(x-2) + x(x-2) + x(x-1)$, $f'(c) = (c-1)(c-2) + c(c-2) + c(c-1)$

$= c^2 - 3c + 2 + c^2 - 2c + c^2 - c$, $f'(c) = 3c^2 - 6c + 2$

According to mean value theorem

$$\Rightarrow f'(c)=\frac{f(b)-f(a)}{b-a} \Rightarrow 3c^2-6c+2=\frac{\left(\frac{3}{8}\right)-0}{\left(\frac{1}{2}\right)-0}=\frac{3}{4} \Rightarrow 3c^2-6c+\frac{5}{4}=0$$

$$c=\frac{6\pm\sqrt{36-15}}{2\times 3}=\frac{6\pm\sqrt{21}}{6}=1\pm\frac{\sqrt{21}}{6}$$

***Illustration 32*:** From value theorem $f(b) - f(a) = (b-a)f'(x_1)$, $a < x_1 < b$ if $f(x) = \frac{1}{x}$ then x_1

(a) $\sqrt{ab}$ (b) $\frac{2ab}{a+b}$ (c) $\frac{a+b}{2}$ (d) $\frac{b-a}{b+a}$

Solution : **(a)** $f'(x_1)=\frac{-1}{x_1^2}$, $\therefore$ $\frac{-1}{x_1^2}=\frac{\frac{1}{b}-\frac{1}{a}}{b-a}=-\frac{1}{ab} \Rightarrow x_1=\sqrt{ab}$

***Illustration 33*:** The abscissae of the points of the curve $y = x^3$ in the interval $[-2, 2]$, where the slope of the tangent can be obtained by mean value theorem for the interval $[-2, 2]$ are

(a) $\pm\frac{2}{\sqrt{3}}$ (b) $\pm\frac{\sqrt{3}}{2}$ (c) $\pm\sqrt{3}$ (d) 0

Solution : **(a)** Given that equation of curve $y = x^3 = f(x)$

So $f(2) = 3x^2 \Rightarrow f'(x) = \frac{f(2)-f(-2)}{2-(-2)}=\frac{8-(-8)}{4}=3x^2$; $\therefore x=\pm\frac{2}{\sqrt{3}}$

Student Task

Q.1 In the function $f(x) = x^3 - 6ax^2 + 5x$ satisfies the conditions of Lagrange's mean value theorem for the interval [1, 2] and the tangent to the curve $y = f(x)$ at $x=\frac{7}{4}$ is parallel to the chord that joins the points of intersection of the curve with the ordinates $x = 1$ and $x = 2$. Then the value of a is

(a) 35/16 (b) 35/48 (c) 7/16 (d) 5/16

Q.2 If $f(x) = \cos x$, $0 \le x \le \frac{\pi}{2}$, then the real number '$c$' of the mean value theorem is

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\sin^{-1}\left(\frac{2}{\pi}\right)$ (d) $\cos^{-1}\left(\frac{2}{\pi}\right)$

Q.3 Let $f(x)$ satisfy all the conditions of mean value theorem in [0, 2]. If $f(0) = 0$ and $|f'(x)| \le \frac{1}{2}$ for all x in [0, 2], then

(a) $f(x) \le 2$ (b) $|f(x)| \le 1$

(c) $f(x) = 2x$ (d) $f(x) = 3$ for at least one x in [0, 2]

Q.4 The function $f(x) = (x-3)^2$ satisfies all the conditions of mean value theorem in [3, 4]. A point on $y = (x-3)^2$, where the tangent is parallel to the chord joining (3, 0) and (4, 1) is

(a) $\left(\frac{7}{2}, \frac{1}{2}\right)$ (b) $\left(\frac{7}{2}, \frac{1}{4}\right)$ (c) (1, 4) (d) (4, 1)

Q.5 Let $f(x)$ and $g(x)$ are defined and differentiable for $x \le x_0$ and $f(x_0) = g(x_0)$, $f'(x) > g'(x)$ for $x > x_0$, then

(a) $f(x) < g(x)$ for some $x > x_0$ (b) $f(x) = g(x)$ for some $x > x_0$

(c) $f(x) > g(x)$ for all $x > x_0$ (d) None of these

Q.6 Let f be differentiable for all x. If $f(1) = -2$ and $f'(x) \le 2$ for all $x \in [1, 6]$ then

(a) $f(6) < 8$ (b) $f(6) \ge 8$ (c) $f(6) \ge 5$ (d) $f(6) \le 5$

Q.7 The value of c in Lagrange's theorem for the function $|x|$ in the interval $[-1, 1]$ is

(a) 0 (b) 1/2

(c) –1/2 (d) Non-existent in the Interval

Answer Key

1. (b) 2. (c) 3. (b) 4. (b) 5. (c) 6. (b) 7. (d)

14. USEFUL TIPS

1. A function is said to be monotonic if it's either increasing or decreasing.

2. The points for which $f'(x)$ is equal to zero or doesn't exist are called critical points. Here it should also be noted that critical points are the interior points of the domain of the function.

3. The stationary points are the points of the domain where $f'(x) = 0$.

4. If $f''(x) = 0$ or dose not exist at points where $f'(x)$ exists and if $f''(x)$ changes sign when passing through $x = x_0$ and $f'(x)$ doesn't change its sign then x_0 is called a point of inflexion.
If $f''(x) < 0$, $x \in (a, b)$ then the curve $y = f(x)$ is convex in (a, b)
If $f''(x) > 0$, $x \in (a, b)$ then the curve $y = f(x)$ is concave in (a, b)
At the point of inflexion, the curve changes its concavity.

5. If the tangent at any point P on the curve is parallel to the axis of x then $dy/dx = 0$ at the point P.

6. If the tangent at any point on the curve is parallel to the axis of y, then $dy/dx = \infty$ or $dy/dx = 0$.

7. If the tangent at any point P on the curve is equally inclined to both the axes then $dy/dx = \pm 1$.

8. If the tangent at any point makes equal intercept on the coordinate axes then $dy/dx = -1$.

9. Tangent to a curve at the point $P(x_1, y_1)$ can be drawn even through dy/dx at P does not exist. eg. $x = 0$ is a tangent to $y = x^{2/3}$ at (0, 0).

10. If a curve passing through the origin be given by a rational integral algebraic equation, the equation of the tangent (or tangents) at the origin is obtained by equating to zero the terms of the lowest degree in the equation, e.g. If the equation of a curve be $x^2 - y^2 + x^3 + 3x^2y - y^3 = 0$, the tangents at the origin are given by $x^2 - y^2 = 0$ i.e. $x + y = 0$ and $x - y = 0$.

11. The curves $x^{2/3} + y^{2/3} = c^{2/3}$ and $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ touch if $c = a + b$.

12. The curves $\frac{x^2}{a^2+K_1}+\frac{y^2}{b^2+K_1}=1$ and $\frac{x^2}{a^2+K_2}+\frac{y^2}{b^2+K_2}=1$ intersect orthogonally.

13. In a triangle *ABC*, the side '*c*' and the angle '*C*' remain constant, while the remaining elements are changed slightly, then $\frac{da}{\cos A}+\frac{db}{\cos B}=0$.

14. If a function is invertible it has to be either increasing or decreasing.

15. If a function is continuous the intervals in which it rises and falls may be separated by points at which its derivative fails to exist.

16. If f is increasing in [*a*, *b*] and is continuous then *f* (*b*) is the greatest and *f* (*c*) is the least value of *f* in [*a*, *b*]. Similarly if *f* is decreasing in [*a*, *b*] then *f* (*a*) is the greatest value and *f* (*b*) is the least value.

17. Given a fixed point *A* (x_1, y_1) and a moving point *P* (*x*, *f* (*x*)) on the curve *y* = *f* (*x*). Then *AP* will be maximum or minimum if it is normal to the curve at *P*.

18. If the sum of two positive numbers *x* and *y* is constant than their product is maximum if they are equal, i.e. $x + y = c$, $x > 0$, $y > 0$, then $xy=\frac{1}{4}[(x+y)^2-(x-y)^2]$

19. Radius of curvature $= \dfrac{d^2y/dx^2}{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}$

20. If the product of two positive numbers is constant then their sum is least if they are equal.
i.e., $(x + y)^2 = (x – y)^2 + 4xy$.

21. The subnormal at any point on the curve $xy^n = a^{n+1}$ is constant for $n = – 2$

22. The lower corner of a least in a book is folded over so as to just reach the inner edge of the page. The fraction of width folded over if the area of the folded part is minimum is 2/3.

23 The least area of circle circumscribing any right triangle of area *S* is πS.

24. The ordinate of all points on the curve $y=\frac{1}{2\sin^2 x+3\cos^2 x}$ where the tangent is horizontal, is 1/2 or 1/3 according as n is an odd or an even integer.

EXERCISE-1

Q.1 Point '*A*' lies on the curve $y = e^{-x^2}$ and has the coordinate $\left(x, e^{-x^2}\right)$ where $x > 0$. Point B has the coordinates (*x*, 0). If *O* is the origin then the maximum area of the triangle *AOB* is

(a) $\frac{1}{\sqrt{2e}}$ (b) $\frac{1}{\sqrt{4e}}$ (c) $\frac{1}{\sqrt{e}}$ (d) $\frac{1}{\sqrt{8e}}$

Q.2 The angle at which the curve $y = Ke^{Kx}$ intersects the y-axis is

(a) $\tan^{-1} K^2$ (b) $\cot^{-1}(K^2)$ (c) $\sec^{-1}\left(\sqrt{1+K^4}\right)$ (d) None of these

Q.3 The angle between the tangent lines to the graph of the function $f(x) = \int_2^x (2t-5)dt$ at the points where the graph cuts the x-axis

(a) $\pi/6$ (b) $\pi/4$ (c) $\pi/3$ (d) $\pi/2$

Q.4 The equation sin x + x cos x = 0 has at least one root in

(a) $\left(-\frac{\pi}{2}, 0\right)$ (b) $(0, \pi)$ (c) $\left(\pi, \frac{3\pi}{2}\right)$ (d) $\left(0, \frac{\pi}{2}\right)$

Q.5 The minimum value of $\frac{\tan\left(x+\frac{\pi}{6}\right)}{\tan x}$ is

(a) 0 (b) 1/2 (c) 1 (d) 3

Q.6 If $a < b < c < d$ and $x \in R$ then the least value of the function, $f(x) = |x-a| + |x-b| + |x-c| + x-d|$ is

(a) $c-d+b-a$ (b) $c+d-b-a$ (c) $c+d-b+a$ (d) $c-d+b+a$

Q.7 If a variable tangent to the curve $x^2y = c^3$ makes intercepts a, b on x andy axis respectively, then the value of a^2b is

(a) $27\,c^3$ (b) $\frac{4}{27}c^3$ (c) $\frac{27}{4}c^3$ (d) $\frac{4}{9}c^3$

Q.8 Let $f(x) = \begin{vmatrix} 1 & 1 & 1 \\ 3-x & 5-3x^2 & 3x^3-1 \\ 2x^2-1 & 3x^5-1 & 7x^8-1 \end{vmatrix}$. Then the equation $f(x) = 0$ has

(a) No real root
(b) Atmost one real roots
(c) Atleast 2 real roots
(d) Exactly one real root in (0, 1) and no other real root

Q.9 Difference between the greatest and the least values of the function $f(x) = x(\ell n\, x - 2)$ on $[1, e^2]$ is

(a) 2 (b) e (c) e^2 (d) 1

Q.10 The function $f : [a, \infty) \to R$ where R denotes the range corresponding to the given domain, with rule $f(x) = 2x^3 - 3x^2 + 6$, will have an inverse provided

(a) $a \geq 1$ (b) $a \geq 0$ (c) $a \leq 0$ (d) $a \leq 1$

Q.11 The graphs $y = 2x^3 - 4x + 2$ and $y = x^3 + 2x - 1$ intersect in exactly 3 distinct points. The slope of the line passing through two of these points

(a) Is equal to 4 (b) Is equal to 6 (c) Is equal to 8 (d) Is not unique

Q.12 In which ofthe following functions Rolle's theorem is applicable?

(a) $f(x) = \begin{cases} x & , \quad 0 \leq x < 1 \\ 0 & , \quad x = 1 \end{cases}$ on [0, 1]

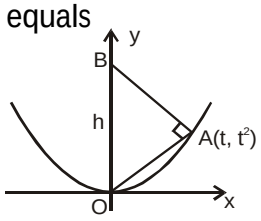
(b) $f(x) = \begin{cases} \frac{\sin x}{x} & , \quad -\pi \leq x < 0 \\ 0 & , \quad x = 0 \end{cases}$ on $[-\pi, 0]$

(c) $f(x) = \frac{x^2-x-6}{x-1}$ on [–2, 3]

(d) $f(x) = \begin{cases} \frac{x^3-2x^2-5x+6}{x-1} & \text{if} \quad x \neq 1 \\ -6 & \text{if} \quad x = 1 \end{cases}$ on [–2, 3]

Q.13 The figure shows a right triangle with its hypotenuse OB along the y-axis and its vertex A on the parabola $y = x^2$. Let h represents the length of the hypotenuse which depends on the x-coordinate

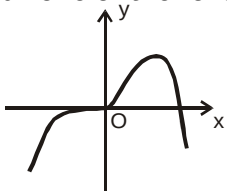
of the point *A*. The value of $\lim_{x\to 0}(h)$ equals

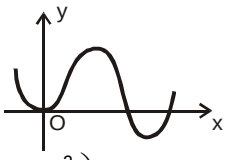


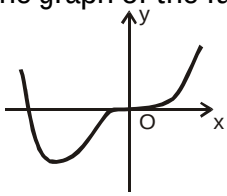
(a) 0 (b) 1/2 (c) 1 (d) 2

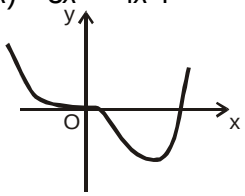
Q.14 Number of positive integral values of '*a*' for which the curve $y = a^x$ intersects the line $y = x$ is
(a) 0 (b) 1 (c) 2 (d) More than 2

Q.15 Which one of the following can best represent the graph of the function $f(x) = 3x^4 - 4x^3$?

(a) 

(b) 

(c) 

(d) 

Q.16 The function $f(x) = \tan^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is
(a) Increasing in its domain
(b) Decreasing in its domain
(c) Decreasing in $(-\infty, 0)$ and increasing in $(0, \infty)$
(d) Increasing in $(-\infty, 0)$ and decreasing in $(0, \infty)$

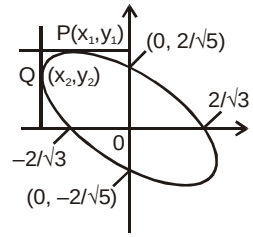
Q.17 The tangent to the graph of the function $y = f(x)$ at the point with abscissa $x = a$ forms with the x-axis an angle of $\pi/3$ and at the point with abscissa $x = b$ at an angle of $\pi/4$, then the value of the integral, $\int_a^b f'(x).f''(x)dx$ is equal to
(a) 1 (b) 0 (c) $-\sqrt{3}$ (d) –1
[assume $f''(x)$ to be continuous]

Q.18 Let *C* be the curve $y = x^3$ (where *x* takes all real values). The tangent at *A* meets the curve again at *B*. If the gradient at *B* is *K* times the gradient at *A* then *K* is equal to
(a) 4 (b) 2 (c) –2 (d) 1/4

Q.19 Which one of the following statements does not hold good for the function $f(x) = \cos^{-1}(2x^2 - 1)$?
(a) *f* is not differentiable at $x = 0$ (b) *f* is monotonic
(c) *f* is even (d) *f* has an extremum

Q.20 The length of the shortest path that begins at the point (2, 5), touches the x-axis and then ends at a point on the circle $x^2 + y^2 + 12x - 20y + 120 = 0$, is
(a) 13 (b) $4\sqrt{10}$ (c) 15 (d) $6 + \sqrt{89}$

Q.21 The lines $y = -\frac{3}{2}x$ and $y = -\frac{2}{5}x$ intersect the curve $3x^2 + 4xy + 5y^2 - 4 = 0$at the points *P* and *Q* respectively. The tangents drawn to the curve at *P* and *Q*



(a) Intersect each other at angle of 45°
(b) Are parallel to each other

(c) are perpendicular to each other
(d) None of these

Q.22 The bottom of the legs of a three legged table are the vertices of an isosceles triangle with sides 5, 5 and 6. The legs are to be braced at the bottom by three wires in the shape of a *Y*. The minimum length of the wire needed for this purpose, is
(a) $4 + 3\sqrt{3}$ (b) 10 (c) $3 + 4\sqrt{3}$ (d) $1 + 6\sqrt{2}$

Q.23 The least value of a for which the equation, $\frac{4}{\sin x} + \frac{1}{1-\sin x} = a$ has atleast one solution on the interval $(0, \pi/2)$ is
(a) 3 (b) 5 (c) 7 (d) 9

Q.24 If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{bmatrix} \min\{f(t) : 0 \le t \le x\} & ; & 0 \le x \le 1 \\ 3 - x & ; & 1 < x \le 2 \end{bmatrix}$, then $g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)$ has the value equal to
(a) $\frac{7}{4}$ (b) $\frac{9}{4}$ (c) $\frac{13}{4}$ (d) $\frac{5}{2}$

Q.25 Given : $f(x) = 4 - \left(\frac{1}{2} - x\right)^{2/3}$ $g(x) = \begin{cases} \frac{\tan[x]}{x} & , \ x \neq 0 \\ 1 & , \ x = 0 \end{cases}$

$h(x) = \{x\}$ $k(x) = 5^{\log_2(x+3)}$

then in [0, 1], Lagranges Mean Value Theorem is not applicable to
(a) *f*, *g*, *h* (b) *h*, *k* (c) *f*, *g* (d) *g*, *h*, *k*
where [*x*] and {*x*} denotes the G.I.F. & F.P.F.

Q.26 If the function $f(x) = x^4 + bx^2 + 8x + 1$ has a horizontal tangent and a point of inflection for the same value of x then the value of b is equal to
(a) –1 (b) 1 (c) 6 (d) –6

Q.27 $f(x) = \int \left(2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{1+x^2}}\right) dx$ then f is
(a) Increasing in $(0, \infty)$ and decreasing in $(-\infty, 0)$
(b) Increasing in $(-\infty, 0)$ and decreasing in $(0, \infty)$
(c) Increasing in $(-\infty, \infty)$
(d) Decreasing in $(-\infty, \infty)$

Q.28 The lower corner of a leaf in a book is folded over so as to just reach the inner edge of the page. The fraction of width folded over if the area of the folded part is minimium is
(a) 5/8 (b) 2/3 (c) 3/4 (d) 4/5

Q.29 A closed vessel tapers to a point both at its top E and its bottom F and is fixed with *EF* vertical when the depth of the liquid in it is x cm, the volume of the liquid in it is, $x^2(15 - x)$ *cu*. cm. The length EF is
(a) 7.5 cm (b) 8 cm (c) 10 cm (d) 12 cm

Q.30 Coffee is draining from a conical filter, height and diameter both 15 cms into a cylinderical coffee pot diameter 15 cm. The rate at which coffee drains from the filter into the pot is 100 cu cm/min. The rate in cms/min at which the level in the pot is rising at the instant when the coffee in the pot is 10 cm, is
(a) $\frac{9}{16\pi}$ (b) $\frac{25}{9\pi}$ (c) $\frac{5}{3\pi}$ (d) $\frac{16}{9\pi}$

Q.31 The true set of real values of x for which the function, $f(x) = x\,\ell n\,x - x + 1$ is positive is
(a) $(1, \infty)$ (b) $(1/e, \infty)$ (c) $[e, \infty)$ (d) $(0, 1) \cup (1, \infty)$

Q.32 A horse runs along a circle with a speed of 20 km/hr. A lantern is at the centre of the circle. A fence is along the tangent to the circle at the point at which the horse starts. The speed with which the shadow of the horse move along the fence at the moment when it covers 1/8 of the circle in km/hr is
(a) 20 (b) 40 (c) 30 (d) 60

Q.33 Give the correct order of initials T and F for following statements. Use T if statement is true andF if it is false.
Statement–1 : If $f : R \to R$ and $c \in R$ is such that f is increasing in $(c - \delta, c)$ and f is decreasing in $(c, c + \delta)$ then f has a local maximum at c. Where δ is a sufficiently small positive quantity.
Statement–2: Let $f : (a, b) \to R, c \in (a, b)$. Then f can not have both a local maximum and a point of inflection at $x = c$.
Statement–3 : The function $f(x) = x^2|x|$ is twice differentiable at $x = 0$.
Statement–4 : Let $f : [c - 1, c + 1] \to [a, b]$ be bijective map such that f is differentiable at c then f^{-1} is also differentiable at $f(c)$.
(a) FFTF (b) TTFT (c) FTTF (d) TTTF

Q.34 A curve is represented by the equations, $x = \sec^2 t$ and $y = \cot t$ where t is a parameter. If the tangent at the point P on the curve where $t = \pi/4$ meets the curve again at the point Q then $|PQ|$ is equal to
(a) $\frac{5\sqrt{3}}{2}$ (b) $\frac{5\sqrt{5}}{2}$ (c) $\frac{2\sqrt{5}}{3}$ (d) $\frac{3\sqrt{5}}{2}$

Q.35 Water runs into an inverted conical tent at the rate of 20 cubic feet per minute and leaks out at the rate of 5 cubic feet per minute. The height of the water is three times the radius of the water's surface. The radius of the water surface is increasing when the radius is 5 feet, is
(a) $\frac{1}{5\pi}$ ft/min (b) $\frac{1}{10\pi}$ ft/min (c) $\frac{1}{15\pi}$ ft/min (d) None of these

Q.36 The set of values of p for which the equation $|\ell n\,x| - px = 0$ possess three distinct roots is
(a) $\left(0, \frac{1}{e}\right)$ (b) (0, 1) (c) (1, e) (d) (0, e)

Q.37 The lateral edge of a regular rectangular pyramid is 'a' cm long. The lateral edge makes an angle α with the plane of the base. The value of α for which the volume of the pyramid is greatest, is
(a) $\frac{\pi}{4}$ (b) $\sin^{-1}\sqrt{\frac{2}{2}}$ (c) $\cot^{-1}\sqrt{2}$ (d) $\pi/3$

Q.38 In a regular triangular prism the distance from the centre of one base to one of the vertices of the other base is ℓ. The altitude of the prism for which the volume is greatest
(a) $\frac{\ell}{2}$ (b) $\frac{\ell}{\sqrt{3}}$ (c) $\frac{\ell}{3}$ (d) $\frac{\ell}{4}$

Q.39 Let $f(x) = \begin{cases} x^{3/5} & \text{if } x \le 1 \\ -(x-2)^3 & \text{if } x > 1 \end{cases}$, then the number of critical points on the graph of the function is
(a) 1 (b) 2 (c) 3 (d) 4

Q.40 Number of roots of the equation $x^2 . e^{2-|x|} = 1$ is
(a) 2 (b) 4 (c) 6 (d) Zero

Q.41 The point(s) at each of which the tangents to the curve $y = x^3 - 3x^2 - 7x + 6$ cut off on the positive semi axis OX a line segment half that on the negative semi axis OY then the co-ordinates the

points is/are given by

(a) (–1, 9) (b) (3, –15) (c) (1, –3) (d) None of these

Q.42 A curve with equation of the form $y = ax^4 + bx^3 + cx + d$ has zero gradient at the point (0, 1) then the values of x for which the curve has a negative gradient are

(a) $x > -1$ (b) $x < 1$ (c) $x < -1$ (d) $-1 \le x \le 1$

Q.43 Consider the function $f(x) = x \cos x - \sin x$, then identify the statement which is correct

(a) f is neither odd nor even (b) f is monotonic decreasing at $x = 0$

(c) f has a maxima at $x = \pi$ (d) f has a minima at $x = -\pi$

Q.44 Let $f(x) = x^3 - 3x^2 + 2x$. If the equation $f(x) = k$ has exactly one positive and one negative solution then the value of k equals

(a) $-\frac{2\sqrt{3}}{9}$ (b) $-\frac{2}{9}$ (c) $\frac{2}{3\sqrt{3}}$ (d) $\frac{1}{3\sqrt{3}}$

Q.45 The x-intercept of the tangent at any arbitrary point of the curve $\frac{a}{x^2} + \frac{b}{y^2} = 1$ is proportional to

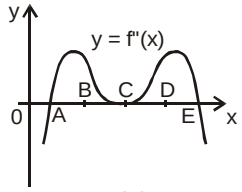
(a) Square of the abscissa of the point of tangency

(b) Square root of the abscissa of the point of tangency

(c) Cube of the abscissa of the point of tangency

(d) Cube root of the abscissa of the point of tangency

Q.46 The graph of $y = f''(x)$ for a function f is shown. Number of points of inflection for $y = f(x)$ is



(a) 4 (b) 3 (c) 2 (d) 1

Q.47 Let h be a twice continuously differentiable positive function on an open interval J. Let suppose $g(x) = \ell n(h(x))$ *for each* $x \in J$ and $(h'(x))^2 > h''(x)h(x)$ *for each* $x \in J$. Then

(a) g is increasing on J

(b) g is decreasing on J

(c) g is concave up on J

(d) g is concave down on J

Q.48 If $f(x)$ is continuous and differentiable over $[-2, 5]$ and $-4 \le f'(x) \le 3$ for all x in $(-2, 5)$ then the greatest possible value of $f(5) - f(-2)$ is

(a) 7 (b) 9 (c) 15 (d) 21

Q.49 Let $f(x)$ and $g(x)$ be two continuous functions defined from $R \to R$, such that $f(x_1) > f(x_2)$ and $g(x_1) < g(x_2)$, $\forall\ x_1 > x_2$, then solution set of $f(g(\alpha^2 - 2\alpha)) > f(g(3\alpha - 4))$ is

(a) R (b) ϕ (c) (1, 4) (d) $R - [1, 4]$

Q.50 A curve is represented parametrically by the equation $x = t + e^{at}$ and $y = -t + e^{at}$ when $t \in R$ and $a > 0$. If the curve touches the axis of x at the point A, then the coordintes of the point A are

(a) (1, 0) (b) $(1/e, 0)$ (c) $(e, 0)$ (d) $(2e, 0)$

Q.51 Let $f(x) = x - \frac{1}{x}$ then which one of the following statement is true

(a) Function is invertible if defined from $R - \{0\} \to R$

(b) $f(x_1) > f(x_2)$, $\forall\ x_1 > x_2$ and $x_1, x_2 \neq 0$

(c) Graph of the function has exactly one asymptote
(d) Function is one-one in every continuous interval [a, b] defined on one side of origin

Q.52 If $f(x) = \int_{x}^{x^2} (t-1)dt$, $1 \le x \le 2$, then global maximum value of $f(x)$ is
(a) 1 (b) 2 (c) 4 (d) 5

Q.53 A right triangle is drawn in a semicircle of radius 1/2 with one of its legs along the diameter. The maximum area of the triangle is
(a) $\frac{1}{4}$ (b) $\frac{3\sqrt{3}}{32}$ (c) $\frac{3\sqrt{3}}{16}$ (d) $\frac{1}{8}$

Q.54 At any two points of the curve represented parametrically by $x = a(2\cos t - \cos 2t)$; $y = (2\sin t - \sin 2t)$ the tangents are parallel to the axis of x corresponding to the values of the parameter t differing from each other by:
(a) $\frac{2\pi}{3}$ (b) $\frac{3\pi}{4}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{3}$

Q.55 If the function $f(x) = \frac{t+3x-x^2}{x-4}$, where '$t$' is a parameter has a minimum and a maximum then the range of values of 't' is
(a) (0, 4) (b) (0, ∞) (c) (–∞, 4) (d) (4, ∞)

Q.56 Let $F(x) = \int_{\sin x}^{\cos x} e^{(1+arc\sin t)^2} dt$ on $\left[0, \frac{\pi}{2}\right]$, then
(a) $F''(c) = 0$ for all $c \in \left(0, \frac{\pi}{2}\right)$ (b) $F''(c) =$ for some $c \in \left(0, \frac{\pi}{2}\right)$
(c) $F'(c) = 0$ for some $c \in \left(0, \frac{\pi}{2}\right)$ (d) $F(c) \neq 0$ for all $c \in \left(0, \frac{\pi}{2}\right)$

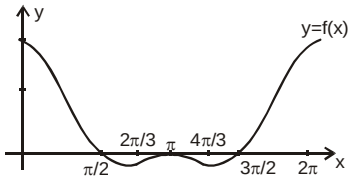
Q.57 The least area of a circle circumscribing any right triangle of area S is
(a) πS (b) $2\pi S$ (c) $\sqrt{2}\pi S$ (d) $4\pi S$

Q.58 Given $f'(1) = 1$ and $\frac{d}{dx}(f(2x)) = f'(x)\ \forall\ x > 0$. If $f'(x)$ is differentiable then there exists a number $c \in (2, 4)$ such that $f''(c)$ equals
(a) –1/4 (b) –1/8 (c) 1/4 (d) 1/8

Q.59 A point is moving along the curve $y^3 = 27x$. The interval in which the abscissa changes at slower rate than ordinate, is
(a) (–3, 3) (b) (–∞, ∞)
(c) (–1, 1) (d) (–∞, –3) ∪ (3, ∞)

Q.60 Let $f(x)$ and $g(x)$ are two function which are defined and differentiable for all $x \ge x_0$. If $f(x_0) = g(x_0)$ and let $f'(x) > g'(x)$ for all $x > x_0$ then
(a) $f(x) < g(x)$ for some $x > x_0$ (b) $f(x) = g(x)$ for some $x > x_0$
(c) $f(x) > g(x)$ only for some $x > x_0$ (d) $f(x) > g(x)$ for all $x > x_0$

Q.61 The graph of $y = f(x)$ is shown. Let $F(x)$ be an antiderivative of $f(x)$. Then $F(x)$ has



(a) Points of inflexion at $x = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$ and 2π, a local maximum at $x = \frac{\pi}{2}$, and a local minimum at $x = \frac{3\pi}{2}$

(b) Points of inflexion at $x = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$ and 2π, a local minimum at $x = \pi/2$ and a local maxima $x = \frac{3\pi}{2}$

(c) Point of inflexion at $x = \pi$, a local maximum at $x = \frac{\pi}{2}$, and a local minimum at $x = \frac{3\pi}{2}$

(d) Point of inflexion at $x = \pi$, a local minimum at $x = \frac{\pi}{2}$, and a local maximum at $x = \frac{3\pi}{2}$

Q.62 *P* and *Q* are two points on a circle of centre *C* and radius α, the angle *PCQ* being 2θ then the radius of the circle inscribed in the triangle CPQ is maximum when

(a) $\sin\theta = \frac{\sqrt{3}-1}{2\sqrt{2}}$ (b) $\sin\theta = \frac{\sqrt{5}-1}{2}$

(c) $\sin\theta = \frac{\sqrt{5}+1}{2}$ (d) $\sin\theta = \frac{\sqrt{5}-1}{4}$

Q.63 Number of critical points of the function, $f(x) = \frac{2}{3}\sqrt{x^3} - \frac{x}{2} + \int_1^x \left(\frac{1}{2} + \frac{1}{2}\cos 2t - \sqrt{t}\right) dt$ which lie in the interval $[-2\pi, 2\pi]$ is

(a) 2 (b) 4 (c) 6 (d) 8

Q.64 Suppose that water is emptied from a spherical tank of radius 10 cm. If the depth of the water in the tank is 4 cm and is decreasing at the rate of 2 cm/sec, then the radius of the top surface of water is decreasing at the rate of

(a) 1 (b) 2/3 (c) 3/2 (d) 2

Q.65 The range of values of *m* for which the line $y = mx$ and the curve $y = \frac{x}{x^2+1}$ enclose a region, is

(a) (–1, 1) (b) (0, 1) (c) [0, 1] (d) $(1, \infty)$

Q.66 For a steamer the consumption of petrol (per hour) varies as the cube of its speed (in km). If the speed of the current is steady at C km/hr then the most economical speed of the steamer going against the current will be

(a) 1.25 C (b) 1.5 C (c) 1.75 C (d) 2 C

Q.67 Let *F* and *g* be increasing and decreasing functions, respectively from $[0, \infty)$ to $[0, \infty]$. Let $h(x) = f[g(x)]$. If $h(0) = 0$, then $h(x) - h(1)$ is

(a) Always zero (b) Strictly increasing

(c) Always negative (d) Always positive

Q.68 A function $y = f(x)$ is given by $x = \frac{x}{1+t^2}$ and $y = \frac{1}{t(1+t^2)}$ for all $t > 0$ then f is

(a) Increasing in (0, 3/2) and decreasing in $(3/2, \infty)$

(b) Increasing in (0, 1)

(c) Increasing in $(0, \infty)$

(d) Decreasing in (0, 1)

Q.69 If the function $f(x) = 2x^2 - kx + 5$ is increasing in [1, 2], then '*k*' lies in the interval

(a) $(-\infty, 4)$ (b) $(4, \infty)$ (c) $(-\infty, 7)$ (d) $(8, \infty)$

Q.70 The set of all values of '*a*' for which the function, $f(x) = (a^2 - 3a + 2)\left(\cos^2\frac{x}{4} - \sin^2\frac{x}{4}\right) + (a-1)x +$

sin 1 does not possess critical points is

(a) $[1, \infty)$ (b) $(0, 1) \cup (1, 4)$ (c) $(-2, 4)$ (d) $(1, 3) \cup (3, 5)$

Q.71 The value of n for which the area of the triangle included between the co-ordinate axes and any tangent to the curve $xy^n = a^{n+1}$ is constant is

(a) –1 (b) 0 (c) 1 (d) a

Q.72 Read the following mathematical statements carefully :

I. A differentiable function 'f' with maximum at $x = c \Rightarrow f'(c) < 0$.

II. Antiderivative of a periodic function is also a periodic function.

III. If f has a period T then for any $a \in R$. $\int_0^T f(x)dx = \int_0^T f(x+a)dx$

IV. If $f(x)$ has a maxima at $x = c$, then 'f' is increasing in $(c - h, c)$ and decreasing in $(c, c + h)$ as $h \to 0$ for $h > 0$.

Now indicate the correct alternative

(a) Exactly one statement is correct

(b) Exactly two statements are correct

(c) Exactly three statements are correct

(d) All the four statements are correct

Q.73 Let a function f be defined as $f(x) = \begin{cases} \frac{|x-1|}{x^2+1} & \text{if } x > -1 \\ x^2 & \text{if } x \le -1 \end{cases}$. Then the number of critical point(s) on the graph of this function is/are

(a) 4 (b) 3 (c) 2 (d) 1

Q.74 Two sides of a triangle are to have lengths 'a' cm and 'b' cm. If the triangle is to have the maximum area, then the length of the median from the vertex containing the sides 'a' and 'b' is

(a) $\frac{1}{2}\sqrt{a^2+b^2}$ (b) $\frac{2a+b}{3}$ (c) $\sqrt{\frac{a^2+b^2}{2}}$ (d) $\frac{a+2b}{3}$

Q.75 Let S be a square with sides of length x. If we approximate the change in size of the area of S by $h.\left.\frac{dA}{dx}\right|_{x=x_0}$, when the sides are changed from x_0 to $x_0 + h$, then the absolute value of the error in our approximation, is

(a) h^2 (b) $2hx_0$ (c) ${x_0}^2$ (d) h

Q.76 Number of critical points on the graph of the function $f(x) = x^{1/3}(x - 4)$ is

(a) 0 (b) 1 (c) 2 (d) 3

Q.77 A rectangle has one side on the positive y-axis and one side on the positive x-axis. The upper right hand vertex of the rectangle lies on the curve $y = \frac{\ell n x}{x^2}$. The maximum area of the rectangle is

(a) e^{-1} (b) $e^{-1/2}$ (c) 1 (d) $e^{1/2}$

Q.78 All roots of the cubic $x^3 + ax + b = 0$ ($a, b \in R$) are real and distinct, and $b > 0$ then

(a) $a = 0$ (b) $a < 0$ (c) $a > 0$ (d) $a \ge b$

Q.79 Let $f(x) = ax^2 - b|x|$, where a and b are constants. Then at $x = 0$, $f(x)$ has

(a) A maxima whenever $a > 0, b > 0$

(b) A maxima whenever $a > 0, b < 0$

(c) Minima whenever $a > 0, b > 0$

(d) Neither a maxima nor minima whenever $a > 0, b < 0$

Q.80 Consider $f(x) = |1 - x| \quad 1 \le x \le 2$ *and* $g(x) = f(x) + b\sin\frac{\pi}{2}x, \quad 1 \le x \le 2$ then which of the following is correct?

(a) Rolles theorem is applicable to both f, g and $b = 3/2$
(b) LMVT is not applicable to f and Rolles theorem if applicable to g with $b = 1/2$
(c) LMVT is applicable to f and Rolles theorem is applicable to g with $b = 1$
(d) Rolles theorem is not applicable to both f, g for any real b

Q.81 Consider $f(x) = \int_1^x \left(t + \frac{1}{t}\right) dt$ and $g(x) = f'(x)$ for $x \in \left[\frac{1}{2}, 3\right]$. If P is a point on the curve $y = g(x)$ such that the tangent to this curve at P is parallel to a chord joining the points $\left(\frac{1}{2}, g\left(\frac{1}{2}\right)\right)$ and $(3, g(3))$ of the curve, then the coordinates of the point P

(a) Can't be found out (b) $\left(\frac{7}{4}, \frac{65}{28}\right)$ (c) (1, 2) (d) $\left(\sqrt{\frac{3}{2}}, \frac{5}{\sqrt{6}}\right)$

Q.82 The angle made by the tangent of the curve $x = a(t + \sin t \cos t)$; $y = a(1 + \sin t)^2$ with the x-axis at any point on it is

(a) $\frac{1}{4}(\pi + 2t)$ (b) $\frac{1 - \sin t}{\cos t}$ (c) $\frac{1}{4}(2t - \pi)$ (d) $\frac{1 + \sin t}{\cos 2t}$

Q.83 If the function $f(x) = 2x^2 + 3x + 5$ satisfies LMVT at $x = 2$ on the closed interval $[1, a]$ then the value of 'a' is equal to

(a) 3 (b) 4 (c) 6 (d) 1

Q.84 Given that $f(x)$ is continuously differentiable on $a \le x \le b$ where $a < b$, $f(a) < 0$ and $f(b) > 0$, which of the following are always true?

(i) $f(x)$ is bounded on $a \le x \le b$
(ii) The equation $f(x) = 0$ has at least one solution in $a < x < b$
(iii) The maximum and minimum values of $f(x)$ on $a \le x \le b$ occur at points where $f'(c) = 0$
(iv) There is at least one point c with $a < c < b$ where $f'(c) > 0$
(v) There is at least one point d with $a < d < b$ where $f'(c) < 0$

(a) Only (ii) and (iv) are true (b) All but (iii) are true
(c) All but (v) are true (d) Only (i), (ii) and (iv) are true

Q.85 Consider the curve represented parametrically by the equation $x = t^3 - 4t^2 - 3t$ and $y = 2t^2 + 3t - 5$ where $t \in R$.
If H denotes the number of point on the curve where the tangent is horizontal and V the number of point where the tangent is vertical then

(a) $H = 2$ and $V = 1$ (b) $H = 1$ and $V = 2$
(c) $H = 2$ and $V = 2$ (d) $H = 1$ and $V = 1$

Q.86 At the point $P(a, a^n)$ on the graph of $y = x^n (n \in N)$ in the first quadrant a normal is drawn. The normal intersects the y-axis at the point $(0, b)$. If $\lim_{a \to 0} b = \frac{1}{2}$, then n equals

(a) 1 (b) 3 (c) 2 (d) 4

Q.87 P is a point on positive x-axis, Q is a point on the positive y-axis and O is the origin. If the line passing through P and Q is tangent to the curve $y = 3 - x^2$ then the minimum area of the triangle OPQ, is

(a) 2 (b) 4 (c) 8 (d) 9

Q.88 Suppose that f is differentiable for all x and that $f'(x) \le 2$ for all x. If $f(1) = 2$ and $f(4) = 8$ then $f(2)$ has the value equal to

(a) 3 (b) 4 (c) 6 (d) 8

Q.89 There are 50 apple trees in an orchard. Each tree produces 800 apples. For each additional tree planted the orchard, the output per additional tree drops by 10 apples. Number of trees that should be added to the existing orchard for maximising the output of the trees, is

(a) 5 (b) 10 (c) 15 (d) 20

Q.90 Range of the function $f(x) = \frac{\ell n x}{\sqrt{x}}$ is

(a) $(-\infty, e)$ (b) $(-\infty, e^2)$ (c) $\left(-\infty, \frac{2}{e}\right]$ (d) $\left(-\infty, \frac{1}{e}\right)$

Q.91 The curve $y = x^3 + x^2 - x$ has two horizontal tangents. The distance between these two horizontal lines, is

(a) $\frac{13}{9}$ (b) $\frac{11}{9}$ (c) $\frac{22}{27}$ (d) $\frac{32}{27}$

ANSWER KEY

1.	(d)	2.	(b)	3.	(d)	4.	(b)	5.	(d)	6.	(b)	7.	(c)
8.	(c)	9.	(b)	10.	(a)	11.	(c)	12.	(d)	13.	(c)	14.	(b)
15.	(d)	16.	(d)	17.	(d)	18.	(a)	19.	(b)	20.	(a)	21.	(c)
22.	(a)	23.	(d)	24.	(d)	25.	(a)	26.	(d)	27.	(c)	28.	(b)
29.	(c)	30.	(d)	31.	(d)	32.	(b)	33.	(c)	34.	(d)	35.	(a)
36.	(a)	37.	(c)	38.	(b)	39.	(c)	40.	(b)	41.	(b)	42.	(c)
43.	(b)	44.	(a)	45.	(c)	46.	(c)	47.	(d)	48.	(d)	49.	(c)
50.	(d)	51.	(d)	52.	(c)	53.	(b)	54.	(a)	55.	(c)	56.	(b)
57.	(a)	58.	(b)	59.	(c)	60.	(d)	61.	(c)	62.	(b)	63.	(b)
64.	(c)	65.	(b)	66.	(b)	67.	(a)	68.	(b)	69.	(a)	70.	(b)
71.	(c)	72.	(a)	73.	(a)	74.	(a)	75.	(a)	76.	(c)	77.	(a)
78.	(b)	79.	(a)	80.	(c)	81.	(d)	82.	(a)	83.	(a)	84.	(d)
85.	(b)	86.	(c)	87.	(b)	88.	(b)	89.	(c)	90.	(c)	91.	(d)

MISCELLANEOUS EXAMPLES

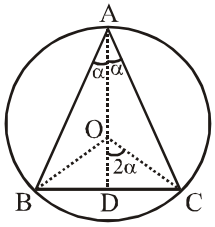
Example 1 : Show that a triangle of maximum area that can be inscribed in a circle of radius '*a*' is an equilateral triangle.

Solution : Let *BC* be one of the sides of the triangle. And the third vertex *A* should be in such a position that the altitude *AD* is maximum (for area of the Δ to be maximum). For that the ΔABC must be symmetric about *AD* i.e. *D* should be the mid point of *BC*.

Let $\angle A = 2\alpha$

$\therefore \angle BOD = \angle COD = 2\alpha$

Thus $AD = AO + OD = a + a\cos 2\alpha$ and $CD = a \sin 2\alpha$



Hence, $\Delta ABC = A = \frac{1}{2} . AD . BC = \frac{1}{2} .a.(1 + \cos 2\alpha) . a \sin 2\alpha$

$= \frac{a^2}{2}\left(\sin 2\alpha + \frac{1}{2}\sin 4\alpha\right)$

$\therefore \frac{dA}{d\alpha} = a^2 [\cos 2\alpha + \cos 4\alpha] = 0 \Rightarrow 2a^2 \cos 3\alpha . \cos \alpha = 0$

$\Rightarrow$ Either $\cos 3\alpha = 0$ or $\cos \alpha = 0 \Rightarrow \alpha = \frac{\pi}{6}, \frac{\pi}{2}$ But $\alpha = \pi/2$ is not possible

Now, $\frac{d^2A}{d\alpha^2} = 2a^2 [-\sin 2\alpha - 4 \sin 4\alpha] =$ –ve at $\alpha = \pi/6 \quad \therefore$ *A* is maximum when $\alpha = \pi/6$

Also, $\angle A = \pi/3$ and triangle is isosceles hence it must be equilateral.

Example 2: Prove that $f(\theta) = \frac{\tan\theta}{\theta}$ is increasing function when $\theta > 0$. Also prove that $\frac{\tan x_2}{x_2} > \frac{\tan x_1}{x_1}$ if $0 < x_1 < x_2 < \frac{\pi}{2}$.

Solution : $f'(\theta) = \dfrac{\theta\sec^2\theta - \tan\theta}{\theta^2} = \dfrac{\theta - \sin\theta\cos\theta}{\theta^2\cos^2\theta} = \dfrac{(2\theta) - \sin 2\theta}{\theta^2\cos^2\theta}$

If $\theta > 0$ then $2\theta > \sin 2\theta$ (Since $\sin x < x$ for $x > 0$)

Hence $f'(\theta) = \dfrac{+ve}{+ve} = +\text{ve} \Rightarrow f(\theta)$ is increasing function when $\theta > 0$.

Now, if we take any two positive values of θ say x_1 and x_2 then we can write

$x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$ i.e. $\dfrac{\tan x_2}{x_2} > \dfrac{\tan x_1}{x_1}$.

Example 3 : Find the minimum value of the function : $f(x) = 3^{(x^2-2)^3+8}$.

Solution : We denote by $\varphi(x)$ the exponent, i.e., $\phi(x) = (x^2-2)^3 + 8$.

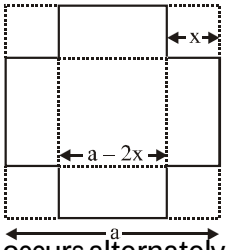
The function $f(x) = 3^{\phi(x)}$ takes on the minimum value at the same point as the function $\phi(x)$.

Hence, $\phi(x) = x^6 - 6x^4 + 12x^2 = x^2[(x^2-3)^2 + 3]$.

Whence it is clear that the function $\varphi(x)$ attains the minimum value (equal to zero) at $x = 0$.

That is why the minimum value of the function $f(x)$ is equal to $3^0 = 1$.

Example 4 : A box of maximum volume with top open is to be made by cutting out four equal squares from four corners of a square tin sheet of side length a ft and then folding up the flaps. Find the side of the square to be cut off.



Solution : Volume of the box is $V = (a-2x)^2x$

$\dfrac{dV}{dx} = (a-2x)^2 + x.2(a-2x).(-2) = (a-2x)(a-6x)$

For V to be extremum $\dfrac{dV}{dx} = 0 \Rightarrow x = \dfrac{a}{2}, \dfrac{a}{6}$

But when $x = \dfrac{a}{2}$; $V = 0$ (minimum) and we know that minimum and maximum occurs alternately in a continuous function. Hence V is maximum at $x = a/6$.

Example 5 : Find the point on the curve $4x^2 + a^2y^2 = 4a^2$; $4 < a^2 < 8$ that is farthest from the point $(0, -2)$.

Solution : The equation of the curve can be expressed as $\dfrac{x^2}{a^2} + \dfrac{y^2}{4} = 1$, which is the equation of an ellipse.

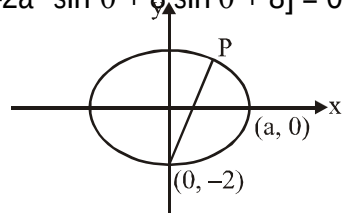
Hence, let us consider a point $P(a\cos\theta, 2\sin\theta)$ on the ellipse.

Let the distance of $P(a\cos\theta, 2\sin\theta)$ from $(0, -2)$ is L

Then $L^2 = (a\cos\theta - 0)^2 + (2\sin\theta + 2)^2$

$\therefore \dfrac{d(L^2)}{d\theta} = a^2 . 2\cos\theta(-\sin\theta) + 4.2(\sin\theta + 1).\cos\theta = \cos\theta[-2a^2\sin\theta + 8\sin\theta + 8] = 0$

$\Rightarrow$ Either $\cos\theta = 0 \Rightarrow \theta = \pi/2$



or, $(8 - 2a^2)\sin\theta + 8 = 0$ i.e. $\sin\theta = \dfrac{4}{a^2-4}$

Since $a^2 < 8$, $a^2 - 4 < 4$

$\therefore \dfrac{4}{a^2-4} > 1 \quad \therefore \quad \sin\theta = \dfrac{4}{a^2-4} > 1$ which is not possible

Further more, $\dfrac{d^2(L^2)}{d\theta^2} = \cos\theta[-2a^2\cos\theta + 8\cos\theta] + (-\sin\theta)[-2a^2\sin\theta + 8\sin\theta + 8]$

At $\theta = \dfrac{\pi}{2}$; $\dfrac{d^2(L^2)}{d\theta^2} = -[16 - 2a^2] = 2(a^2 - 8) < 0$

Hence, L is maximum at $\theta = \pi/2$ and the farthest point is $(0, 2)$.

Example 6 : Determine the points of maxima and minima of the function $f(x) = \dfrac{1}{8}\log x - bx + x^2$, $x > 0$ when $b \geq 0$ is a constant.

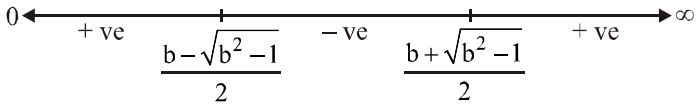
Solution : Here $f(x)$ is defined and continuous for all $x > 0$.

$f'(x) = \frac{1}{8x} - b + 2x = \frac{16x^2 - 8bx + 1}{8x}$

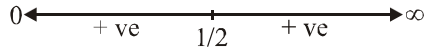
For extrema $f'(x) = 0 \Rightarrow 16x^2 - 8bx + 1 = 0 \Rightarrow x = \frac{8b \pm \sqrt{64(b^2-1)}}{16} = \frac{b \pm \sqrt{b^2-1}}{2}$

Obviously, roots are real if $b^2 - 1 \geq 0 \Rightarrow b \geq 1$ (As $b \geq 0$)

Hence when $b > 1$, Sign scheme for $f'(x)$ is as below

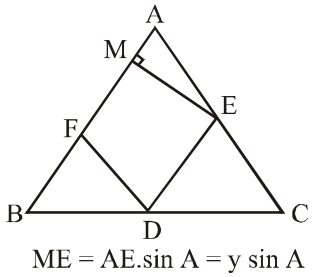


Thus $f(x)$ is maximum at $x = \frac{b-\sqrt{b^2-1}}{2}$ and minimum at $x = \frac{b+\sqrt{b^2-1}}{2}$.

But when $b = 1$; the sign scheme is 

i.e. There is neither maximum nor minimum.

Example 7 : Let $A(p^2, -p)$, $B(q^2, -q)$ and $C(r^2, -r)$ be the vertices of the ΔABC. A parallelogram $AFDE$ is drawn with vertices D, E, F on the line segments BC, CA and AB respectively. Show that maximum area of such a parallelogram is $\frac{1}{4}(p-q)(q-r)(r-p)$.



ME = AE.sin A = y sin A

Solution : Let $AF = x = DE$ and $AE = y = DF$.

Since $\Delta^S CAB$ and CED are similar,

we have, $\frac{CE}{CA} = \frac{DE}{AB} \Rightarrow \frac{b-y}{b} = \frac{x}{c}$

(Here $BC = a$, $AC = b$ & $AB = c$)

Now the area of the parallelogram

$S = \text{AF.EM} = xy \sin A = x.b\left(1 - \frac{x}{c}\right) \sin A$ .... (i)

$\frac{dS}{dx} = \frac{b}{c}(c - 2x) \sin A = 0 \Rightarrow x = \frac{c}{2}$ Also, $\frac{d^2S}{dx^2} = -\frac{2b}{c} < 0$; hence, S is maximum.

Now, $S_{max.} = \frac{1}{4}bc \sin A$ [from (i)]

$= \frac{1}{2}(\Delta ABC) \Rightarrow S_{max.} = \frac{1}{4}\begin{vmatrix} p^2 & -p & 1 \\ q^2 & -q & 1 \\ r^2 & -r & 1 \end{vmatrix} = \frac{1}{4}(p-q)(q-r)(r-p).$

Example 8 : On the curve $x^3 = 12y$, find the interval of values of x for which the abscissa changes at a faster rate than the ordinate?

Solution : Given $x^3 = 12y$, differentiating with respect to y

$3x^2 \frac{dx}{dy} = 12 \quad \therefore \frac{dx}{dy} = \frac{12}{3x^2}$

The interval in which the abscissa changes at a faster rate than the ordinate, we must have

$\Rightarrow \left|\frac{dx}{dy}\right| > 1$ or $\left|\frac{12}{3x^2}\right| > 1$ or $\frac{4}{x^2} > 1 \Rightarrow \frac{4-x^2}{x^2} > 0 \Rightarrow x \in (-2, 2) - \{0\}$.

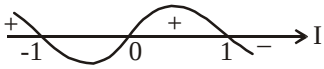
Thus $x \in (-2, 2) - \{0\}$ is the required interval in which abscissa changes at a faster rate than the ordinate.

Example 9 : Using the first derivative, find the extrema of the following functions :

(a) $f(x) = 3\sqrt[3]{x^2} - x^2$; (b) $f(x) = \sqrt[3]{(x-1)^2} + \sqrt[3]{(x+1)^2}$.

Solution : (a) The function is defined and continuous throughout the number scale.

Let us find the derivative : $f'(x) = 2\left(\frac{1}{\sqrt[3]{x}} - x\right)$.



From the equation $f'(x) = 0$ we find the roots of the derivative : $x = \pm 1$.
Furthermore, the derivative goes to infinity at the point $x = 0$.
Thus, the critical points are $x_1 = -1$, $x_2 = 1$.
The results of investigating the sign of the derivative in the neighbourhood of these points are given in Figure.
The investigation show that the function has two maximum :
$f(-1) = 2$; $f(1) = 2$ and minimum $f(0) = 0$.

Example 10: Find the largest term in the sequence $a_n = \frac{n^2}{n^3 + 200}$.

Solution : Consider the function $f(x) = \frac{x^2}{x^3 + 200}$ in the interval $[1, \infty)$.

Since the derivative $f'(x) = \frac{x(400 - x^3)}{(x^3 + 200)^2}$ is positive at $0 < x < \sqrt[3]{400}$ and negative at $x > \sqrt[3]{400}$, the function $f(x)$ increases at

$0 < x < \sqrt[3]{400} < 8$ it follows that the largest term in the sequence can be either a_7 or a_8. Since $a_7 = 49/543 > a_8 = 8/89$, the largest term in the given sequence is $a_7 = \frac{49}{543}$.

Example 11: Show that $\tan^2 x + 6 \log \sec x + 2\cos x + 4 \geq 6 \sec x$ for $0 \leq x \leq \pi/2$

Solution : Let $f(x) = \tan^2 x + 6 \log (\sec x) + 2\cos x + 4 - 6 \sec x$

$f'(x) = 2\tan x \sec^2 x + \frac{6}{\sec x} . \sec x \tan x - 2\sin x - 6\sec x \tan x$

$= \frac{2\sin x}{\cos^3 x}\left\{1 + 3\cos^2 x - \cos^3 x - 3\cos x\right\} = \frac{2\sin x}{\cos^3 x}(1 - \cos x)^3$

Now, in $\left(0, \frac{\pi}{2}\right)$ $f'(x)$ is positive and hence $f(x)$ increasing.

Besides, $f(0) = 0$. Hence, $f(x)$ is positive in $\left(0, \frac{\pi}{2}\right)$.

Example 12: Evaluate the expression $\frac{[1 + (y')^2]^{3/2}}{y''}$ at a point of the curve whose parametric equations are $x = a\cos^3\theta$, $y = a\sin^3\theta$ and show that it attains a maximum of $3a/2$ at $\theta = \pi/4$.

Solution : $y' = \frac{dy}{d\theta} . \frac{d\theta}{dx} = -\frac{3a\sin^2\theta\cos\theta}{3a\cos^2\theta\sin\theta} = -\tan\theta$

$y'' = \frac{d}{d\theta}(-\tan\theta)\frac{d\theta}{dx} = \frac{-\sec^2\theta}{-3a\cos^2\theta\sin\theta} = \frac{1}{3a\cos^4\theta\sin\theta}$

$\therefore \frac{[1 + (y')^2]^{3/2}}{y''} = \sec^3\theta . 3a\cos^4\theta\sin\theta = \frac{3a}{2}\sin 2\theta.$

This attains its maximum value of $3a/2$ at $\theta = \pi/4$.

Example 13: Let (x, y) be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $A > 0$, $B > 0$. Show that $Ax^3 - By^3$ attains a maximal value $-\frac{ABa^3b^3}{\sqrt{B^2b^6 - A^2a^6}}$

Solution : With $x = a\sec\theta$, $y = b\tan\theta$, $Ax^3 - By^3 = Aa^3\sec^3\theta - Bb^3\tan^3\theta \equiv E(\theta)$.

$E'(\theta) = 3\sec^2\theta\tan\theta\,(Aa^3\sec\theta - Bb^3\tan\theta)$ $E'(\theta) = 0$ gives $\theta = 0$, $\theta = \pi$, $\sin^{-1}\frac{Aa^3}{Bb^3}$, $\pi - \sin^{-1}\frac{Aa^3}{Bb^3}$ for the last two of which it is necessary that $Aa^3 < Bb^3$.

It can be verified that $E''(0) = 3Aa^3 > 0$, $E''(\pi) = -Aa^3 < 0$

$E''\left(\sin^{-1}\frac{Aa^3}{Bb^3}\right) < 0$ and $E''\left(\pi - \sin^{-1}\frac{Aa^3}{Bb^3}\right) > 0$

$\theta = 0$ and $\theta = \pi - \sin^{-1}$ are points of minimum.

$\theta = \pi$ and $\theta = \sin^{-1}\frac{Aa^3}{Bb^3}$ are points of maximum.

E (θ) attains a maximum value of $\frac{ABa^3b^3}{\sqrt{B^3b^6 - A^2a^6}}$

Example 14: The tangent to $y = ax^2 + bx + \frac{7}{2}$ at (1, 2) is parallel to the normal at (–2, 2) on $y = x^2 + 6x + 10$ Find a and b.

Solution : (1, 2) satisfies the equation of the curve

$\therefore\ 2 = a + b + \frac{7}{2} \Rightarrow a + b = -\frac{3}{2}$ ..... (1)

$\frac{dy}{dx} = 2ax + b$, at (1, 2) $\frac{dy}{dx} = 2a + b$; $\frac{dy}{dx} = 2x + 6$, at (–2, 2) $\frac{dy}{dx} = 2$

Since, the tangent at (1, 2) is parallel to the normal at (–2, 2), $2a + b = -1/2$ ..... (2)

Solving equations (1) and (2) we get $a = 1$, $b = 5/2$.

Example 15: Prove that all the points on the curve $y^2 = 4a(x + a\sin\frac{x}{a})$ at which the tangents are parallel to the x-axis lies on the parabola $y^2 = 4ax$.

Solution : $y^2 = 4a\left(x + a\sin\frac{x}{a}\right)$ .....(1)

$\therefore\ 2y\frac{dy}{dx} = 4a\left(1 + \cos\frac{x}{a}\right)$.

Hence at the point where the tangent is parallel to the x-axis, $\frac{dy}{dx} = 0$ and so $\cos\frac{x}{a} = -1$.

Hence $\sin\frac{x}{a} = 0$. Hence from equation (1), these points lie on the curve $y^2 = 4ax$.

Example 16: For $a, b > 0$; $0 \le p < 1$, prove that $(a + b)^p \le a^p + b^p$.

Solution : We have to prove $(a + b)^p \le a^p + b^p$

By dividing both sides by b^p , we have to prove $\left(1 + \frac{a}{b}\right)^p \le \left(\frac{a}{b}\right)^p + 1$ i.e., $(1 + x)^p \le x^p + 1$, where $x = \frac{a}{b} > 0$

Let $f(x) = x^p + 1 - (1 + x)^p$, $x > 0$

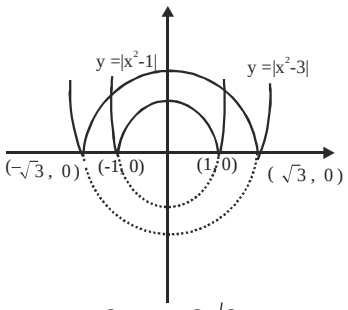
$\therefore\ f'(x) = px^{p-1} - p(1 + x)^{p-1} = p\left\{\frac{1}{x^{1-p}} - \frac{1}{(1+x)^{1-p}}\right\} > 0$ always as $1 - p > 0$ and $x > 0$

Hence the function increases in $(0, \infty)$

$\therefore\ f(x) = 1 + x^p - (1 + x)^p > f(0) = 0$, which proves the proposition.

Example 17: Find the acute angle between the curves $y = |x^2 - 1|$ and $y = |x^2 - 3|$ at their points of intersection.

Solution : The points of intersection are $(\pm\sqrt{2}, 1)$ Since the curves are symmetrical about y-axis, the angle of intersection at $(-\sqrt{2}, 1)$ = the angle of intersection at $(\sqrt{2}, 1)$.



At $(\sqrt{2},1)$, $m_1 = 2x = 2\sqrt{2}$, $m_2 = -2x = -2\sqrt{2}$.

$\therefore \tan\theta = \left|\dfrac{4\sqrt{2}}{1-8}\right| = \dfrac{4\sqrt{2}}{7} \Rightarrow \theta = \tan^{-1}\dfrac{4\sqrt{2}}{7}$

Example 18: The square of the subtangent to the curve $by^2 = (x + a)^2$ is proportional to –
(a) (subnormal)$^{3/2}$ (b) Subnormal (c) (subnormal)$^{1/2}$ (d) None of these

Solution : **(b)** $by^2 = (x + a)^3 \Rightarrow 2by\dfrac{dy}{dx} = 3(x + a)^2 \Rightarrow y\dfrac{dy}{dx} = \dfrac{3(x+a)^2}{2b}$ i.e. $\dfrac{dy}{dx} = \dfrac{3(x+a)^2}{2by}$

Length of the subnormal $= y\dfrac{dy}{dx} = \dfrac{3(x+a)^2}{2b}$

Again, subtangent $= \dfrac{y}{dy/dx} = \dfrac{y}{\frac{3(x+a)^2}{2by}} = \dfrac{2by^2}{3(x+a)^2}$

$\Rightarrow$ (sub-tangent)$^2 = \dfrac{4b^2y^2}{9(x+a)^4} = \dfrac{4b^2}{9(x+a)^4}\cdot\dfrac{(x+a)^6}{b^2} = \dfrac{4}{9}(x + a)^2$

$= \dfrac{4}{9}\cdot\dfrac{2b}{3}\cdot\dfrac{3(x+a)^2}{2b} = \dfrac{8b}{27}$ (subnormal)

$\therefore$ (sub-tangent)^2 is proportional to subnormal

Example 19: If the tangents at (x_1, y_1) to the curve $x^3 + y^3 = a^3$ meets the curve again at (x_2, y_2), then prove that $\dfrac{x_2}{x_1} + \dfrac{y_2}{y_1} + 1 = 0$.

Solution : $x^3 + y^3 = a^3 \Rightarrow \dfrac{dy}{dx} = -\dfrac{x^2}{y^2}$

$\therefore$ equation of the tangent at (x_1, y_1) is $y - y_1 = -\dfrac{x_1^2}{y_1^2}(x - x_1)$

(x_2, y_2) lies on the tangent $\Rightarrow y_2 - y_1 = -\dfrac{x_1^2}{y_1^2}(x_2 - x_1) \Rightarrow -\dfrac{x_1^2}{y_1^2} = \dfrac{y_2 - y_1}{x_2 - x_1}$ ....(1)

Also (x_1, y_1), (x_2, y_2) lie on the curve.

$\therefore x_1^3 + y_1^3 = x_2^3 + y_2^3 = a^3$

$\Rightarrow x_2^3 - x_1^3 = -\left(y_2^3 - y_1^3\right) \Rightarrow \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{-\left(x_2^2 + x_1x_2 + x_1^2\right)}{\left(y_2^2 + y_1y_2 + y_1^2\right)}$ ....(2)

From equations (1) and (2), $\dfrac{x_2^2 + x_1x_2 + x_1^2}{x_1^2} = \dfrac{y_2^2 + y_1y_2 + y_1^2}{y_1^2}$

$\Rightarrow \dfrac{x_2^2}{y_1^2} + \dfrac{x_2}{x_1} + 1 = \dfrac{y_2^2}{y_1^2} + \dfrac{y_2}{y_1} + 1$

$$\Rightarrow \frac{x_2^2}{x_1^2} - \frac{y_2^2}{y_1^2} = \frac{y_2}{y_1} - \frac{x_2}{x_1} \Rightarrow \frac{x_2}{x_1} + \frac{y_2}{y_1} = -1 \Rightarrow \frac{x_2}{x_1} + \frac{y_2}{y_1} + 1 = 0.$$

Example 20: Find the equation of the normal to the curve $y = (1 + x)^y + \sin^{-1}(\sin^2 x)$ at the point $x = 0$.

Solution : When $x = 0$, $y = (1 + 0)^y + \sin^{-1}(\sin^2 0) = 0$.

Differentiating the given function, $\frac{dy}{dx} = y(1 + x)^{y-1} + (1 + x)^y \log(1 + x) + \frac{2\sin x \cos x}{\sqrt{1 - \sin^4 x}}$

$\therefore$ at (0, 1), $\frac{dy}{dx} = 1 + 0 + 0 = 1$

Hence the equation of the normal at (0, 1) is $y - 1 = -1(x - 0) \Rightarrow x + y = 1$.

Example 21: The area of an expanding rectangle is increasing at the rate of 48 cm^2/sec. The length of the rectangle is always equal to the square of the breadth. At what rate the length is increasing at the instant when the breadth is 4.5 cm?

Solution : Let ℓ and b be respectively the length and breadth of the rectangle at time t.

$\therefore \ell = b^2$

Let A be the area of the rectangle at time t

$\therefore A = \ell \times b = b^2 \times b = b^3$

The area is increasing at the rate of 48 cm^2 /sec.

$$\Rightarrow \frac{dA}{dt} = 48 \Rightarrow \frac{d}{dt}(b^3) = 48 \Rightarrow 3b^2 \frac{db}{dt} = 48 \Rightarrow \frac{db}{dt} = \frac{16}{b^2}$$

Rate of change of length w.r.t t = $\frac{d\ell}{dt} = \frac{d}{dt}(b^2) = 2b\frac{db}{dt} = 2b \times \frac{16}{b^2} = \frac{32}{b}$ (a + ve quantity)

$\therefore$ When $b = 4.5$, the rate of increase ($\because d\ell/dt$ is + ve) of length = $\frac{32}{4.5}$ = 7.11 cm./sec.

Example 22: The length x of a rectangle is decreasing at the rate of 2 cm/sec and the width y is increasing at the rate of 2 cm/sec. When x = 12 cm and y = 5 cm, find the rate of change of (i) the perimeter and (ii) the area of the rectangle.

Solution : Let P be the perimeter and A be the area of the rectangle at any time t.

Then, $P = 2(x + y)$ and $A = xy$.

It is given that $\frac{dx}{dt} = -2$ cm/sec and $\frac{dy}{dt}$ = 2cm/sec.

(i) We have, $P = 2(x + y) \Rightarrow \frac{dP}{dt} = 2\left(\frac{dx}{dt} + \frac{dy}{dt}\right) = 2(-2 + 2) = 0$ cm/sec i.e. the perimeter remains constant.

(ii) We have, $A = xy \Rightarrow \frac{dA}{dt} = \left(\frac{dx}{dt}\right)y + x\left(\frac{dy}{dt}\right) \Rightarrow \frac{dA}{dt} = -2 \times 5 + 12 \times 2$ [$\because x$ = 12 cm and y = 5 cm (given)]

$\Rightarrow \frac{dA}{dt}$ = 14 cm^2/sec

Example 23: Find the points on the curve $ay^2 = x^3$ where normal to the curve makes equal intercepts with the axes.

Solution : Let the point at which normal is drawn be (x_1, y_1). Then it must satisfy $ay^2 = x^3$,

i.e., $ay_1^2 = x_1^3$ or $y_1 = \pm\sqrt{\frac{x_1^3}{a}}$

Now, differentiating both sides of the given curve with respect to x we get,

$$2ay\frac{dy}{dx}=3x^2 \Rightarrow \left(\frac{dy}{dx}\right)_{(x_1,y_1)}=\frac{3x_1^2}{2ay_1}=\frac{3x_1^2}{\pm 2a\sqrt{\frac{x_1^3}{a}}}=\pm\frac{3}{2}\sqrt{\frac{x_1}{a}} \quad \text{.... (1)}$$

Thus, slope of the normal $\Rightarrow -\left(\frac{dx}{dy}\right)_{(x_1,y_1)}=\pm\frac{2}{3}\sqrt{\frac{a}{x_1}}$.

We know that the slope of the line making equal intercept with the axes = ± 1.

$\Rightarrow \pm\frac{2}{3}\sqrt{\frac{a}{x_1}}=\pm 1 \Rightarrow x_1=\frac{4a}{9}$

Hence, the required points are $\left(\frac{4a}{9},\frac{8a}{27}\right)$ *and* $\left(\frac{4a}{9},\frac{-8a}{27}\right)$

Example 24: Prove that all the normals to the curve $x = a\cos t + at\sin t$ and $y = a\sin t - at\cos t$ are at a distance a from the origin $(a \in R^+)$.

Solution : $x = a\cos t + at\sin t \Rightarrow \frac{dx}{dt} = -a\sin t + at\cos t + a\sin t = at\cos t$

$y = a\sin t - at\cos t \Rightarrow \frac{dy}{dt} = a\cos t + at\sin t - a\cos t = at\sin t \Rightarrow \frac{dy}{dx}=\frac{at\sin t}{at\cos t}=\tan t$

Hence the equation of the normal at any point 't' on the curve is

$(y - (a\sin t - at\cos t)) = -\frac{1}{\tan t}(x - (a\cos t + at\sin t))$

$\Rightarrow y\sin t - a\sin^2 t + at\sin t\cos t = -x\cos t + a\cos^2 t + at\sin t\cos t$ or $x\cos t + y\sin t = a$

Distance of the normal from (0, 0) $= \frac{|a|}{\sqrt{\sin^2 t+\cos^2 t}}=a$, as $a \in R^+$.

Example 25: Show that the radius of the right circular cylinder of greatest curved surface which can be inscribed in a given cone is half that of the cone.

Solution : Let 'b' be the height of the cone and α be its semivertical angle.

$LD = x$ = radius of the inscribed cylinder and LM = h be its height

$LM = OM - OL = b - x\cot\alpha$

Now, $S = 2\pi rh$ = curved surface

$S = 2\pi x(b - x\cot\alpha)$

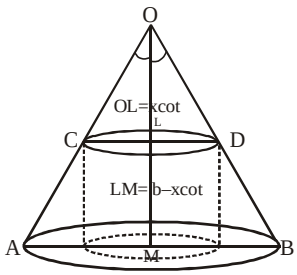
or $S = 2\pi(bx - x^2\cot\alpha)$

$\therefore dS/dx = 2\pi(b - 2x\cot\alpha) = 0$

$\therefore x = (b/2)\tan\alpha$

or $x = \frac{1}{2}(b\tan\alpha) = \frac{1}{2}(r_1)$

or Radius of cylinder $\left(\frac{1}{2}\right)$. (radius of cone)



Example 26: The parametric equation of curve is $x = \sec^2 t$, $y = \cot t$. The tangent at P $(t = \pi/4)$ meets the curve again at Q. Prove that $PQ = \frac{3\sqrt{5}}{2}$

Solution : $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-\cos ec^2 t}{2\sec^2 t\tan t} = -\frac{1}{2}$, at $t = \frac{\pi}{4}$

Coordinates of P are (2, 1).

Equation of the curve in cartesian form is $y^2 = \cot^2 t = \frac{\cos^2 t}{1-\cos^2 t} = \frac{\frac{1}{x}}{1-\frac{1}{x}} = \frac{1}{x-1}$ .... (1)

Equation of tangent at P is

$y - 1 = -\frac{1}{2}(x - 2)$ (or) $2y - 2 = -x = 2$ or $x + 2y = 4$ .... (2)

This meets the curve, $y^2 = \frac{1}{x-1}$, at the point given by

$\left(\frac{4-x}{2}\right)^2 = \frac{1}{x-1}$ (or) $x^3 - 9x^2 + 24x - 20 = 0$

$(x-2)^2 (x-5) = 0 \quad \therefore \quad x = 2$ or 5

Hence, the coordinates of Q are $\left(5, -\frac{1}{2}\right)$ $\therefore PQ = \sqrt{(5-2)^2 + \left(1+\frac{1}{2}\right)^2} = \frac{3\sqrt{5}}{2}$

Example 27: (i) Find the intervals in which the function $f(x) = 3\cos^4 x + 10\cos^3 x + 6\cos^2 x - 3$, $0 \le x \le \pi$, is monotonically increasing or decreasing.

(ii) Let S be the non-empty set containing all a for which $f(x) = \frac{4a-7}{3}x^3 + (a-3)x^2 + x + 5$ is monotonic for all $x \in R$. Find S.

Solution : (i) $f(x) = 3\cos^4 x + 10\cos^3 x + 6\cos^2 x - 3$, $0 \le x \le \pi$

$\Rightarrow f'(x)\, f'(x) = 12\cos^3 x(-\sin x) + 30\cos^2 x(-\sin x) + 12\cos x(-\sin x) = -6\cos x \sin x(2\cos^2 x + 5\cos x + 2) = -3\sin 2x(\cos x + 2)(2\cos x + 1) < 0$ when $x \in \left(0, \frac{\pi}{2}\right) > 0$ when $x \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$; < 0 when $x \in \left(\frac{2\pi}{3}, \pi\right)$

$\therefore f(x)$ is increasing in $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ and decreasing in $\left(0, \frac{\pi}{2}\right)$ and $\left(\frac{2\pi}{3}, \pi\right)$

(ii) Let $f(x) = \frac{4a-7}{3}x^3 + (a-3)x^2 + x + 5$

$\Rightarrow f'(x) = (4a-7)x^2 + 2(a-3)x + 1$

For $f(x)$ to be monotonic, $f'(x) \ge 0$ or $f'(x) \le 0$

$\Rightarrow D \le 0 \Rightarrow (a-3)^2 - (4a-7) \le 0 \Rightarrow a^2 - 10a + 16 \le 0$

$\Rightarrow (a-2)(a-8) \le 0 \Rightarrow a \in [2, 8]$. Also $4a - 7 \ne 0 \Rightarrow a \ne 7/4$

So $a \in [2, 8] - \left\{\frac{7}{4}\right\} \Rightarrow S = [2, 8] - \left\{\frac{7}{4}\right\}$.

Example 28: (i) Find the point on the hyperbola $3x^2 - 4y^2 = 72$ which is nearest to the line $3x + 2y + 1 = 0$

(ii) Find the shortest distance between the curves $9x^2 + 9y^2 - 30y + 16 = 0$ and $y^2 = x^3$.

Solution : (i) Slope of the given line $= -3/2$. First of all we try to locate the points on the curve at which the tangent is parallel to the given line.

Differentiating $3x^2 - 4y^2 = 72$ w.r.t. x, we get $6x - 8y\frac{dy}{dx} = 0$

$\Rightarrow \frac{dy}{dx} = \frac{3x}{4y} = -\frac{3}{2} \Rightarrow \frac{x}{y} = -2$

Also, $3\left(\frac{x}{y}\right)^2 - 4 = \frac{72}{y^2} \Rightarrow \frac{72}{y^2} = 3.4 - 4 = 8 \Rightarrow y^2 = 9 \Rightarrow y = 3, -3$

The required points are $(-6, 3)$ and $(6, -3)$

Distance of $(-6, 3)$ from the given line $= \frac{|-18+6+1|}{\sqrt{13}} = \frac{11}{\sqrt{13}}$

and the distance of $(6, -3)$ from the given line $= \frac{|18-6+1|}{\sqrt{13}} = \frac{13}{\sqrt{13}} = \sqrt{13}$

Clearly the required point is $(-6, 3)$.

(ii) $9x^2 + 9y^2 - 30y + 16 = 0$ can be rewritten as $x^2 + \left(y - \frac{5}{3}\right)^2 = 1$

Any point on the curve $y^2 = x^3$ can be taken as (t^2, t^3).

Let L be the distance between the centre of the given circle and the point (t^2, t^3), then $K = L^2 = t^4 + (t^3 - 5/3)^2$

Now, we calculate the minimum value of L. Required distance = L – radius of given circle.

Now, $\frac{dK}{dt}=4t^3+2\left(t^3-\frac{5}{3}\right)3t^2=0$

for maximum or minimum, $t = 0$ or 1

Now, $\frac{d^2K}{dt^2}=12t^2+30t^4-20t$; $\left.\frac{d^2K}{dt^2}\right|_{t=0}$ = 0

But, $\left.\frac{d^3K}{dt^3}\right|_{t=0}\neq 0 \Rightarrow$ There is neither maxima nor minima at $t = 0$.

Also, $\frac{d^2K}{dt^2}>0$ at t = 1 $\Rightarrow$ L^2 is minimum at $t = 1$ i.e., L is minimum at $t = 1$

So, shortest distance = (value of L at $t = 1$) – (Radius of the circle) = $\frac{\sqrt{13}}{3}-1$

Example 29: Find the values of a, if the equation $x - \sin x = a$ has a unique root in $\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$

Solution : Consider the function $f(x) = x - \sin x - a$, $x\in\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$

Then $f'(x)=1-\cos x=2\sin^2\left(\frac{x}{2}\right)>0 \quad \forall \quad x\in\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$

$\Rightarrow f(x)$ strictly increases in $\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$

Also, we have $f\left(\frac{-\pi}{2}\right)=\frac{-\pi}{2}+1-a$ and $f\left(\frac{\pi}{2}\right)=\frac{\pi}{2}-1-a$

The curve $y = f(x)$ will cuts the x-axis exactly once, if $f(-\pi/2)$ is negative or zero and $f(\pi/2)$ is positive or zero.

i.e., $\frac{-\pi}{2}+1-a\leq 0$ and $\frac{\pi}{2}-1-a>0$ i.e., $a\geq\frac{-\pi}{2}+1$ and $a<\frac{\pi}{2}-1$

Hence, we have , $a\in\left[1-\frac{\pi}{2},\frac{\pi}{2}-1\right]$

Example 30: The ordinate of $y = (a/2)(e^{x/a} + e^{-x/a})$ is the geometric mean of the length of the normal and the quantity.

(a) $a/2$ (b) a (c) e (d) None of these

Solution : **(b)** $y = (a/2)(e^{x/a} + e^{-x/a})$

$L_N = y_1\sqrt{1+m^2}$

Let the other quantity be b

$y^2 = L_N \times b = y_1 b\sqrt{1+m^2}$

$y_1 = b\sqrt{1+(dy/dx)^2} = b\sqrt{1+\frac{1}{4}(e^{x/a}-e^{-x/a})^2}$; $y_1=\frac{b}{2}\sqrt{e^{2x/a}+e^{-2x/a}+2}$

$y_1=\frac{a}{2}(e^{x/a}+e^{-x/a})$; $\frac{4y_1^2}{a^2}=e^{2x/a}+e^{-2x/a}+2$

Example 31: Water is poured into an inverted conical vessel of which the radius of the base is 2m and height 4m, at the rate of 77 litre/minute. Find the rate at which the water level is rising at the instant when the depth is 70 cm. (use $\pi = 22/7$)

Solution : Volume of $ABDE = \frac{1}{3}\pi R^2 h-\frac{1}{3}\pi r^2 h=-\frac{\pi}{3}(R^2H-r^2h)$

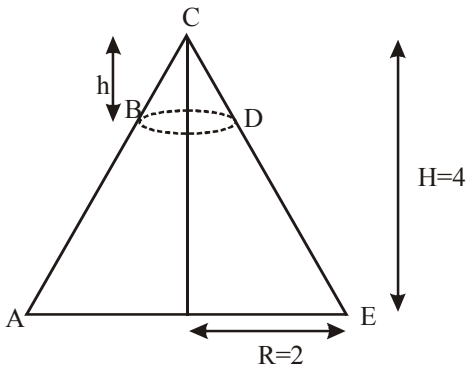
As $\frac{r}{h}\leq\frac{R}{H}\Rightarrow r=\frac{hR}{H}=\frac{h}{2}$

$$v = \frac{\pi}{3}\left(R^2H - \frac{h^3}{4}\right)$$

$$\frac{dv}{dt} = \frac{\pi}{3}\left(0 - \frac{3h^2}{4}\frac{dh}{dt}\right)$$

$\frac{dv}{dt} = 77\ \ell/\min$; h = 70 cm.

substitute and get $\frac{dh}{dt} = 20\ cm/\min$



Example 32: Find the maximum and the minimum values, if any of the following function $f(x) = \sin(\sin x)$

Solution : We have $f(x) = \sin x$, $x \in R$

$\because -1 \le \sin x \le 1$ for all $x \in R$

$\therefore \sin(-1) \le \sin(\sin x) \le \sin 1$ for all $x \in R$ [$\because$ $\sin x$ is an increasing function on $[-1, 1]$]

$\Rightarrow -\sin 1 \le f(x) \le \sin 1$ for all $x \in R$

This shows that the maximum value of $f(x)$ is $\sin 1$ and the minimum value is $-\sin 1$.

Example 33: Find all possible values of the parameter 'a' so that $x^3 - 3x + a = 0$ has three real and distinct roots.

Solution : Let $f(x) = x^3 - 3x + a$ $\Rightarrow f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$

Clearly $x = -1$ is the point of maxima and $x = 1$ is the point of minima.

Now, $f(1) = a - 2$, $f(-1) = a + 2$

The roots of $f(x) = 0$ would be real and distinct if $f(1)\,f(-1) < 0$

$\Rightarrow (a-2)(a+2) < 0 \Rightarrow -2 < a < 2$

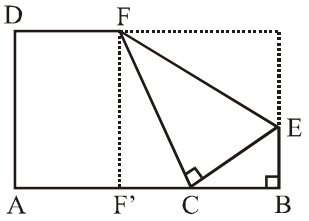
$\Rightarrow$ Thus given equation would have real and distinct roots if $a \in (-2, 2)$.

Example 34: A rectangular sheet of paper $ABCD$ is folded such that the corner C of the sheet lies on the edge AB. If the width $BC = a$, show that the minimum length of the crease will be $\frac{3\sqrt{3}}{4}a$.

Solution : If EF is the crease and $BE = x$, $CE = a - x$ and $BC^2 = (a-x)^2 - x^2 = a^2 - 2ax$. $F'\hat{F}C = B\hat{C}E$ and $F\hat{C}F' = B\hat{E}C$.

Δ^s $F'FC$ and BCE are similar.

$\frac{F'F}{FC} = \frac{BC}{CE}$ gives



$$FC^2 = \frac{a^2(x-x)^2}{a^2 - 2ax} = \frac{a(a-x)^2}{a-2x}.$$

$$EF^2 = FC^2 + CE^2 = \frac{a(a-x)^2}{a-2x} + (a-x)^2 = \frac{2(a-x)^3}{a-2x}$$

$\frac{d}{dx}(EF^2) = \frac{2(a-x)^2(4x-a)}{(a-2x)^2}$ vanishes for $x = \frac{a}{4}$ and its sign changes from –ve to +ve as x passes through $\frac{a}{4}$ from left to right. Thus the crease is minimum for $x = \frac{a}{4}$ and minimum $EF = \frac{3\sqrt{3}}{4}a$.

Example 35: For the curve $xy = c^2$, prove that the portion of the tangent intercepted between the coordinate axes is bisected at the point of contact.

Solution : Let the point at which tangent is drawn be (α, β) on the curve $xy = c^2$.

$$\Rightarrow \left(\frac{dy}{dx}\right)_{(\alpha,\beta)} = -\frac{\beta}{\alpha}$$

Thus, the equation of tangent is, $y - \beta = -\frac{\beta}{\alpha}(x-\alpha)$

$\Rightarrow y\alpha - \alpha\beta = -x\beta + \alpha\beta \Rightarrow x\beta + y\alpha = 2\alpha\beta \Rightarrow \frac{x}{2\alpha} + \frac{y}{2\beta} = 1$

It is clear that the tangent line cuts x and y-axis at A $(2\alpha, 0)$ and B $(0, 2\beta)$ and the point (α, β) bisects AB.

Example 36: Which of the following statements is/are correct –
(a) $x + \sin x$ is increasing function
(b) $\sec x$ is neither increasing nor decreasing function
(c) $x + \sin x$ is decreasing function
(d) $\sec x$ is an increasing function

Solution : **(ab) (a)** $f(x) = x + \sin x$
$f'(x) = 1 + \cos x$
$\therefore f'(x) > 0$ for all $x \neq (2n + 1)\pi$
$f'(x) = 0$ for all $x = (2n + 1)\pi$ $\therefore f(x)$ is increasing
(b) Consider $f(x) = \sec x$
$f'(x) = \sec x \cdot \tan x = \sin x \sec^2 x$
$f'(x)$ takes both positive and negative values
$\therefore$ neither increasing nor decreasing.

Example 37: Find the maximum and the minimum values, if any of the following function : $f(x) = 3x^2 + 6x + 8, x \in R$

Solution : We have, $f(x) = 3x^2 + 6x + 8 = 3(x^2 + 2x + 1) + 5 = 3(x + 1)^2 + 5$.
$\because 3(x + 1)^2 \geq 0$ for all $x \in R$
$\therefore 3(x + 1)^2 + 5 \geq 5$ for all $x \in R$ or $f(x) \geq 5$ for all $x \in R$
Thus, 5 is the minimum value of $f(x)$ which it attains at $x = -1$.
Since $f(x)$ can be made as large as we please, therefore the maximum value does not exist.

Example 38: Prove that $y = e^{|x|}$ cannot have a unique tangent line at the point $x = 0$. Find the angle between the one sided tangents to the curve at the point $x = 0$.

Solution : $y = e^{|x|}$ can be written as $y = e^{-x}$, $x \leq 0 = e^{x}$, $x \geq 0$
$\therefore$ Left hand derivative at $x = 0$ is $\left(\frac{dy}{dx}\right)_{x=0} = \left(-e^{-x}\right)_{x=0} = -1$
Right hand derivative at $x = 0$ is $\left(\frac{dy}{dx}\right)_{x=0+} = \left(\frac{dy}{dx}\right)_{x=0+} = 1$
It has two tangents to the curve at (0, 1) and they are
$y - 1 = -x$ i.e., $x + y = 1$ and $y - 1 = x$ i.e., $x - y = -1$
Clearly they cut each other at right angles.

Example 39: Show that $\left(a - \frac{1}{a}\right)(4 - 3x^2)$ has just one maximum and one minimum value. Show also that the difference between them is $\frac{4}{9}\left(a + \frac{1}{a}\right)^3$ (assume $a > 0$).

Solution : If $y = \left(a - \frac{1}{a} - x\right)(4 - 3x^2)$

$\frac{dy}{dx} = \left(a - \frac{1}{a} - x\right)(-6x) - 1\,(4 - 3x^2) \Rightarrow \frac{d^2y}{dx^2} = 18x - 6\left(a - \frac{1}{a}\right)$

$\frac{dy}{dx} = \left\{3x - \left(a - \frac{1}{2}\right)\right\}^2 - \left(a + \frac{1}{2}\right)^2 = \left(3x + \frac{2}{a}\right)(3x - 2a)$

Now, $\frac{dy}{dx} = 0$, when $x = \frac{2a}{3}$ or when $x = -\frac{2}{3a}$

When $x = \frac{2a}{3}$, y is minimum and when $x = -\frac{2}{3a}$, y is maximum.

Maximum value $= \left(a - \frac{1}{a} + \frac{2}{3a}\right)\left(4 - \frac{4}{3a^2}\right) = \left(a - \frac{1}{3a}\right)4\left(1 - \frac{1}{3a^2}\right) = 4a\left(1 - \frac{1}{3a^2}\right)^2$

Minimum value $= \left(a - \frac{1}{a} - \frac{2}{3a}\right)\left(4 - \frac{4a^2}{3}\right) = \frac{4(a^2-3)^2}{9a}$

Difference between these two values $= \frac{4a(3a^2-1)^2}{9a^4} + \frac{4(a^2-3)^2}{9a}$

$= \frac{4}{9}\left(\frac{a^2+1}{a}\right)^3 = \frac{4}{9}\left(a + \frac{1}{a}\right)^3$

Example 40: Find the coordinates of the point on the parabola $y = x^2 + 7x + 2$ which is the closest to the straight line $y = 3x - 3$. Find also this distance.

Solution : The point on the curve can be taken as $(t, t^2 + 7t + 2)$.
The perpendicular distance from this point to the line $y = 3x - 3$ is given by

$f(t) = \left|\frac{3t - (t^2 + 7t + 2) - 3}{\sqrt{10}}\right| = \left|\frac{-t^2 - 4t - 5}{\sqrt{10}}\right| = \frac{(t+2)^2 + 1}{\sqrt{10}}$

$\therefore f'(t) = \frac{2(t+2)}{\sqrt{10}}$, $f''(t) = \frac{2}{\sqrt{10}}$

$f'(t) = 0 \Rightarrow t = -2$. Then $f''(t) > 0$. Hence f(t) is minimum when $t = -2 = x$
$\therefore y = 4 - 14 + 2 = -8$ i.e., when the point is $(-2, -8)$ and the distance $= 1/\sqrt{10}$ units.

Example 41: A truck is to be driven 300 km. on a highway at a constant speed of x kmph. Speed rules of the highway required that $30 \le x \le 60$. The fuel costs Rs. 10 per litre and is consumed at the rate of $2 + \frac{x^2}{600}$ litres per hour. The wages of the driver are Rs. 200 per hour. The most economical speed to drive the truck, in kmph, is –

(a) 30 (b) 60 (c) $30\sqrt{3.3}$ (d) $20\sqrt{3.3}$

Solution : **(b)** Time taken by the truck $= \frac{300}{x}$ hours

$\therefore$ petrol consumed $= \left(2 + \frac{x^2}{600}\right)\frac{300}{x}$ litre

$\therefore$ Expenses on travelling, $e = 200 \times \frac{300}{x} + \left(2 + \frac{x^2}{600}\right)\frac{3000}{x}$

$= \frac{60000}{x} + \frac{6000}{x} + 5x = \frac{66000}{x} + 5x$

$\therefore \frac{dE}{dx} = -\frac{66000}{x^2} + 5 < 0$ for all $x \in [30, 60]$

$\therefore$ most economical speed is 60 kmph.

Example 42: If $y = 2x + \cot^{-1} x + \log(\sqrt{1+x^2} - x)$, then y

(a) decreases on $(-\infty, \infty)$ (b) decreases on $[0, \infty)$

(c) neither decreases nor increases on $[0, \infty)$ (d) increases on $(-\infty, \infty)$

Solutoin : **(d)** $y'(x) = 2 - \frac{1}{1+x^2} + \frac{1}{\sqrt{1+x^2} - x} \frac{d}{dx}(\sqrt{1+x^2} - x)$

$= 2 - \frac{1}{1+x^2} + \frac{1}{\sqrt{1+x^2} - x}\left(\frac{x}{\sqrt{1+x^2}} - 1\right)$

$= 2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{1+x^2}} > 0$, for all $x \in R$, except at $x = 0$ and $y'(0) = 0$

$\therefore$ $f(x)$ increases on $(-\infty, \infty)$

Example 43: If $f(x) = xe^{x(1-x)}$, then $f(x)$ is

(a) increasing in $[-1/2, 1]$ (b) decreasing in R

(c) increasing in R (d) decreasing in $[-1/2, 1]$

Solution : **(a)** $f(x) = xe^{x(1-x)}$

$\Rightarrow f'(x) = e^{x(1-x)} + xe^{x(1-x)}(1 - 2x)$

$= e^{x(1-x)}[1 + x - 2x^2] = -e^{x(1-x)}(x - 1)(2x + 1)$

Now $f(x)$ is increasing when $f'(x) > 0$

$\Rightarrow -e^{x(1-x)}(x - 1)(2x + 1) > 0$

$\Rightarrow (x - 1)(2x + 1) < 0$ $[\because e^{x(1-x)} > 0]$

$\Rightarrow -\frac{1}{2} < x < 1$

Also $f(x)$ is decreasing when $f'(x) < 0$

$\Rightarrow (x - 1)(2x + 1) > 0 \Rightarrow x < -1/2$ or $x > 1$

Example 44: Find the maximum and the minimum values, if any of the following function $f(x) = x^3 + 1, x \in R$

Solution : We have $f(x) = x^3 + 1, x \in R$

Here we observe that the values of $f(x)$ increase when the values of x are increased and $f(x)$ can be made as large as we please by giving large values to x. So $f(x)$ does not have the maximum value. Similarly, $f(x)$ can be made as small as we please by giving smaller values to x. So $f(x)$ does not have the minimum value also.

Example 45: A place of wire of length 60 cm is cut into two pieces. Each piece is bent to form a rectangle which is twice as long as it is wide. Find the lengths of the two pieces of wire if the sum of the areas of the rectangles is a minimum.

Solution : $60 = 2(x + 2x) + 2(y + 2y)$ $\therefore$ $x + y = 10$

Area $= A_1 + A_2 = 2x^2 + 2y^2$; $A = 2\{x^2 + (10 - x)^2\}$

$\frac{dA}{dx} = 2\{2x - 2(10 - x)\}$; $\frac{d^2A}{dx^2} = 4(1 + 1) =$ positive

Now, A is minimum when $\frac{dA}{dx} = 0$ i.e. when $2x - 10 = 0$ (or) $x = 5$.

$\therefore$ each piece must be of length 30 cm.

Example 46: Find the interval of monotonicity of $f(x) = \frac{1}{4x^2 - 21x^2 + 36x}$. Find the points of minimum and maximum. Also find the global minimum and global maximum (the lowest the highest values) of $f(x)$ in $1 \le x \le 3$.

Solution : $f(x) = \dfrac{1}{4x^3 - 21x^2 + 36x}$. Denominator $= x\left[4\left(x - \dfrac{21}{8}\right)^2 - \dfrac{441}{16} + 36\right]$

$f(x)$ is defined for all $x \neq 0$ since the quadratic factor is positive for all x.

$f'(x) = \dfrac{-6(2x-3)(x-2)}{(4x^3 - 21x^2 + 36x)^2}$

$f'(x) < 0$ for all x such that $(2x - 3)(x - 2) > 0$ i.e., for $x < \dfrac{3}{2}$ and $x > 2$.

$\therefore$ f(x) is decreasing in $(-\infty, 0) \cup \left(0, \dfrac{3}{2}\right) \cup (2, \infty)$.

f (x) > 0 for all x such that (2x – 3) (x – 2) < 0 i.e., for $\dfrac{3}{2} < x < 2$.

$\therefore$ $f(x)$ is increasing in $\left(\dfrac{3}{2}, 2\right)$.

As $f'(x) > 0$ for all x such that $(2x - 3)(x - 2) < 0$ i.e., for $\dfrac{3}{2} < x < 2$

$\therefore$ $f(x)$ is increasing in $\left(\dfrac{3}{2}, 2\right)$.

As $f'(x)$ changes from negative to positive values as x passes through $\dfrac{3}{2}$ from left to right $f(x)$ attains a minimum value of $\dfrac{4}{81}$ at $\dfrac{3}{2}$. $f(x)$ attains a maximum value of $\dfrac{1}{20}$ at $x = 2$.

To find the global minimum and global maximum in $1 \le x \le 3$, $f(1) = \dfrac{1}{19}$, $f(3) = \dfrac{1}{27}$.

Local extremes in $1 \le x \le 3$: $\dfrac{4}{81}$, $\dfrac{1}{20}$.

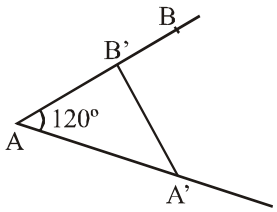
As $\dfrac{1}{19} > \dfrac{1}{20} > \dfrac{4}{81} > \dfrac{1}{27}$, global minimum $= \dfrac{1}{27}$, global maximum $= \dfrac{1}{19}$.

Example 47: Two cars moving with the same uniform speed of 60 km/h are travelling along two straight roads crossing at 120°. At a certain moment, one car is at the junction of the roads while the other car on the road travelling towards the junction, is at a distance 240 km from the junction. When will be the distance between them will be the least and what will be that distance ?

Solution : When the first car is at A, let the second car be at B. After t hours, the positions of the two cars are A' and B' as shown in the Figure. $AB = 240$ km, $AB' = 240 - 60t$, $AA' = 60t$

From $\Delta AA'B$, $\ell = A'B'^2 = AA'^2 + AB'^2 - 2AA' \cdot AB' \cos 120°$

i.e. $\ell = 3600\,t^2 + (240 - 60t)^2 + 60t\,(240 - 60t) = 3600t^2 - 14400t + 57600$



$\dfrac{d\ell}{dt} = 7200t - 14400$ $\therefore$ $t = 2$ hours.

$\dfrac{d\ell}{dt}$ is positive here.

$\therefore$ ℓ is least when $t = 2$ hours and the least distance $= \sqrt{\ell} = \sqrt{43200} = 120\sqrt{3}$ km.

EXERCISE – 2

Q.1 Let $h(x) = f(x) - (f(x))^2 + (f(x))^3$ for every real number x. Then
(a) h is increasing whenever f is increasing
(b) h is increasing whenever f is decreasing
(c) h is decreasing whenever f is decreasing
(d) Nothing can be said in general

Q.2 If $f(x) = \dfrac{x^2-1}{x^2+1}$, for every real number x, then the minimum value of f
(a) Does not exist because f is unbounded (b) Is not attained even though f is bounded
(c) Is equal to 1 (d) Is equal to –1

Q.3 The number of values of x where the function $f(x) = \cos x + \cos(\sqrt{2}x)$ attains its maximum is
(a) 0 (b) 1 (c) 2 (d) infinite

Q.4 Suppose $f(x)$ is a functions satisfying the following conditions :
(a) $f(0) = 2$, $f(1) = 1$
(b) f has a minimum value at $x = 5/2$
(c) For all x, $f'(x) = \begin{vmatrix} 2ax & 2ax-1 & 2ax+b+1 \\ b & b+1 & -1 \\ 2(ax+b) & 2ax+2b+1 & 2ax+b \end{vmatrix}$
When a, b are some constants. Determine the constants a, b and the function $f(x)$.

Q.5 A curve C has the property that if the tangent drawn at any point P on C meets the co-ordinate axes at A and B, then P is the mid-point of AB. The curve passes through the point (1, 1). Determine the equation of the curve.

Q.6 The function $f(x) = \sin^4 x + \cos^4 x$ increases if
(a) $0 < x < \pi/8$ (b) $\pi/4 < x < 3\pi/8$ (c) $3\pi/8 < x < 5\pi/8$ (d) $5\pi/8 < x < 3\pi/4$

Q.7 Find the coordinates of all the points P on the ellipse $\left(\dfrac{x^2}{a^2}\right)+\left(\dfrac{y^2}{b^2}\right)=1$ for which the area of the triangle PON is maximum, where O denotes the origin and N the foot of the perpendicular from O to the tangent at P.

Q.8 Find the normal to the ellipse $\left(\dfrac{x^2}{9}\right)+\left(\dfrac{y^2}{4}\right)=1$ which are farthest from its centre.

Q.9 If the normal to the curve, $y = f(x)$ at the point (3, 4) makes an angle $\dfrac{3\pi}{4}$ with the positive x-axis. Then $f'(3) =$
(a) – 1 (b) – 3/4 (c) 4/3 (d) 1

Q.10 Let $f(x) = \begin{cases} |x| & \text{for } 0 < |x| \le 2 \\ 1 & \text{for } x = 0 \end{cases}$. Then at $x = 0$, 'f' has :
(a) A local maximum (b) No local maximum (c) A local minimum (d) No extremum.

Q.11 For all $x \in (0, 1)$:
(a) $e^x < 1 + x$ (b) $\log_e(1 + x) < x$ (c) $\sin x > x$ (d) $\log_e x > x$

Q.12 Consider the following statements in S and R :

S : Both sin x and cos x are decreasing functions in the interval $(\pi/2, \pi)$.

R : If a differentiable function decreases in an interval (a, b), then its derivative also decreases in (*a*, *b*).

Which of the following is true ?

(a) Both S and R are wrong

(b) Both S and R are correct, but R is not the correct explanation for S

(c) S is correct and R is the correct explanation for S

(d) S is correct and R is wrong.

Answers Key

1. (a, c)
2. (d
3. (a)
4. $a=\frac{1}{4} b=-\frac{5}{4}; f(x)=\frac{1}{4}(x^2-5x+8)$
5. log *xy* = 1
6. (b)
7. $\left(\pm\frac{a^2}{\sqrt{a^2+b^2}}, \pm\frac{b^2}{\sqrt{a^2+b^2}}\right)$
8. $\pm\sqrt{3}x \pm \sqrt{2}y = 5$
9. (d)
10. (d)
11. (b)
12. (b)

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

1. Equation of Tangent & Normal Based Question

Q.1 The curve represented parametrically by the equations $x = 2 \ln \cot t + 1$ & $y = \tan t + \cot t$
(a) tangent and normal intersect at the point (2, 1)
(b) normal at $t = \pi/4$ is parallel to y - axis
(c) tangent at $t = \pi/4$ is parallel to the line y = x
(d) tangent at $t = \pi/4$ is parallel to x - axis

Q.2 The two curves $x^3 - 3xy^2 + 2 = 0$ and $3x^2y - y^3 - 2 = 0$
(a) Cut at right angles
(b) Touch each other
(c) Cut at an angle $\pi/3$
(d) Cut at an angle $\pi/4$

Q.3 A curve $y = f(x)$ passes through the point (4, 3) and the normal to the curve at the point happens to be a tangent to the circle $x^2 + y^2 = 25$. The value of $f'(D)$ is
(a) –3/4 (b) 3/4
(c) 4/3 (d) – 4/3

Q.4 The triangle formed by the tangent to the curve $f(x) = x^2 + bx - b$ at the point (1, 1) and the coordinate axes, lies in the first quadrant. If its area is 2, then the value of b is –
(a) – 1 (b) – 3
(c) 3 (d) 1

Q.5 Tangent of acute angle between the curves $y = |x^2 - 1|$ and $y = \sqrt{7 - x^2}$ at their points of intersection is –
(a) $\frac{5\sqrt{3}}{2}$ (b) $\frac{3\sqrt{5}}{2}$
(c) $\frac{5\sqrt{3}}{4}$ (d) $\frac{3\sqrt{5}}{4}$

Q.6 For the curves, $x^3 + 2 = 3xy^2$ and $y^3 + 2 = 3x^2y$ which of the following are true?
(i) They are orthogonal.
(ii) They are symmetric with respect to the axes of coordinates.
(iii) They are reflections of each other with respect to $y = x$.
(a) (i) Only
(b) (ii) and (iii) Only
(c) (i) and (iii) Only
(d) (i), (ii) and (iii)

Q.7 A curve $y = f(x)$ passes through the point $P(1, 1)$. The normal to the curve at P is a $(y - 1) + (x - 1) = 0$. If the slope of the tangent at any point on the curve is proportional to the ordinate of the point, then the equation of the curve is-
(a) $y = e^{a(x-1)}$
(b) $y = e^{a(1-x)}$
(c) $y = e^{a/2(x-1)}$
(d) $e^{a/2(x+1)}$

Q.8 The distance between the origin and the tangent to the curve $y = e^{2x} + x^2$ drawn at the point $x = 0$ is
(a) $\frac{1}{\sqrt{5}}$ (b) $\frac{2}{\sqrt{5}}$
(c) $\frac{-1}{\sqrt{5}}$ (d) $\frac{2}{\sqrt{3}}$

2. Increasing & Decreasing Function Based Question

Q.9 For all values of x, function $f(x) = 2x^3 + 6x^2 + 7x - 19$ is
(a) Monotonic increasing
(b) Monotonic decreasing
(c) Not monotonic
(d) None of these

Q.10 Function $f(x) = \cos x - 2\lambda x$ is monotonic decreasing when
(a) $\lambda > 1/2$ (b) $\lambda < 1/2$
(c) $\lambda < 2$ (d) $\lambda > 2$

Q.11 The interval in which $f(x) = 2 \sin x + \tan x - 3x$ increases is
(a) $\left(-\frac{\pi}{2}, 0\right)$ (b) $\left(0, \frac{\pi}{2}\right)$
(c) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (d) $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

Q.12 Let $f'(x) < 0$ and $g'(x) > 0$ for all real x, then
(a) $f(g(x + 1)) > g(g(x + 5))$
(b) $f(g(x)) < f(g(f(x + 2)))$

(c) $g(f(x)) < g(f(x+2))$
(d) $g(f(x)) > g(f(x-2))$

Q.13 Function $f(x) = \frac{a\sin x + b\cos x}{c\sin x + d\cos x}$ is monotonic decreasing if –
(a) $ad - bc < 0$ (b) $ad - bc > 0$
(c) $ab - cd < 0$ (d) $ab - cd > 0$

Q.14 A right triangle is drawn in a semicircle of radius 1/2 with one of its legs along the diameter. The maximum area of the triangle is
(a) 1/4 (b) $\frac{3\sqrt{3}}{32}$
(c) $\frac{3\sqrt{3}}{16}$ (d) 1/8

Q.15 The function $f(x) = 2x^2 - \log|x|$, $x \neq 0$ is increasing in the interval
(a) $\left(-\frac{1}{2},0\right)\cup\left(\frac{1}{2},\infty\right)$
(b) $\left(-\infty,-\frac{1}{2}\right)\cup\left(0,\frac{1}{2}\right)$
(c) $\left(-\frac{1}{2},\frac{1}{2}\right)$
(d) None of these

Q.16 The values of x for which $1 + x\ln\left(x+\sqrt{x^2+1}\right) \geq \sqrt{1+x^2}$ are-
(a) $x \leq 0$
(b) $0 \leq x \leq 1$
(c) $x \geq 0$
(d) None of these

Q.17 If function $f(x) = kx^3 - 9x^2 + 9x + 3$ is monotonic increasing in every interval then
(a) $k < 3$ (b) $k \leq 3$
(c) $k > 3$ (d) $k \geq 3$

3. Maxima and Minima Based Question

Q.18 If $a^2x^4 + b^2y^4 = c^4$, then the maximum value of xy is
(a) $\frac{c}{\sqrt{ab}}$ (b) $\frac{c^2}{2\sqrt{ab}}$
(c) $\frac{c}{2\sqrt{ab}}$ (d) $\frac{c^2}{2ab}$

Q.19 The lateral edge of a regular hexagonal pyramid is 1 cm. If the volume is maximum, then its height must be equal to –
(a) 1/3 (b) 2/3
(c) $1/\sqrt{3}$ (d) 1

Q.20 Maximum value of $[\sin x] + [\cos x]$ is (where [.] represents greatest integer function)
(a) 0 (b) 1
(c) 2 (d) 3

Q.21 The minimum value of $64\sec x + 27\operatorname{cosec} x$, $0 < x < \frac{\pi}{2}$ is -
(a) 91 (b) 25
(c) 125 (d) None of these

Q.22 The value of θ, $\theta \in [0, \pi/2]$ for which the sum of intercepts on co-ordinate axes by tangent at point $(3\sqrt{3}\cos\theta, \sin\theta)$ of ellipse $\frac{x^2}{27} + y^2 = 1$ is minimum, is :
(a) $\pi/6$ (b) $\pi/4$
(c) $\pi/3$ (d) $\pi/2$

Q.23 Let $f(x)$ be defined as $f(x) = \begin{cases} \tan^{-1}\alpha - 5x^2, & 0 < x < 1 \\ -6x, & x \geq 1 \end{cases}$; $f(x)$ can have a maximum at $x = 1$ if value of α is
(a) 0 (b) – 1
(c) – tan 1 (d) – 2

Q.24 For the function $f(x) = x^4(12\ln x - 7)$
(a) the point (1, –7) is the point of inflection
(b) $x = e^{1/3}$ is the point of minima
(c) the graph is concave downwards in (0, 1)
(d) the graph is concave upwards in $(1, \infty)$

Q.25 $f(x) = \begin{cases} \sin\frac{\pi x}{2}, & 0 \leq x < 1 \\ 3 - 2x, & x \geq 1 \end{cases}$, then
(a) $f(x)$ has a local minimum at $x = 1$
(b) $f(x)$ has a local maximum at $x = 1$
(c) $f(x)$ does not have any local maximum or minimum at $x = 1$
(d) $f(x)$ has a global minimum at $x = 1$

Q.26 The greatest value of $f(x) = (x+1)^{1/3} - (x-1)^{1/3}$ on [0, 1] is
(a) 1 (b) 2
(c) 3 (d) 1/3

Q.27 If x and y are real numbers satisfying the relation $x^2 + y^2 - 6x + 8y + 24 = 0$ then minimum value of $f(x) = \log_2(x^2 + y^2)$ is
(a) 1 (b) 2
(c) 3 (d) 4

Q.28 If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, where $a > 0$, attains its maximum and minimum at p and q respectively such that $p^2 = q$, then a equals-
(a) 3 (a) 1
(c) 2 (d) 1/2

Q.29 The greatest value of $f(x) = (x+1)^{1/3} - (x-1)^{1/3}$ on [0, 1] is-
(a) 1 (b) 2
(c) 3 (d) 1/3

Q.30 The largest possible value of the expression $y = \sqrt{x-2} + 2\sqrt{3-x}$ is –
(a) 3 (b) $\sqrt{5}$
(c) 2 (d) 1

Q.31 The function $f(x) = \begin{vmatrix} x-1 & x+1 & 2x+1 \\ x+1 & x+3 & 2x+3 \\ 2x+1 & 2x-1 & 4x+1 \end{vmatrix}$ has –
(a) one point of maximum and one point of minimum
(b) one point of maximum only
(c) one point of minimum only
(d) none of the above

Q.32 The difference between greatest and least value of $f(x) = 2\sin x + \sin 2x$, $x \in \left[0, \frac{3\pi}{2}\right]$ is –
(a) $\frac{3\sqrt{3}}{2}$ (b) $\frac{3\sqrt{3}}{2} - 2$
(c) $\frac{3\sqrt{3}}{2} + 2$ (d) None of these

Q.33 The minimum value of $e^{(2x^2-2x-1)\sin^2 x}$ is
(a) e (b) $1/e$
(c) 1 (d) 0

Q.34 Consider the locus of a moving point $P(x, y)$ in the plane which satisfies the condition $2x^2 = r^2 + r^4$, where $r^2 = x^2 + y^2$. Then which one of the following statement is true ?
(a) for every $0 < r < 1$, there are exactly four points on the curve
(b) for every $9 < r \le 1$, there are exactly four points on the curve
(c) the locus is ellipse
(d) none of these

Q.35 The real number x when added to its inverse gives the minimum value of the sum at x equal to-
(a) 2 (b) 1
(c) –1 (d) –2

Section-B
(Previous Year Question)

Q.1 On the ellipse $4x^2 + 9y^2 = 1$, the point at which the tangents are parallel to the line $8x = 9y$, are
(a) $\left(\frac{2}{5}, \frac{1}{5}\right)$ (b) $\left(-\frac{2}{5}, \frac{1}{5}\right)$
(c) $\left(-\frac{2}{5}, -\frac{1}{5}\right)$ (d) $\left(\frac{2}{5}, -\frac{1}{5}\right)$

Q.2 If the line ax + by + c = 0 is a normal to the curve xy = 1then
(a) $a > 0, b > 0$ (b) $a > 0, b < 0$
(c) $a < 0, b > 0$ (d) $a < 0, b < 0$

Q.3 Let f and g be increasing and decreasing functions respectively from $[0, \infty)$ to $[0, \infty)$ and $h(x) = f\{f(x)\}$. If $h(0) = 0$, then $h(x) - h(1)$ is
(a) always negative
(b) always positive
(c) strictly increasing
(d) None of the above

Q.4 If $f : (0, \infty) \to R$ be given by $f(x) = \int_{1/x}^{x} e^{-\left(t + \frac{1}{t}\right)} \frac{dt}{t}$ Then,
(a) $f(x)$ is monotonically increasing on $[1, \infty)$
(b) $f(x)$ is monotonically decreasing on $[0, 1)$
(c) $f(x) + f\left(\frac{1}{x}\right) = 0, \forall x \in (0, \infty)$
(d) $f(2^x)$ is an odd function of x on R

Q.5 If $h(x) = f(x) - f(x)^2 + f(x)^3$ for every realnumber x. Then,
(a) h is increasing , whenever f is increasing
(b) h is icnreasing, whenver f is decreasing
(c) h is decreasing, whenver f is decreasng
(d) Nothing can be said in general

Q.6 Let f(x) be a polynomial of degree four haivng extreme values at x = 1 and x = 2. If $\lim_{x \to 0}\left[1 + \frac{f(x)}{x^2}\right] = 3$, then f(b)is equal to
(a) –8 (b) –4
(c) 0 (d) 4

Q.7 If x = –1 and x = 2are extreme points of $f(x) = \alpha \log |x| + \beta x^2 + x$, then
(a) $\alpha = -6, \beta = \frac{1}{2}$ (b) $\alpha = -6, \beta = -\frac{1}{2}$
(c) $\alpha = 2, \beta = -\frac{1}{2}$ (d) $\alpha = 2, \beta = \frac{1}{2}$

Q.8 The number of points in $(-\infty,\infty)$ for which $x^2 - x\sin x - \cos x = 0$, is
(a) 6 (b) 4
(c) 2 (d) 0

Q.9 Let *f*, g and h be real-valued functions defined on the interval $[0,1]$ by $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = xe^{x^2} + e^{-x^2}$ and $h(x) = x^2e^{x^2} + e^{-x^2}$. If *a, b* and c denote respectively, the absolute maximum of *f*, g and *h* on [0, 1], then
(a) a = b and $c \neq b$
(b) a = c and $a \neq b$
(c) $a \neq b$ and $c \neq b$
(d) a= b = c

Q.10 The total number of local maxima and local minima of the function
$f(x) \neq b = \begin{cases} (2+x)^3, & -3 < x \leq -1 \\ x^3, & -1 < x < 2 \end{cases}$ is
(a) 0 (b) 1
(c) 2 (d) 3

Q.11 The second degree polynomial f(x), satisfying f(0) = 0, $f'(x) > 0 \forall x \in (0,1)$
(a) $f(x) = \phi$
(b) $f(x) = ax + (1-a)x^2, \forall a \in (0,\infty)$
(c) $f(x) = ax + (1-a)x^2, a \in (0,2)$
(d) No such polynomial

Q.12 If $f(x) = x^2 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$, such that min $f(x) > \max g(x)$, then the relation between b and c, is
(a) No real value of b and c
(b) $0 < c < b\sqrt{2}$
(c) $|c| < |b|\sqrt{2}$
(d) $|c| < |b|\sqrt{2}$

Q.13 The maximum value of $(\cos\alpha_1).(\cos\alpha_2).\ldots.(\cos\alpha_n)$, under the restrictions $0 \leq \alpha_1, \alpha_2 \ldots\ldots, \alpha_n \leq \frac{\pi}{2}$ and $(\cot\alpha_1).(\cot\alpha_2).\ldots.(\cot_n) = 1$ is
(a) $\frac{1}{2^{n/2}}$ (b) $\frac{1}{2^n}$
(c) $\frac{1}{2n}$ (d) 1

Q.14 If $f(x) = \begin{cases} |x|, & \text{for } 0 < |x| \leq 2 \\ 1, & \text{for } x = 0 \end{cases}$
(a) a local maximum
(b) no local maximum
(c) a local minimum
(d) no extremum

Q.15 If $f(x) = \frac{x^2-1}{x^2+1}$, for every real number x, then the minimum value of *f*
(a)does not exist because *f* is unbounded
(b) is not attained even though *f* is bounded
(c) is 1
(d) is –1

Q.16 the number of values of x, where the function $f(x) = \cos x + \cos(\sqrt{2}x)$ attains its maximum, is
(a) 0 (b) 1
(c) 2 (d) infinite

Q.17 find the coordinates of all the points P on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, for which the area of the ΔPON is maximum, where O denotes the origin and N is the foot of the perpendicular from O to the tangents at P.
(a) $\left(\frac{\pm a^2}{\sqrt{a^2+b^2}}, \frac{\pm b^2}{\sqrt{a^2+b^2}}\right)$
(b) $\left(\frac{\pm a^2}{\sqrt{a^2-b^2}}, \frac{\pm b^2}{\sqrt{a^2-b^2}}\right)$
(c) $\left(\frac{\pm a^2}{\sqrt{a^2+b^2}}, \frac{\pm b^2}{\sqrt{a^2-b^2}}\right)$
(d) $\left(\frac{\pm a^2}{\sqrt{a^2-b^2}}, \frac{\pm b^2}{\sqrt{a^2+b^2}}\right)$

Q.18 If $y = a\log x + bx^2 + x$ has its extremum values at x = –1 and x = 2 then
(a) $a = 2, b = -1$
(b) $a = 2, b = -\frac{1}{2}$
(c) $a = -2, b = \frac{1}{2}$
(d) None of the above

Q.19 The function $f(x) = 2|x| + |x+2| - \|x+2| - 2| x\|$ has a local minimum or a local maximum at x is equal to
(a) –2 (b) $\frac{-2}{3}$
(c) 2 (d) 2/3

Q.20 A rectangular sheet of fixed perimeter with sides having their lengths in the ratio 8: 15 is converted into an open reactangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. The lenggths of the sides of the rectangualr sheet are
(a) 24 (b) 32
(c) 45 (d) 60

Q.21 If $f(x)=\begin{cases} e^x & ,0\le x\le 1 \\ 2-e^{x-1} & ,1<x\le 2 \\ x-e & ,2<x\le 3 \end{cases}$ and $g(x)=\int_0^x f(t)\,dt,\ x\in[1,3]$, then

(a) g(x) has local maxima at $x=\log_e 2$ and local minima at x = e

(b) f(x) has local maxima at $x = 1 + \log_e 2$ and local minima at x=e

(c) g(x) has no local minima

(d) f(x) has no local maxima

Q.22 If f(x) is a cubic polynomial which has local maximum at x = –1.If f(b) = 18, f(a) = –1 and f′(x) has local minimum at x = 0, then

(a) The distance between (–1, 3) and (a, f(a)), where x = a is the point of local minima, is $2\sqrt{5}$

(b) f(x) is increasing for $x\in\left[1,2\sqrt{5}\right]$

(c) f(x) has local minima at x = 1

(d) the value of f(0) = 5

Q.23 If $f(x)=\begin{cases} 3x^2+12x-1, & -1\le x\le 2 \\ 37-x, & 2<x\le 3 \end{cases}$, then

(a) f(x) is increasing on [–1, 2]

(b) f(x) is continuous on [–1, 3)

(c) f'(b) does not exist

(d) f(x) has the maximum value at x =2

Q.24 If f: $R\to R$ is a differentiable function such that f'(x) > 2f(x) for all $x\in R$ and f(0) = 1, then

(a) $f'(x) < e^{2x}$ in $(0,\infty)$

(b) $f(x) > e^{2x}$ in $(0,\infty)$

(c) f(x) is increasing in $(0,\infty)$

(d) f(x) is decreasing in $(0,\infty)$

Q.25 If $f(x)=\begin{vmatrix} \cos 2x & \cos 2x & \sin 2x \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix}$ then

(a) f'(x) = 0 at exactly three points in $(-\pi,\pi)$

(b) f(x) attains its minimum at x = 0

(c) f(x) attains its maximum at x = 0

(d) f'(x) = 0 at more than three points in $(-\pi,\pi)$

Q.26 The total number of distinct $x\in[0,1]$ for which $\int_0^x \frac{t^2}{1+t^4}dt = 2x-1$ is

(a) 2 (b) 1

(c) 3 (d) 0

Q.27 Let $f:R\to(0,\infty)$ and $g: R\to R$ be twice differentiable functions such that f" and g" are continuous functions on R. Suppose f'(2) = g(2) = 0, $f''(2)\neq 0$ and $g'(2)\neq 0$. If $\lim_{x\to 2}\frac{f(x)g(x)}{f'(x)g'(x)}=1$, then

(a) f has a local minimum at x = 2

(b) f has a local maximum at x = 2

(c) f" (2) > f(2)

(d) f(x) – f"(x) = 0 for at least one $x\in R$

Q.28 A cylindrical container is to be made from certain solid material with the following constraints : It has a fixed inner volume of Vmm^3, has a 2 mm thick solid wall and is open at the top. The bottom of the container is a solid circular disc of thickness 2 mm and is of radius equal to the outer radius of the container.

If the volume of the material used to make the container is minimum when the inner radius of the container is 10 mm, then the value of $\frac{V}{250\pi}$ is

(a) 1/250 (b) 1000

(c) 5 (d) 4

Directions (Q. 29 to 31): by appropriately matching the information given in the three columns of the following table.

Let $f(x) = x + \log_e x - \log_e x,\ x\in(0,\infty)$

- Column 1 contains information about zeros of f(x), f'(x) and f"(x).
- Column 2 contains information about the limiting behaviour of f(x), f'(x) and f"(x) at infinity.
- Column 3 contains information about increasing / decreasing nature of f(x) and f'(x).

Column 1

(I) f(x) = 0 for some $x\in(1,e^2)$

(II) f'(x) = 0 for some $x\in(1,e)$

(III) f'(x) = 0 for some $x\in(0,1)$

(IV) f"(x) = 0 for some $x\in(1,e)$

Column 2

(i) $\lim_{x\to\infty} f(x)=0$

(ii) $\lim_{x\to\infty} f(x)=-\infty$

(iii) $\lim_{x\to\infty} f'(x)=-\infty$

(iv) $\lim_{x\to\infty} f''(x)=0$

Column 3

(P) f is increasing in (0,1)
(Q) f is decreasing in (e, e^2)
(R) f' is increasing in (0,1)
(S) f' is decreasing in (e, e^2)

Q.29 Which of the following options is the only CORRECT combination?

(a) (II) (ii) (Q) (b) (III) (iii) (R)
(c) (IV) (iv) (S) (d) (I) (i) (P)

Q.30 Which of the following options is the only CORRECT combination?

(a) (III) (iv) (P) (b) (II) (iii) (S)
(c) (I) (ii) (R) (d) (IV) (i) (S)

Q.31 Which of the following options is the only INCORRECT combination?

(a) (I) (iii) (P) (b) (II) (iii) (P)
(c) (III) (i) (R) (d) (II) (iv) (Q)

Section-D
(School / Board pattern)

Q.1 Find the semi-vertical angle of the cone of maximum curved surface that can be inscribed in a sphere of radius *R*.

Q.2 A cylinder is such that the sum of its height and circumference of its base is 10 metres. Find the maximum volume of the cylinder.

Q.3 Show that the maximum volume of a cylinder which can be inscribed in a cone of height *h* and semi-vertical angle 30° is $\frac{4}{81}\pi h^3$.

Q.4 Find the minimum distance from the point (4, 2) to the parabola $y^2 = 8x$.

Q.5 The perimeter of a sector of a circle is *p*. Show that its area is maximum when its radius *p*/4.

Q.6 Find the dimensions of the right circular cone of minimum volume that can be circumscribed about a sphere of radius 8 cm.

Q.7 Find the maximum surface area of a circular cylinder that can be inscribed in a sphere of radius *R*.

Q.8 A window of fixed primeter (including the base of the arch) is in the form of a rectangle surmounted by a semicircle. The circular portion is fitted with colour glass while the rectangular part is fitted with clear glass. The clear glass transmits three times as much light per square unit as the coloured glass does. What is the ratio of the sides of the rectangle so that the window transmits maximum light?

Q.9 A box of constant volume *c* is to be twice as long it is wide. The cost per unit area of the material on the top and four side faces is three times the cost for bottom. What are the most economical dimensions of the box?

Q.10 A swimmer *P* is in a sea at a distance *d* km from the nearest point *A* on a straight shore. The house of the swimmer is on the shore at a distance *c* km from *A*. The ratio of his rate of walking to the rate of swimming is sec α. At what point on the shore should he land so that he reaches his house in shortest possible time?

Q.11 Show that the triangle of maximum area that can inscribed in a given circle is an equilateral triangle.

Q.12 The sum of the surface areas of a rectangular parallelopiped with sides *x*, 2*x* and *x*/3 and a sphere is given to be constant. Prove that the sum of their volumes is minimum if *x* is equal to three times the radius of the sphere. Also find the minimum value of sum of their volumes.

Q.13 Find the altitude of a right circular cone of maximum curved surface which can be inscribed in a sphere of radius *r*.

Q.14 Find the volume of the largest cone that can be inscribed in a sphere of radius *r*.

Q.15 Show that the height of a cylinder of maximum volume that can be inscribed in a sphere of radius *R* is $\frac{2R}{\sqrt{3}}$.

Q.16 Assuming that the stiffness of a beam of rectangular corss-section varies as the breadth and as the cube of the depth, what must be the breadth of the stiffest beam that can be cut from a log of diameter *a*?

Q.17 An enemy vehicle is moving along the curve $y = x^2 + 2$. Find the shortest distance between the vehicle and our artillery located at (3, 2). Also find the co-ordinates of the vehicle when the distance is shortest.

Q.18 The cost of fuel for running a train is proportional to the square of the speed

generated in km/hr. It costs Rs. 48 per hour when the train is moving at a speed of 16 km/hr. What is the most economical speed if the fixed charges are Rs 300 per hour, over and above the running cost?

Q.19 At 8 a.m. ship *A* is 65 km due east of another ship *B*. Ship *A* is sailing due west at 10 km/h and ship *B* is sailing due south at 15 km/h. If they continue sailing on their respective paths, when will they be nearest to one another, and how near?

Q.20 A window is in the form of a rectangle surmounted by a semicircle. The total area of the window is fixed. What should be the ratio of the areas of the semicircular part and the rectangular part so that the total perimeter is minimum.

Q.21 A window is in the form of a rectangle surmounted by a semi-circular opening. If the perimeter of the window is 20 m, find the dimensions of the window so that the maximum possible light may be admitted through the whole opening.

Q.22 The length of the perimeter of a sector of a circle is 20 cm. Give an expression for the area of the sector in terms of *r* (the radius of the circle) and hence find the maximum area of the sector.

Q.23 Show that the right circular cone of least curved surface area and given volume has an altitude equal to $\sqrt{2}$ times the radius of the base.

Q.24 Show that a cylindrical vessel of given volume has the least surface area when its height is twice its radius.

Q.25 If the surface area of an open cylindrical can is 100 cm^2, find its maximum capacity.

Q.26 A closed rectangular box is made of aluminium sheet of negligible thickness and the length of the box is twice its breadth. If the capacity of the box is 243 cm^3, compute the dimensions of the box of least surface area.

Q.27 An open box with a square base is to be made out of a given quantity of cardboard whose area is c^2 square units. Show that the maximum volume of the box is $\frac{c^3}{6\sqrt{3}}$ cubic units.

Q.28 *ABC* is a right-angled triangle of given area *S*. Find the sides of the triangle for which the area of the circumscribed circle is least.

Q.29 If the sum of the surface areas of a cube and a sphere is constant, what is the ratio of an edge of the cube to the diameter of the sphere, when the sum of their volumes is minimum?

Q.30 A cone is circumscribed about a sphere of radius *R*. Show that the volume of the cone is minimum if its height is 4*R*.

Q.31 Find the co-ordinates of the point on the parabola $y^2 = x$ whose distance from the line $y - x = 1$ is shortest. Also find the shrotest distance.

Q.32 Find the maximum area of an isosceles triangle inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with its vertex at one end of the major axis.

Q.33 The combined resistance *R* of two resistors R_1 and R_2 ($R_1, R_2 > 0$) is given by $\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}$. If $R_1 + R_2 = C$ (a constant), show that the maximum resistance *R* is obtained by choosing $R_1 = R_2$.

Q.34 Prove that the semi-vertical angle of a right circular cone of maximum volume and of given slant height is $\tan^{-1}\sqrt{2}$.

Q.35 Prove that a conical tent of given capacity will require the least amount of canvas when the height is $\sqrt{2}$ times the radius of the base.

Q.36 If f(x)is twice differentiable function such that f(a) = 0, f(b) = 2, f(d) = 2, f(e) = 0,where a < b < c < d, e then the minimum number of zeroes of $g(x)=\{f'(x)\}^2+f''(x).f(x)$ in the interval [a, e] is.

Q.37 For the circle $x^2 + y^2 = r^2$ find the value of r for which the area enclosed by the tangents drawn from the point P(6, 8) to the circle and the chord of contact is maximum.

Q.38 Find a point on th curve $x^2 + 2y^2 = 6$ whose distance from thel ine x + y = 7 is minimum.

Q.39 Determine the points of maxima and minima of the function $f(x)=\frac{1}{8}\ln x - bx + x^2$, x > 0, where $b \geq 0$ is a constant.

Q.40 Let (h, k) be a fixed point, where h >0, k>0. A straight line passing through this point cuts

the positive dirctions of the coordinate axes at he points P and Q. find the minimum area of the ΔOPQ, O being the origin.

Q.41 The circle $x^2 + y^2 = 1$ cuts the X -axis at P and Q . Another circle with centre at Q and variable radius intersects the first circle at R above the X-axis and the line segment PQ at S. find the maximum area of the ΔQSR.

Q.42 What normal to the curve $y = x^2$ forms the shortest chord?

Q.43 Let $f(x) = \sin^3 x + \lambda \sin^2 x, -\frac{\pi}{2} < x < \frac{\pi}{2}$. Find the intervals in which λ should lie in the order that f(x) has exactly one minimum and exactly one maximum.

Q.44 Let $f : R \to R$ be defined as $f(x) = |x| + |x^2 - 1|$. The total number of points at which f attains either a local maximum or a local minimum is

Q.45. Let p(x) be a real polynomial of least degree which has a local maximum at x = 1 and a local minimum at x = 3. If p(1) = 6 and p(3) =2, then p'(0) is equal to

Q.46 Let f be a fucntion defined on R (the set of all real numbers) such that $f'(x) = 2010(x - 2009)(x - 2010)^2 (x - 2011)^3 (x - 2012)^4, \forall x \in R$. If g is a function defined on R with values in the interval $(0, \infty)$ such that $f(x) = \ln(g(x))$, $\forall x \in R$, then the number of points in R at which g has a local maximum is.....

Q.47 The maximum value of the expression $\frac{1}{\sin^2\theta + 3\sin\theta + 5\cos^2\theta}$ is.....

Q.48 The maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x - 48$ on the set $A = \{x | x^2 + 20 \le 9x\}$ is...

CHAPTER
2

INDEFINITE INTEGRATION

1. INDEFINITE INTEGRAL

Introduction :

A function $\phi(x)$ is called a primitive or an antiderivative of a function $f(x)$ if $\phi'(x) = f(x)$.

For example, $\frac{x^5}{5}$ is a primitive of x^4 because $\frac{d}{dx}\left(\frac{x^5}{5}\right) = x^4$

Let $\phi(x)$ be a primitive of a function $f(x)$ and let c be any constant.

Then $\frac{d}{dx}[\phi(x)+c] = \phi'(x) = f(x)$ $\quad [\because\ \phi'(x) = f(x)]$

$\Rightarrow$ $\phi(x) + c$ is also a primitive of $f(x)$

Thus, if a function $f(x)$ possesses a primitive, then it possesses infinitely many primitives which are cntained in the expression $f(x) + c$, where c is a constant.

For example $\frac{x^5}{5}, \frac{x^5}{5}+2, \frac{x^5}{5}-1$ etc. are primitive of x^4.

Note :- *Let f(x) be a function. Then the collection of all its primitives is called the indefinite integral of f(x) and is denoted by* $\int f(x)dx$.

Thus, $\frac{d}{dx}(\phi(x)+c) = f(x) \Rightarrow \int f(x)dx = \phi(x)+c$

where $f(x)$ is primitive of $f(x)$ and c is an arbitrary constant knwon as the constant of integration. Here $\int$ is the integral sign, $f(x)$ is the integrand, x is the variable of integration and dx is the element f integration.

The process of finding an idefinite integral of a given function is called integration of the function. It follows from the above discussion that integrating a function $f(x)$ means finding a function $\phi(x)$ such that $\frac{d}{dx}(\phi(x)) = f(x)$.

2. COMPARISION BETWEEN DIFFERENTIATION AND INTEGRATION

(1) Differentiation and integration both are operations on functions and each gives rise to a function.
(2) Each function is not differentiable or integrable.
(3) The derivative of a function, if it exists, is unique. The integral of a function, if it exists, is not unique.
(4) The derivative of a polynomial function decreases its degree by 1, but the integral of a polynomial function increases its degree by 1.
(5) The derivative has a geometrical meaning, namely, the slope of the tangent to a curve at a point on it. The integral has also a geometrical meaning, namely, the area of some region.
(6) The derivative is used in obtaining some physical quantities like velocity, acceleration etc. of a particle. The integral is used in obtaining some physical quantities like centre of mass, momentum etc.
(7) Differentiation and integration are inverse of each other.

Properties of Integrals

(1) The differentiation of an integral is the integrand itself or the process of differentiation and integration neutralize each other, i.e., $\frac{d}{dx}\left[\int f(x)dx\right] = f(x)$.

(2) The integral of the product of a constant and a function is equal to the product of the constant and the integral f the function, i.e., $\int cf(x)dx = c\int f(x)dx$.

(3) Integral of the sum or difference of two functions is equal to the sum or difference of their integrals, i.e., $\int\{f_1(x) \pm f_2(x)\}dx = \int f_1(x)dx \pm \int f_2(x)dx$

In the general form,

$\int\{k_1.f_1(x) \pm k_2.f_2(x) \pm k_3.f_3(x) \pm ....\}dx = k_1\int f_1(x)dx \pm k_2\int f_2(x)dx \pm k_3\int f_3(x)dx \pm .....$

3. FUNDAMENTAL INTEGRATION FORMULAE

1. (i) $\int x^n dx = \frac{x^{n+1}}{n+1} + c, n \pm 1 \quad \because \frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = x^n$

(ii) $\int dx = x + c$

(iii) $\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + c$,

(iv) $\int (ax+b)^n dx = \frac{1}{a}.\frac{(ax+b)^{n+1}}{n+1} + c$

2. (i) $\int \frac{1}{x} dx = \log|x| + c \quad \because \frac{d}{dx}(\log|x|) = \frac{1}{x}$

(ii) $\int \frac{1}{ax+b} dx = \frac{1}{a}(\log|ax+b| + c$

3. $\int e^x dx = e^x + c \qquad \because \frac{d}{dx}(e^x) = e^x$

4. $\int a^x dx = \frac{a^x}{\log_e a} + c \qquad \because \frac{d}{dx}\left(\frac{a^x}{\log_e a}\right) = a^x$

5. $\int \sin x\, dx = -\cos x + c \qquad \because \frac{d}{dx}(-\cos x) = \sin x$

6. $\int \cos x\, dx = \sin x + c \qquad \because \frac{d}{dx}(\sin x) = \cos x$

7. $\int \sec^2 x\, dx = \tan x + c \qquad \because \frac{d}{dx}(\tan x) = \sec^2 x$

8. $\int \text{cosec}^2 x\, dx = -\cot x + c \qquad \because \frac{d}{dx}(-\cot x) = \text{cosec}^2 x$

9. $\int \sec x \tan x\, dx = \sec x + c \qquad \because \frac{d}{dx}(\sec x) = \sec x \tan x$

10. $\int \text{cosec}\, x \cot x\, dx = -\text{cosec}\, x + c \qquad \because \frac{d}{dx}(-\text{cosec}\, x) = \text{cosec}\, x \cot x$

11. $\int \tan x\, dx = -\log|\cos x| + c = \log|\sec x| + c \qquad \because \frac{d}{dx}(\log\cos x) = -\tan x$

12 $\int \cot x\, dx = \log|\sin x| + c = -\log|\cos ec\, x| + c \qquad \because \frac{d}{dx}(\log\sin x) = \cot x$

13 $\int \sec x\, dx = \log|\sec x + \tan x| + c = \log\tan\left(\frac{\pi}{4} + \frac{x}{2}\right) + c \quad \because \frac{d}{dx}\log(\sec x + \tan x) = \sec x$

14. $\int \text{cosec } x\,dx = \log|\text{cosec } x - \cot x| + c = \log\tan\frac{x}{2} + c \quad \because \frac{d}{dx}(\log|\text{cosec } x - \cot x|) = \text{cosec } x$

15. $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1}x + c = -\cos^{-1}x + c \quad \because \frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}, \frac{d}{dx}(\cos^{-1}x) = \frac{-1}{\sqrt{1-x^2}}$

16. $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + c = -\cos^{-1}\frac{x}{a} + c \quad \because \frac{d}{dx}\left(\sin^{-1}\frac{x}{a}\right) = \frac{1}{\sqrt{a^2-x^2}}, \frac{d}{dx}(\cos^{-1}\frac{x}{a}) = \frac{-1}{\sqrt{a^2-x^2}}$

17. $\int \frac{dx}{1+x^2} = \tan^{-1}x + c = -\cot^{-1}x + c \quad \because \frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}, \frac{d}{dx}(\cot^{-1}x) = \frac{-1}{1+x^2}$

18. $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + c = \frac{-1}{a}\cot^{-1}\frac{x}{a} + c \quad \because \frac{d}{dx}\left(\tan^{-1}\frac{x}{a}\right) = \frac{a}{a^2+x^2} \frac{d}{dx}\left(\cot^{-1}\frac{x}{a}\right) = \frac{-a}{a^2+x^2}$

19. $\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1}x + c = -\text{cosec}^{-1}x + c \quad \because \frac{d}{dx}(\sec^{-1}x) = \frac{1}{x\sqrt{x^2-1}} \frac{d}{dx}(\text{cosec}^{-1}x) = \frac{-1}{x\sqrt{x^2-1}}$

20. $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\frac{x}{a} + c = \frac{-1}{a}\text{cosec}^{-1}\frac{x}{a} + c \quad \because \frac{d}{dx}\left(\sec^{-1}\frac{x}{a}\right) = \frac{a}{x\sqrt{x^2-a^2}} \frac{d}{dx}(\text{cosec}^{-1}\frac{x}{a}) = \frac{-a}{x\sqrt{x^2-a^2}}$

Note :- *In any of the fundamental integration formulae , If x is replaced by ax + b, then the same formulae is applicable but we must divide by coefficient of x or derivative f (ax + b) i.e., a. In general, if* $\int f(x)dx = \phi(x) + c$, then $\int f(ax+b)dx = \frac{1}{a}\phi(ax+b) + c$

Some more results :

(i) $\int \frac{1}{x^2-a^2} = \frac{-1}{a}\cosh^{-1}\frac{x}{a} + c = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + c$, when $x > a$

(ii) $\int \frac{1}{a^2-x^2}dx = \frac{1}{a}\tanh^{-1}\frac{x}{a} + c = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + c$, when $x < a$

(iii) $\int \frac{dx}{\sqrt{x^2-a^2}} = \log\{|x + \sqrt{x^2-a^2}|\} + c = \sinh^{-1}\left(\frac{x}{a}\right) + c$

(iv) $\int \frac{dx}{\sqrt{x^2+a^2}} = \log\{|x + \sqrt{x^2+a^2}|\} + c = \sinh^{-1}\left(\frac{x}{a}\right) + c$

(v) $\int \sqrt{a^2-x^2}dx = \frac{1}{2}x\sqrt{a^2-x^2} + \frac{1}{2}a^2\sin^{-1}\left(\frac{x}{a}\right) + c$

(vi) $\int \sqrt{x^2-a^2}dx = \frac{1}{2}x\sqrt{x^2-a^2} - \frac{1}{2}a^2\log\{x + \sqrt{x^2-a^2}\} + c = \frac{1}{2}x\sqrt{x^2-a^2} - \frac{1}{2}a^2\cosh^{-1}\left(\frac{x}{a}\right) + c$

(vii) $\int \sqrt{x^2+a^2}dx = \frac{1}{2}x\sqrt{x^2+a^2} + \frac{1}{2}a^2\log\{x + \sqrt{x^2+a^2}\} + c = \frac{1}{2}x\sqrt{x^2+a^2} + \frac{1}{2}a^2\sinh^{-1}\left(\frac{x}{a}\right) + c$

Note :-

- *The signum function has an antiderivative on any interval which doesn't contain the point x = 0, and does not possess an antiderivative on any interval which contains the point.*
- *The antiderivative of every odd function is an even function and vice-versa.*

***Illustration 1*:** Evaluate : $\int(\sqrt{3}\sin x - \cos x)dx$

Solution : $\int(\sqrt{3}\sin x - \cos x)dx = \sqrt{3}\int \sin x dx - \int \cos x\,dx$

$= -\sqrt{3}\cos x - \sin x + c = -2\left[\cos x.\cos\frac{\pi}{6} + \sin x.\sin\frac{\pi}{6}\right] + c = -2\cos\left(x - \frac{\pi}{6}\right) + c$

Student Task

If in illustration 1, we write $\sqrt{3}\sin x - \cos x$ as $-2\cos\left(x+\frac{\pi}{3}\right)$, then what will be the integral?

***Illustration 2*:** Evaluate : $\int \sec^2(3x+5)dx$

Solution : We know that $\int \sec^2 x\,dx = \tan x + c$

so, $\int \sec^2(3x+5)dx = \frac{1}{3}\tan(3x+5)+c$

4. INTEGRATION USING TRIGONOMETRIC IDENTITIES

When the integrand consists of trigonometric functions, we use known identities to convert it into a form which can easily be integrated. Some of the identities useful for this purpose are given below :

(i) $2\sin^2\left(\frac{x}{2}\right) = (1-\cos x)$ (ii) $2\cos^2\left(\frac{x}{2}\right) = (1+\cos x)$

(iii) 2 sin *a* cos *b* = sin (*a* + *b*) + sin (*a* – *b*) (iv) 2 cos *a* sin *b* = sin (*a* + *b*) – sin (*a* – *b*)

(v) 2 cos *a* cos *b* = cos (*a* + *b*) + cos (*a* – *b*) (vi) 2 sin *a* sin *b* = cos (*a* – *b*) – cos (*a* + *b*)

***Illustration 3*:** Evaluate : $\int \sin 3x \sin 2x\,dx$

Solution : Using 2 sin *a* sin *b* = cos (*a* – *b*) – cos (*a* + *b*), we have

$$= \frac{1}{2}\int(\cos x - \cos 5x)dx = \frac{1}{2}\int\cos x dx - \frac{1}{2}\int\cos 5x\,dx = \frac{1}{2}\sin x - \frac{\sin 5x}{10} + c$$

***Illustration 4*:** Evaluate : $\int \cos x\cos 2x\cos 3x\,dx$

Solution :

$$\int\cos x\cos 2x\cos 3x\,dx = \frac{1}{2}\int(2\cos x\cos 2x)\cos 3x\,dx$$

$$= \frac{1}{2}\int(\cos 3x+\cos x)\cos 3x\,dx = \frac{1}{2}\int(\cos^2 3x+\cos x\cos 3x)dx$$

$$= \frac{1}{4}\int(2\cos^2 3x)dx + \frac{1}{4}\int(2\cos x\cos 3x)dx = \frac{1}{4}\int(1+\cos 6x)dx + \frac{1}{4}\int(\cos 4x+\cos 2x)dx$$

$$= \frac{1}{4}\int dx + \frac{1}{4}\int\cos 6x\,dx + \frac{1}{4}\int\cos 4x\,dx + \frac{1}{4}\int\cos 2x\,dx$$

$$= \frac{1}{4}x + \frac{1}{4}.\frac{\sin 6x}{6} + \frac{1}{4}.\frac{\sin 4x}{4} + \frac{1}{4}.\frac{\sin 2x}{2} + c = \frac{x}{4} + \frac{\sin 6x}{24} + \frac{\sin 4x}{16} + \frac{\sin 2x}{8} + c$$

***Illustration 5*:** Evaluate : $\int \frac{dx}{4\cos x + 3\sin x}$

Solution : Put 4 = *r* sin θ and 3 = *r* cos θ so that $r^2 = 25$ and $\theta = \tan^{-1}\left(\frac{4}{3}\right)$

$$\therefore \int\frac{dx}{4\cos x+3\sin x} = \int\frac{dx}{r\sin\theta\cos x + r\cos\theta\sin x} = \frac{1}{r}\int\frac{dx}{\sin(\theta+x)} = \frac{1}{r}\int\text{cosec}\,(\theta+x)dx$$

$$= \frac{1}{r}\log\left\{\tan\left(\frac{\theta+x}{2}\right)\right\} + c = \frac{1}{5}\log\left[\tan\left\{\frac{1}{2}\tan^{-1}\left(\frac{4}{3}\right)+\frac{x}{2}\right\}\right] + c$$

$$I = \int\sin x\sin 2x\sin 3x\,dx$$

$\sin x \sin 2x \sin 3x = \frac{1}{2}\{\cos 2x - \cos 4x\}\sin 2x = \frac{1}{4}\{2\sin 2x\cos 2x - 2\sin 2x\cos 4x\}$

$= \frac{1}{4}\{\sin 4x - (\sin 6x - \sin 2x)\} = \frac{1}{4}\{\sin 2x + \sin 4x - \sin 6x\}$.

$$\therefore\ I = \frac{1}{4}\int(\sin 2x+\sin 4x-\sin 6x)dx = -\frac{1}{4}\left\{\frac{\cos 2x}{2}+\frac{\cos 4x}{4}-\frac{\cos 6x}{6}\right\}+c$$

***Illustration 6*:** $I = \int \cos^4 x \sin^2 x\, dx$

$$\cos^4 x \sin^2 x = \frac{2}{4}\sin^2 2x \cos^2 x = \frac{1}{16}[(1-\cos 4x)(1+\cos 2x)]$$

$$= \frac{1}{16}\{1 + \cos 2x - \cos 4x - \cos 2x \cos 4x\} = \frac{1}{32}\{2 + 2\cos 2x - 2\cos 4x - \cos 6x - \cos 2x\}$$

$$= \frac{1}{32}\{2 + \cos 2x - 2\cos 4x - \cos 6x\} \therefore I = \frac{1}{32}\left\{2x + \frac{\sin 2x}{2} - \frac{\sin 4x}{2} - \frac{\sin 6x}{6}\right\} + c$$

(ii) $\int \sin^p x \cos^q x\, dx$, at least one of p and q is odd.

If $p = 2k + 1$, put $\cos x = t$,
If $q = 2k + 1$, put $\sin x = t$.

***Illustration 7*:** $I = \int \cos^4 x \sin^3 x\, dx$

$$\therefore I = \int \cos^4 x \sin^2 x . \sin x\, dx$$

Put $\cos x = t$, $-\sin x\, dx = dt$

$$\therefore I - \int t^4(1-t^2)dt = \int (t^6 - t^4)dt = \frac{1}{7}t^7 - \frac{1}{5}t^5 + c = \frac{1}{7}\cos^7 x - \frac{1}{5}\cos^5 x + c$$

(iii) $\int \sin^p x \cos^q x\, dx, p + q = -2n, n \in Z$

***Illustration 8*:** $I = \int \sin^{-3/2} x \cos^{-5/2} x\, dx$

$$I = \int \frac{1}{\left(\frac{\sin^{3/2} x}{\cos^{3/2} x}\right)\cos^4 x} dx = \int \frac{\sec^2 x, \sec^2 x}{\tan^{3/2} x}$$

Put $\tan x = t$, $\sec^2 x\, dx = dt$

$$\therefore I = \int \frac{(1+t^2)dt}{t^{3/2}} = \int (t^{-3/2})dt = -2t^{-1/2} + \frac{2}{3}t^{3/2} + c \ = -2\sqrt{\cot x} + \frac{2}{3}\tan^{3/2} x + c$$

(iv) Integration of $\int \frac{dx}{a + b\cos x}, \int \frac{dx}{a + b\sin x}, \int \frac{dx}{a + b\cos x + c\sin x}$

Replace, $\sin x = \frac{(2\tan\frac{1}{2}x)}{(1 + \tan^2\frac{1}{x}x)}$, $\cos x = \frac{(1 - \tan^2\frac{1}{x}x)}{(1 + \tan^2\frac{1}{2}x)}$. Then, put $\tan\frac{1}{2}x = t$

***Illustration 9*:** $I = \int \frac{dx}{5 + 7\cos x + \sin x}$

$$= \int \frac{dx}{5 + 7\frac{(1 - \tan^2 x/2)}{1 + \tan^2 x/2} + \frac{2\tan x/2}{1 + \tan^2 x/2}} = \int \frac{(1 + \tan^2 x/2)dx}{-2\tan^2 x/2 + 2\tan x/2 + 12}$$

Put $\tan(x/2) = t, \frac{1}{2}\sec^2(x/2)dx = dt$

$$\therefore I = 2\int \frac{dt}{-2(t^2 - t - 6)} = -\int \frac{dt}{(t-3)(t+2)} = -\frac{1}{5}\int\left(\frac{1}{t-3} - \frac{1}{t+2}\right)dt$$

$$= -\frac{1}{5}\{\log|t-3| - \log|t+2|\} + c = \frac{1}{5}\log\left|\frac{\tan(x/2)+2}{\tan(x/2)-3}\right| + c$$

(v) Integration of $\int \frac{p\cos x + q\sin x}{a + b\cos x + b\sin x} dx$

Put $p\cos x + q\sin x = A(a + b\cos x + b\sin x) + B$.[Differential of $(a + b\cos x + b\sin x)$]+ C. A, B and C can be calculated by equating the coefficient of $\cos x$, $\sin x$ and the constant terms.

***Illustration 10*:** $I = \int \frac{3\cos x + 2}{\sin x + 2\cos x + 3} dx$

Let $3\cos x + 2 = A(\sin x + 2\cos x + 3) + B(\cos x - 2\sin x) + C$.

Then comparing coefficients of sin x, cos x and constnat terms, we get

$A - 2B = 0$, $2A + B = 3$, $3A + C = 2$

Solving these equaitons, we get $A = 6/5$, $B = 3/5$, $C = -8/5$

Then $I = \frac{6}{5}\int dx + \frac{3}{5}\int \frac{\cos x - 2\sin x}{\sin x + 2\cos x + 3} dx - \frac{8}{5}\int \frac{dx}{\sin x + 2\cos x + 3}$

$\Rightarrow I = \frac{6}{5}x + \frac{3}{5}\log|\sin x + 2\cos x + 3| - \frac{8}{5}I_1$

$I_1 = \int \frac{dx}{\frac{2\tan(x/2)}{1+\tan^2(x/2)} + \frac{2(1-\tan^2(x/2))}{1+\tan^2(x/2)} + 3} = \int \frac{\sec^2(x/2)dx}{\tan^2(x/2) + 2\tan(x/2) + 5}$

Putting $\tan(x/2) = t$, $\frac{1}{2}\sec^2(x/2) = dt$

$\Rightarrow I_1 = 2\int \frac{dt}{t^2 + 2t + 5} = 2\int \frac{dt}{(t+1)^2 + 2^2} = \frac{2}{2}\tan^{-1}\left(\frac{t+1}{2}\right) = \tan^{-1}\{\frac{1}{2}\tan(x/2) + 1\}$.

Hence, $I = \frac{6}{5}x + \frac{3}{5}\log|\sin x + 2\cos x + 3| - \frac{8}{5}\tan^{-1}\{\frac{1}{2}\tan x/2 + 1)\} + c$

5. INTEGRATION BY SUBSTITUTION

It is not always possible to find the integral of a complicated function only by observation, so we need some methods of integration and integration by substitution is one of them. This method has 3 parts :

(i) Direct substitution
(ii) Standard substitution
(iii) Indirect substitution

DIRECT SUBSTITUTION

If $\int f(x)dx = g(x) + c$ then in $I = \int f(h(x))h'(x)dx$,

we put $h(x) = t$

$\Rightarrow$ $h'(x)dx = dt$

so $I = \int f(t)dt = g(t) + c = g(h(x)) + c$

***Illustration 11*:** Evaluate : $\int \cot x\, dx$

Solution : $I = \int \cot x\, dx = \int \frac{\cos x\, dx}{\sin x}$ put $\sin x = t$

$\Rightarrow \cos x\, dx = dt$

so $I = \int \frac{dt}{t} = \ln|t| + c = \ln|\sin x| + c$

***Illustration 12*:** Evaluate : $\int \frac{dx}{2\sqrt{x}(x+1)}$

Solution : Put $x = t^2 \Rightarrow dx = 2t\, dt$

So $I = \int \frac{dx}{2\sqrt{x}(x+1)} = \int \frac{2t\, dt}{2t(t^2+1)} = \int \frac{dt}{1+t^1} = \tan^{-1} t + c = \tan^{-1}(\sqrt{x}) + c$

STANDARD SUBSTITUTION

In some standard integrand or a part of it, we have standard substitution. List of standard substitution is as follows :

Integrand	**Substitution**
$x^2 + a^2$ or $\sqrt{x^2+a^2}$	$x = a \tan\theta$
$x^2 - a^2$ or $\sqrt{x^2-a^2}$	$x = a \sec\theta$
$a^2 - x^2$ or $\sqrt{a^2-x^2}$	$x = a \sin\theta$ or $x = a\cos\theta$
$\sqrt{a+x}$ and $\sqrt{a-x}$	$x = a \cos 2\theta$
$\left(x \pm \sqrt{x^2 \pm a^2}\right)^n$	expression inside the bracket = t
$\frac{2x}{a^2-x^2}, \frac{2x}{a^2+x^2}, \frac{a^2-x^2}{a^2+x^2}$	$x = a\tan\theta$
$2x^2 - 1$	$x = \cos\theta$
$\frac{1}{(x+a)^{1-\frac{1}{n}}(x+b)^{1+\frac{1}{n}}} (n \in N, n > 1)$	$\frac{x+a}{x+b} = t$

***Illustration 13*:** Evaluate : $\int \frac{dx}{(x+3)^{15/16}(x-4)^{17/16}}$

Solution : $I = \int \frac{dx}{(x+3)^{15/16}(x-4)^{17/16}} = \int \frac{dx}{\left(\frac{x+3}{x-4}\right)^{15/16}(x-4)^2}$

Put $\frac{x+3}{x-4} = t \Rightarrow \left(\frac{(x-4)-(x+3)}{(x-4)^2}\right)dx = dt \Rightarrow \frac{dx}{(x-4)^2} = \frac{dt}{-7}$

So $I = \frac{-1}{7}\int \frac{dt}{t^{15/16}} = \frac{-1}{7}\int t^{-15/16}dt$

$= \frac{-1}{7}\frac{t^{1/16}}{1/16} + c = \frac{-16}{7}t^{1/16} + c = \frac{-16}{7}\left(\frac{x+3}{x-4}\right)^{1/16} + c$

***Illustration 14*:** Evaluate : $\int \frac{dx}{(x+\sqrt{x^2-4})^{5/3}}$

Put $x + \sqrt{x^2-4} = t \Rightarrow \left(1 + \frac{x}{\sqrt{x^2-4}}\right)dx = dt$

$x + \sqrt{x^2-4} = t \Rightarrow \sqrt{x^2-4} = t - x$

$\Rightarrow x^2 - 4 = t^2 + x^2 - 2tx \Rightarrow x = \frac{t^2+4}{2t}$

$x^2 - 4 = \left(\frac{t^2+4}{2t}\right)^2 - 4 = \frac{t^4+16+8t^2-16t^2}{4t^2} = \left(\frac{t^2-4}{2t}\right)^2$

$\sqrt{x^2-4} = \frac{t^2-4}{2t} \Rightarrow dx = \left(\frac{t^2 2 - 4}{2t^2}\right)dt$

so $I = \int\left(\frac{t^2-4}{2t^2}\right)\frac{1}{t^{5/3}}dt = \frac{1}{2}\int t^{-5/3}dt - 2\int t^{-11/3}dt = \frac{1}{2}\frac{t^{-2/3}}{(-2/3)} - 2\frac{t^{-8/3}}{(-8/3)} + c$

$= \frac{-3}{4}t^{-2/3} + \frac{3}{4}t^{-8/3} + c = \frac{3}{4}t^{-8/3}[1-t^2] + c$ where $t = (x + \sqrt{x^2-4})$

Student Task

Q. Can we put $x = a \sin\theta$ in $\sqrt{x^2 - a^2}$?

INDIRECT SUBSTITUTION

If integrand $f(x)$ can be rewritten as product of two functions. $f(x) = f_1(x)\, f_2(x)$, where $f_2(x)$ is a function of integral of $f_1(x)$, then put inegral of $f_1(x) = t$.

***Illustration 15*:** Evaluate : $\int \sqrt{\frac{x}{4-x^3}}dx$

Solution : $I = \int \sqrt{\frac{x}{4-x^3}}dx = \int \frac{\sqrt{x}\,dx}{\sqrt{4-x^3}}$

Here integral of $\sqrt{x} = \frac{2}{3}x^{3/2}$ and $4 - x^3 = 4 - (x^{3/2})^2$

Put $x^{3/2} = t \Rightarrow \sqrt{x}\,dx = \frac{2}{3}dt$

So $I = \frac{2}{3}\int \frac{dt}{\sqrt{4-t^2}} = \frac{2}{3}\sin^{-1}\left(\frac{t}{2}\right) + c = \frac{2}{3}\sin^{-1}\left(\frac{x^{3/2}}{2}\right) + c$

***Illustration 16*:** Evaluate : $\int (\cos x - \sin x)(3 + 4\sin 2x)dx$

Solution : $I = \int (\cos x - \sin x)(3 + 4\sin 2x)dx$

Here integration of $\cos x - \sin x = \sin x + \cos x$

and $3 + 4 \sin 2x = 3 + 4((\sin x + \cos x)^2 - 1)$

Put $\sin x + \cos x = t \Rightarrow (\cos x - \sin x)\,dx = dt$

so $I = \int (3 + 4(t^2 - 1))dt = \int (4t^2 - 1)dt = \frac{4t^3}{3} - t + c$

$= \frac{t}{3}[4t^2 - 3] + c = \left(\frac{\sin x + \cos x}{3}\right)\left[4(\sin x + \cos x)^2 - 3\right] = \left(\frac{\sin x + \cos x}{3}\right)(1 + 4\sin 2x) + c$

***Illustration 17*:** Evaluate : $\int \frac{x^2 + 4}{x^4 + 16}dx$

Soltion : $\int \frac{x^2 + 4}{x^4 + 16}dx = \int \frac{\left(1 + \frac{4}{x^2}\right)}{\left(x^2 + \frac{16}{x^2}\right)}dx$ [dividing numerator and denominator by x^2]

$= \int \frac{\left(1 + \frac{4}{x^2}\right)}{\left(x - \frac{4}{x}\right)^2 + 8}dx = \int \frac{dt}{(t^2 + 8)}$ [putting $\left(x - \frac{4}{x}\right) = t$ and $\left(1 + \frac{4}{x^2}\right)dx = dt$]

$= \int \frac{dt}{t^2 + \left(\sqrt{8}\right)^2} = \frac{1}{\sqrt{8}}\tan^{-1}\left(\frac{t}{\sqrt{8}}\right) + c = \frac{1}{\sqrt{8}}\tan^{-1}\left(\frac{x - \frac{4}{x}}{\sqrt{8}}\right) + c = \frac{1}{2\sqrt{2}}\tan^{-1}\left(\frac{x^2 - 4}{2\sqrt{2}x}\right) + c$

***Illustration 18*:** Evaluate : $\int \frac{x^2-1}{x^4+x^2+1}dx$

Solution : $\int \frac{x^2-1}{x^4+x^2+1}dx = \int \frac{\left(1-\frac{1}{x^2}\right)}{\left(x^2+\frac{1}{x^2}+1\right)}dx$ [dividing numerator and denominator by x^2]

$= \int \frac{1-\frac{1}{x^2}}{\left[\left(x+\frac{1}{x}\right)^2-1\right]}dx = \int \frac{dt}{(t^2-1)}$ [putting $\left(x+\frac{1}{x}\right)=t$ and $\left(1-\frac{1}{x^2}\right)dx=dt$]

$= \frac{1}{2}\log\left|\frac{t-1}{t+1}\right|+c = \frac{1}{2}\log\left|\frac{x+\frac{1}{x}-1}{x+\frac{1}{x}+1}\right|+c = \frac{1}{2}\log\left|\frac{x^2-x+1}{x^2+x+1}\right|+c$

Student Task

Q. What will be substitution in $\int \frac{\left(1+\frac{1}{x^2}\right)dx}{x^2+\frac{1}{x^2}+3}$?

EXERCISE -1

Q.1 $\int\left(\sqrt{x}+\frac{1}{x}\right)dx$

Q.2 $\int \frac{\sin x-\cos x}{\sin x+\cos x}dx$

Q.3 $\int \frac{1}{e^x+1}dx$

Q.4 $\int \frac{1}{x\ln x}dx$

Q.5 $\int \sec^2(4x-7)dx$

Q.6 $\int \frac{\cos(\tan^{-1}x)}{(1+x^2)\sqrt{\sin(\tan^{-1}x)}}dx$

Q.7 $\int \frac{1-\frac{1}{x^2}}{x^2+\frac{1}{x^2}+3}dx$

Q.8 $\int \tan x \ln(\cos x)dx$

Q.9 $\int \frac{4\cos x-6\sin x}{3\cos x+2\sin x}dx$

Q.10 $\int \frac{dx}{x^{3/2}\left(1+\frac{1}{\sqrt{x}}\right)}$

ANSWERS

1. $\frac{2}{3}x^{3/2}+\ln|x|+c$
2. $\ln\left|\frac{1}{\sin x+\cos x}\right|+c$
3. $x - \ln(e^x+1)+c$
4. $\ln|\ln x|+c$
5. $\frac{1}{4}\tan(4x-7)+c$
6. $2\sqrt{\sin(\tan^{-1}x)}+c$
7. $\tan^{-1}\left(x+\frac{1}{x}\right)+c$
8. $-\frac{(\ln(\cos x))^2}{2}+c$
9. $2\ln|2\sin x+3\cos x|+c$
10. $-2\ln\left|1+\frac{1}{\sqrt{x}}\right|+c$

6. INTEGRATION BY PARTS

If integrand can be expressed as product of two function, then we use the following formula.

$$\int f_1(x) f_2(x)dx = f_1(x)\int f_2(x)dx - \int((f_1'(x)\int f_2(x)dx)dx$$

where $f_1(x)$ and $f_2(x)$ are known as first and second function respectively.

Remarks :

(i) We do not put constant of integration in 1st integral, we put this only once in the end.

(ii) Order of $f_1(x)$ and $f_2(x)$ is normally decided by the rule ILATE, where I $\rightarrow$ Inverse, L $\rightarrow$ Logarithms, A $\rightarrow$ Algebric, T $\rightarrow$ Trigonometric and E $\rightarrow$ Exponential.

Student Task

Q. Find 1st and 2nd functions in $\int e^{\sin x} \sin 2x\, dx$?

***Illustration 19*:** Evaluate : $\int x^2 \sin x\, dx$

Solution : $\int x^2 \sin x\, dx = x^2\int \sin x\, dx - \int(2x\int \sin x\, dx)dx$

$= -x^2\cos x + 2\int x\cos x\, dx = -x^2\cos x + 2[x\int \cos x\, dx - \int(1\int \cos x\, dx)dx$

$= -x^2\cos x + 2[s x\sin x - \int \sin x\, dx] = -x^2\cos x + 2x\sin x - 2\cos x + c$

***Illustration 20*:** Evaluate : $\int \sin^{-1}\left(\frac{2x+2}{\sqrt{4x^2+8x+13}}\right)dx$

Solution : Let $I = \int \sin^{-1}\left(\frac{2x+2}{\sqrt{4x^2+8x+13}}\right)dx$

Here $I = \int \sin^{-1}\left(\frac{2x+2}{\sqrt{4x^2+8x+13}}\right)dx = \int \sin^{-1}\left(\frac{2x+2}{\sqrt{(2x+2)^2+9}}\right)dx$

Put $2x + 2 = 3\tan\theta$

$dx = \frac{3}{2}\sec^2\theta\, d\theta$

Also $\frac{2x+2}{\sqrt{(2x+2)^2+9}} = \frac{3\tan\theta}{3\sec\theta} = \sin\theta$

So $I = \frac{3}{2}\int \theta\sec^2\theta\, d\theta = \frac{3}{2}\left[\theta\int \sec^2\theta - \int(1\int \sec^2\theta\, d\theta)d\theta\right]$

$= \frac{3}{2}\left[\theta.\tan\theta - \int \tan\theta\, d\theta\right] = \frac{3}{2}[\theta\tan\theta + \ln(\cos\theta)] + c$

$= \frac{3}{2}\left[\frac{2x+2}{3}\tan^{-1}\left(\frac{2x+2}{3}\right) + \ln\left(\frac{3}{\sqrt{4x^2+8x+13}}\right)\right] + c$

SPECIAL USE OF INTEGRATION BY PARTS

(i) $\int f(x)dx = \int (f(x)).1dx$

Now integrate taking $f(x)$ as 1st function and 1 as 2nd function. In most of the cases, $f(x)$ is an inverse or logarithmic function.

(ii) $\int \frac{f(x)}{[g(x)]^n}dx = \int \frac{f(x)}{g'(x)}.\frac{g'(x)}{[g(x)]^n}dx$

Now integrate taking $\frac{f(x)}{g'(x)}$ as 1st function and $\frac{g'(x)}{[g(x)]^n}$ as 2nd function.

(iii) If integrand is of the form $e^x f(x)$, then rewrite $f(x)$ as sum of two functions in which one is derivative of other.

$$\int e^x f(x)dx = \int e^x (g(x) + g'(x))dx = e^x g(x) + c$$

Some Special integrals :

(i) $\int e^{ax}\cos(bx+c)dx = \frac{e^{ax}}{r}\cos(bx+c-\theta)$

(ii) $\int e^{ax}\sin(bx+c)dx = \frac{e^{ax}}{r}\sin(bx+c-\theta)$

(iii) $\int xe^{ax}\cos(bx+c)dx = \frac{xe^{ax}}{r}\cos(bx+c-\theta) - \frac{e^{ax}}{r^2}\cos(bx+c-2\theta)$

(iv) $\int xe^{ax}\sin(bx+c)dx = \frac{xe^{ax}}{r}\sin(bx+c-\theta) - \frac{e^{ax}}{r^2}\sin(bx+c-2\theta)$

where $a = r\cos\theta$, $b = r\sin\theta$, $r = \sqrt{a^2+b^2}$

Example :

(i) $\int e^x \cos\left(\sqrt{3}\,x\right)dx, a=1, b=\sqrt{3}, r=2, \theta=\frac{\pi}{3} = \frac{e^x}{2}\cos\left(\sqrt{3}x - \frac{\pi}{3}\right) + c$

(ii) $\int xe^x \cos\left(\sqrt{3}x\right)dx = \frac{xe^x}{2}\cos\left(\sqrt{3}x - \frac{\pi}{3}\right) - \frac{e^x}{4}\cos\left(\sqrt{3}x - \frac{2\pi}{3}\right) + c$

(iii) $\int e^{-x}\sin x\,dx, a=-1, b=1, r=\sqrt{2}, \theta=\frac{3\pi}{4} = \frac{e^{-x}}{\sqrt{2}}\sin\left(x - \frac{3\pi}{4}\right) + c$

(iv) $\int xe^{-x}\sin x\,dx = \frac{xe^{-x}}{\sqrt{2}}\sin\left(x - \frac{3\pi}{4}\right) - \frac{e^{-x}}{2}\sin\left(x - \frac{3\pi}{2}\right) + c$

***Illustration 21*:** Evaluate : $\int \ln x\,dx$

Solution : $I = \int \ln x\,dx = \int (\ln x.1)dx = \ln x.x - \int \frac{1}{x}.x\,dx$

$= x\ln x - x + c = x(\ln x - 1) + c$

***Illustration 22*:** Evaluate : $\int \cos^{-1} x\,dx$

Solution : Let 1 is the second function and $\cos^{-1} x$ is the 1st function.

Then $\int \cos^{-1} x.1\,dx = \cos^{-1} x.x - \int \frac{-x}{\sqrt{1-x^2}}dx$

$= x\cos^{-1} x + \int \frac{x}{\sqrt{1-x^2}}dx$

Let $1 - x^2 = t^2 \Rightarrow x\,dx = -t\,dt$

$$\Rightarrow \int \frac{x}{\sqrt{1-x^2}}dx = \int \frac{-t}{t}dt = \int -1\,dt = -t + c$$

$$= -\sqrt{1-x^2} + c \Rightarrow \int \cos^{-1} x\,dx = x\cos^{-1} x - \sqrt{(1-x^2)} + c$$

***Illustration 23*:** Evaluate : $\int \frac{x^2}{(x\sin x + \cos x)^2}$

Solution : $I = \int \frac{x^2}{(x\sin x + \cos x)^2} = \int x.\sec x \left(\frac{x\cos x}{(x\sin x + \cos x)^2}\right)dx$

$$= x\sec x\left(\frac{-1}{(x\sin x + \cos x)}\right) - \int \frac{(\sec x + x\sec x.\tan x)(-1)}{x\sin x + \cos x}dx$$

$$= \frac{-x\sec x}{x\sin x + \cos x} + \tan x + c$$

***Illustration 24*:** Evaluate : $\int \left(\frac{x-1}{x^2+1}\right)^2 e^x dx$

Solution : $\left(\frac{x-1}{x^2+1}\right)^2 = \frac{x^2-2x+1}{(x^2+1)^2} = \frac{1}{(x^2+1)} - \frac{2x}{(x^2+1)^2}$

Here derivative of $\frac{1}{x^2+1}$ is $\frac{-2x}{(x^2+1)^2}$

So $\int e^x \left(\frac{x-1}{x^2+1}\right)^2 dx = \int e^x \left(\frac{1}{(x^2+1)} + \frac{(-2x)}{(x^2+1)^2}\right)dx$

$$= \frac{1}{(x^2+1)}e^x - \int \frac{-2x}{(x^2+1)^2}e^x dx + \int \frac{-2x}{(x^2+1)^2}e^x\,dx = \frac{e^x}{(x^2+1)} + c$$

7. INTEGRATION USING PARTIAL FRACTIONS

When integrand is a rational function i.e., of the form $\frac{f(x)}{g(x)}$, where $f(x)$ and $g(x)$ are the polynomials function of x, we use the method of partial fraction. For example we can rewrite

$\frac{1}{(3x-1)(3x+2)}$ as $\frac{1}{3(3x-1)} - \frac{1}{3(3x+2)}$

If the degree of $f(x)$ is less than degree of $g(x)$ and $g(x) = (x-a_1)^{\alpha_1}$ ..... $(x^2 + b_1x + c_1)^{\beta_1}$ ..... then we can put

$$\frac{f(x)}{g(x)} = \frac{A_1}{(x-a_1)} + \frac{A_2}{(x-a_1)^2} + ... + \frac{A_{\alpha_1}}{(x-a_1)^{\alpha_1}} .....$$

$$+ \frac{B_1x + C_1}{(x^2+b_1x+c_1)} + \frac{B_2x + C_2}{(x^2+b_1x+c_1)^2} + ... + \frac{B_{\beta_1}x + C_{\beta_1}}{(x^2+b_1x+c_1)^{\beta_1}} .....$$

Here A_1, A_2 ....., A_{α_1} ....., B_1, B_2 ....., B_{β_1} .....C_1, C_2 .....C_{β_1} ..... are the real constants and these can be calculated by reducing both sides of the above equation as identity in polynomial form and then by comparing the coefficients of like powers. The constants can also be obtained by putting some suitable numerical values of x in both sides of the identity.

If degree of $f(x)$ is more than or equal to degree of $g(x)$, then divide $f(x)$ by $g(x)$ so that the remainder has degree less than of $g(x)$.

Form of the rational function	**Form of the partial fraction**
(i) $\frac{px+q}{(x-a)(x-b)}$	$\frac{A}{x-a}+\frac{B}{x-b}$
(ii) $\frac{px^2+qx+r}{(x-a)(x-b)^2}$	$\frac{A}{x-a}+\frac{B}{x-b}+\frac{C}{(x-b)^2}$
(iii) $\frac{px^2+qx+r}{(x-a)(bx^2+cx+d)}$	$\frac{A}{x-a}+\frac{Bx+C}{bx^2+cx+d}$
(iv) $\frac{px^4+qx^3+rx^2+sx+t}{(x-a)(bx^2+cx+d)^2}$	$\frac{A}{x-a}+\frac{Bx+C}{bx^2+cx+d}+\frac{Dx+E}{(bx^2+cx+d)^2}$

Note :-

In partial fraction decomposition of a rational function, numerator polynomila must be of degre less than the degree of the denominator polynomial. If the degree of the numerator polynomial is greater than or equal to the degree of the denominator, then we have to carry out the division befroe finding partial fractions. We explain the method of partial fraction decomposition through some examples.

***Illustration 25*:** Evaluate : $\int \frac{dx}{(x-1)(x-2)(x-3)}$

Solution : Put $\frac{1}{(x-1)(x-2)(x-3)}=\frac{A}{(x-1)}+\frac{B}{(x-2)}+\frac{C}{(x-3)}$

$\Rightarrow 1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$

Put $x = 1$, we get, $A=\frac{1}{2}$

$x = 2$, we get, $B = -1$

$x = 3$, we get, $C=\frac{1}{2}$

So Integral $=\frac{1}{2}\int\frac{dx}{x-1}-\int\frac{dx}{x-2}+\frac{1}{2}\int\frac{dx}{x-3}$

$=\frac{1}{2}\ln|x-1|-\ln|x-2|+\frac{1}{2}\ln|x-3|+c=\ln\left(\frac{\sqrt{x^2-4x+3}}{|x-2|}\right)+c$

Student Task

Q. In the illustration 20, can we put $\frac{1}{(x-1)(x-2)(x-3)}=\frac{A}{(x-1)}+\frac{Bx+C}{(x^2-5x+6)}$?

***Illustration 26*:** Evaluate : $\int \frac{dx}{(x+2)(x^2+1)}$

Solution : Let $\frac{1}{(x+2)(x^2+1)} = \frac{A}{x+2} + \frac{Bx+c}{(x^2+1)}$

$\Rightarrow 1 = A(x^2 + 1) + (Bx + C)(x + 2)$

Put $x = -2$, we get $A = \frac{1}{5}$

Now compare the coefficients of x^2 and constant term we get

$0 = A + B$ and $1 = A + 2C$

$\Rightarrow B = -\frac{1}{5}, C = \frac{2}{5}$

So $I = \frac{1}{5}\int \frac{dx}{x+2} - \frac{1}{5}\int \frac{x}{x^2+1}dx + \frac{2}{5}\int \frac{dx}{x^2+1}$

$= \frac{1}{5}\ln|x+2| - \frac{1}{10}\ln(x^2+1) + \frac{2}{5}\tan^{-1}x + C$

***Illustration 27*:** Evaluate : $\int \frac{x^4 dx}{(x-1)(x+1)^2}$

Solution : Here degree of numerator is more than the degree of denominator so first we have to divide it to reduce it to proper fraction.

$\frac{x^4}{(x-1)(x+1)^2} = (x-1) + \frac{2x^2-1}{(x-1)(x+1)^2}$

Put $\frac{2x^2-1}{(x-1)(x+1)^2} = \frac{A}{(x-1)} + \frac{B}{(x+1)} + \frac{C}{(x+1)^2}$

$\Rightarrow x = 1$, we get $A = \frac{1}{2}$, Put $x = -1$, we get $C = -\frac{1}{2}$

comparing the coefficient of x^2, we get $2 = A + B$

$\Rightarrow \quad B = \frac{3}{2}$

So $I = \int (x-1)dx + \frac{1}{2}\int \frac{dx}{(x-1)} + \frac{3}{2}\int \frac{dx}{(x+1)} - \frac{1}{2}\int \frac{dx}{(x+2)^2}$

$= \frac{x^2}{2} - x + \frac{1}{2}\ln|x-1| + \frac{3}{2}\ln|x+1| + \frac{1}{2(x+2)} + C$

EXERCISE -2 (INTEGRATE)

Q.1 $\frac{2\tan x \sec^2 x}{\tan^2 x + 3\tan x + 2}$

Q.2 $\frac{x^3}{(x-1)(x-2)}$

Q.3 $\frac{x}{(x-2)(x+5)}$

Q.4 $\frac{(\log_{ex} e)(\log_{e^2 x} e)}{x}$

Q.5 $\frac{2}{(1-x)(1+x^2)}$

Q.6 $\tan^{-1} x$

Q.7 $x\ln\sqrt{x}$

Q.8 $x\, e^x$

Q.9 $e^x (\cos x - \sin x)$

Q.10 $\frac{2x^2}{(x^2+1)^2}$

ANSWERS

1. $\ln\left(\frac{(\tan x+2)^4}{(\tan x+1)^2}\right)+c$
2. $\frac{x^2}{2}+3x+\ln(x-2)^8-\ln|x-1|+c$
3. $\frac{2}{7}\ln|x-2|+\frac{5}{7}\ln|x+5|+c$
4. $\ln\left|\frac{\ln x+1}{\ln x+2}\right|+c$
5. $\frac{1}{2}\ln(1+x^2)-\ln|1-x|+\tan^{-1}x+c$
6. $x\tan^{-1}x-\frac{1}{2}\ln(1+x^2)+c$
7. $\frac{x^2}{8}(2\ln x-1)+c$
8. $e^x(x-1)+c$
9. $e^x\cos x+c$
10. $\frac{-x}{x^2+1}+\tan^{-1}x+c$

8. ALGEBRAIC INTEGRALS

Using the technique of standard substitution and integration by parts, we can derive the following formulae.

$$\int\frac{dx}{x^2+a^2}=\frac{1}{a}\tan^{-1}\frac{x}{a}+c$$

$$\int\frac{dx}{x^2-a^2}=\frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right|+c$$

$$\int\frac{dx}{a^2-x^2}=\frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right|+c$$

$$\int\frac{dx}{\sqrt{a^2-x^2}}=\sin^{-1}\frac{x}{a}+c$$

$$\int\frac{dx}{\sqrt{x^2-a^2}}=\ln\left|x+\sqrt{x^2-a^2}\right|+c$$

$$\int\frac{dx}{\sqrt{x^2+a^2}}=\ln\left|x+\sqrt{x^2+a^2}\right|+c$$

$$\int\sqrt{x^2+a^2}\,dx=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\ln\left|x+\sqrt{x^2+a^2}\right|+c$$

$$\int\sqrt{x^2-a^2}\,dx=\frac{x}{2}\sqrt{x^2-a^2}-\frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right|+c$$

$$\int\sqrt{a^2-x^2}\,dx=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}+c$$

INTEGRAL OF THE FORM

$$\int\frac{dx}{ax^2+bx+c},\int\frac{dx}{\sqrt{ax^2+bx+c}},\int\sqrt{ax^2+bx+c}\,dx$$

Here in each case write $ax^2+bx+c=a\left(x+\frac{b}{2a}\right)^2+\frac{4ac-b^2}{2a}=t$ and use the standard formulae.

***Illustration 28*:** Evaluate : $\int \frac{dx}{\sqrt{-x^2+4x+6}}$

Solution : $-x^2 + 4x + 6 = -[x^2 - 4x - 6 + 4 - 4]$

$= -[x^2 - 4x + 4 - 10] = 10 - (x-2)^2$

$I = \int \frac{dx}{\sqrt{10-(x-2)^2}}$ put $x - 2 = t \Rightarrow dx = dt$

$I = \int \frac{dt}{\sqrt{10-t^2}} = \sin^{-1}\frac{t}{\sqrt{10}} + c = \sin^{-1}\left(\frac{x-2}{\sqrt{10}}\right) + c$

***Illustration 29*:** Evaluate : $\int \sqrt{3x^2-6x+10}\, dx$

Solution : $3x^2 - 6x + 10 = 3(x-1)^2 + 7$ put $x - 1 = t$

$\Rightarrow dx = dt$

$I = \sqrt{3}\int \sqrt{t^2 + \frac{7}{3}}\, dt = \sqrt{3}\left[\frac{t}{2}\sqrt{t^2+\frac{7}{3}} + \frac{7}{6}\ln\left|t + \sqrt{t^2+\frac{7}{3}}\right|\right] + c$

where $t = x - 1$

INTEGRALS OF THE FORM

$$\int \frac{(ax+b)dx}{\sqrt{cx^2+ex+f}}, \int \frac{(ax+b)dx}{cx^2+ex+f}, \int (ax+b)\sqrt{cx^2+ex+f}\, dx$$

Here write $ax + b = A(2cx + e) + B$

Find A and B by comparing, the coefficients of x and constant term.

***Illustration 30*:** Evaluate : $\int \frac{(3x+5)dx}{\sqrt{x^2+4x+3}}$

Solution : Write $3x + 5 = A(2x + 4) + B \Rightarrow A = \frac{3}{2}, B = -1$

So $I = \frac{3}{2}\int \frac{(2x+4)dx}{\sqrt{x^2+4x+3}} - \int \frac{dx}{\sqrt{x^2+4x+3}}$

In 1st integral put $x^2 + 4x + 3 = t \Rightarrow (2x + 4)dx = dt$

$I = \frac{3}{2}\int \frac{dt}{\sqrt{t}} - \int \frac{dx}{\sqrt{(x+2)^2-1}} = 3\sqrt{x^2+4x+3} - \ln\left|(x+2) + \sqrt{x^2+4x+3}\right| + c$

INTEGRALS OF THE FORM

$$\int \frac{(ax^2+bx+c)dx}{(ex^2+fx+g)}, \int \frac{(ax^2+bx+c)dx}{(ex^2+fx+g)}, \int (ax^2+bx+c)\sqrt{(ex^2+fx+g)}\, dx$$

Here put $ax^2 + bx + c = A(ex^2 + fx + g) + B(2ex + f) + C$

Find the values of A, B and C by comparing the coefficient of x^2, x and constant term.

***Illustration 31*:** Evaluate : $\int \frac{(x^2+4x+7)}{\sqrt{x^2+x+1}}$

Solution : Let $x^2 + 4x + 7 = A(x^2 + x + 1) + B(2x + 1) + C$

Comparing the coefficients of x^2, x and constant term, we get

$A = 1, A + 2B = 4, A + B + C = 7 \Rightarrow A = 1, B = \frac{3}{2}, C = \frac{9}{2}$

$$\text{So } I = \int \sqrt{x^2+x+1}\,dx + \frac{3}{2}\int \frac{(2x+1)dx}{\sqrt{x^2+x+1}} + \frac{9}{2}\int \frac{dx}{\sqrt{x^2+x+1}}$$

$$\text{Now } x^2+x+1 = \left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2$$

$$\text{So } I = \left(\frac{x+\frac{1}{2}}{2}\right)\sqrt{x^2+x+1} + \frac{3}{8}\ln\left(x+\frac{1}{2}+\sqrt{x^2+x+1}\right)$$
$$+3\sqrt{x^2+x+1} + \frac{9}{2}\ln\left(x+\frac{1}{2}+\sqrt{x^2+x+1}\right) + c$$

INTEGRALS OF THE FORM

$$\int \frac{dx}{(ax^2+b)\sqrt{cx+d}}, \frac{dx}{(ax+b)\sqrt{cx+d}}$$. Here, put $cx + d = t^2$

INTEGRALS OF THE FORM

$$\int \frac{dx}{(ax+b)\sqrt{ex^2+fx+g}}$$. Here put $ax+b=\frac{1}{t}$

***Illustration 32*:** Evaluate : $\int \frac{dx}{(x+1)\sqrt{x+2}}$

Solution : Put $\sqrt{x+2} = t$, i.e., $x + 2 = t^2$

So that $dx = 2t\,dt$

$$\therefore \int \frac{dx}{(x+1)\sqrt{x+2}} = \int \frac{2t}{(t^2-1)t}dt = 2\int \frac{dt}{(t^2-1)}$$

$$= 2.\frac{1}{2}\log\left|\frac{t-1}{t+1}\right| + c = \log\left|\frac{\sqrt{x+2}-1}{\sqrt{x+2}+1}\right| + c$$

***Illustration 33*:** Evaluate : $\int \frac{dx}{(x^2+1)\sqrt{x}}$

Solution : Put $\sqrt{x} = t$, i.e. $x = t^2$ so that $dx = 2t\,dt$

$$\therefore \int \frac{dx}{(x^2+1)\sqrt{x}} = \int \frac{2t}{(t^4+1)t}dt$$

$$= \int \frac{2dt}{(t^4+1)} = \int \frac{(t^2+1)-(t^2-1)}{(t^4+1)}dt$$

$$= \int \frac{(t^2+1)}{(t^4+1)}dt - \int \frac{(t^2-1)}{(t^4+1)}dt = \int \frac{\left(1+\frac{1}{t^2}\right)}{\left(t^2+\frac{1}{t^2}\right)}dt - \int \frac{\left(1-\frac{1}{t^2}\right)}{\left(t^2+\frac{1}{t^2}\right)}dt$$

$$= \int \frac{\left(1+\frac{1}{t^2}\right)}{\left(t-\frac{1}{t}\right)^2+2}dt - \int \frac{\left(1-\frac{1}{t^2}\right)}{\left(t+\frac{1}{t}\right)^2-2}dt$$

$$= \int \frac{du}{u^2+\left(\sqrt{2}\right)^2} - \int \frac{dv}{v^2-\left(\sqrt{2}\right)^2}$$

where $t-\frac{1}{t}=u$ in the 1st integral and $t+\frac{1}{t}=v$ in the 2nd integral

$$= \frac{1}{\sqrt{2}}\tan^{-1}\frac{u}{\sqrt{2}} - \frac{1}{2\sqrt{2}}\log\left|\frac{v-\sqrt{2}}{v+\sqrt{2}}\right| + c$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left\{\frac{\left(t-\frac{1}{t}\right)}{\sqrt{2}}\right\} - \frac{1}{2\sqrt{2}}\log\left|\frac{t+\frac{1}{t}-\sqrt{2}}{t+\frac{1}{t}+\sqrt{2}}\right| + c$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{t^2-1}{\sqrt{2}\,t}\right) - \frac{1}{2\sqrt{2}}\log\left|\frac{t^2-\sqrt{2}\,t+1}{t^2+\sqrt{2}\,t+1}\right| + c$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x-1}{\sqrt{2x}}\right) - \frac{1}{2\sqrt{2}}\log\left|\frac{x-\sqrt{2x}+1}{x+\sqrt{2x}+1}\right| + c$$

***Illustration 34*:** Evaluate : $\int \frac{dx}{(x+2)\sqrt{x^2+4x+8}}$

Solution : Put $x+2=\frac{1}{t} \Rightarrow dx=\frac{-dt}{t^2}$

Now $x^2 + 4x + 8 = (x + 2)^2 + 4$

So $I = \int \frac{-dt}{t\sqrt{\frac{1}{t^2}+4}} = -\int \frac{dt}{\sqrt{1+4t^2}}$

$$= -\frac{1}{2}\int \frac{dt}{\sqrt{t^2+\frac{1}{4}}} = -\frac{1}{2}\ln\left|t+\sqrt{t^2+\frac{1}{4}}\right| + c$$

$$= -\frac{1}{2}\ln\left|\frac{1}{x+2}+\sqrt{\frac{1}{(x+2)^2}+\frac{1}{4}}\right| + c$$

INTEGRALS OF THE FORM

$\int \frac{(ax+b)dx}{(cx+e)\sqrt{ex^2+fx+g}}$. Here put $(ax + b) = A(cx + e) + B$

Find the values of A and B by comparing the coefficients of x and constant term.

***Illustration 35*:** Evaluate : $\int \frac{(4x+7)}{(x+2)\sqrt{x^2+4x+8}}dx$

Solution : Let $4x + 7 = A(x + 2) + B$

$\Rightarrow A = 4, B = -1$

So $I = 4\int \frac{dx}{\sqrt{x^2+4x+8}} - \int \frac{dx}{(x+2)\sqrt{x^2+4x+8}}$

$= 4\ln\left(x+2+\sqrt{x^2+4x+8}\right) + \frac{1}{2}\ln\left|\frac{1}{x+2} + \sqrt{\frac{1}{(x+2)^2}+\frac{1}{2}}\right| + c$

INTEGRALS OF THE FORM

$$\int \frac{(ax^2+bx+c)dx}{(ex+f)\sqrt{gx^2+hx+i}}$$

Here put $ax^2 + bx + c = A(ex + f)(2gx + h) + B(ex + f) + C$

Find the value sof A, B and C by comparing the coefficients of x^2, x and constnat term.

Student Task

Q. Can we put $ax^2 + bx + c = A(gx^2 + hx + i) + B(ex + f) + C$ in above form?

***Illustration 36*:** Evaluate : $\int \frac{2x+7x+11}{(x+2)\sqrt{x^2+4x+8}}dx$

Solution : Put $2x^2 + 7x + 11 = A(x + 2)(2x + 4) + B(x + 2) + C$

Compare the coefficients of x^2, x and constnat term, we get

$A = 1, 7 = 8A + B$

$C + 2B + 8A = 11$

$\Rightarrow \quad B = -1, C = 5$

So $\int \frac{2x+4}{\sqrt{x^2+4x+8}} - \int \frac{dx}{\sqrt{x^2+4x+8}} + 5\int \frac{dx}{(x+2)\sqrt{x^2+4x+8}}$

$= 2\sqrt{x^2+4x+8} - \ln\left|(x+2)+\sqrt{x^2+4x+8}\right|$

$-\frac{5}{2}\ln\left|\frac{1}{(x+2)} + \sqrt{\frac{1}{(x+2)^2}+\frac{1}{4}}\right| + c$

INTEGRALS OF THE FORM

$$\int \frac{x\,dx}{(ax^2+b)\sqrt{(cx^2+e)}}.$$

Here put $cx^2 + e = t^2$

***Illustration 37*:** Evaluate : $\int \frac{x\,dx}{(2x^2+3)\sqrt{x^2-1}}$

Solution : Put $x^2 - 1 = t^2 \Rightarrow x\,dx = t\,dt$

So $I = \int \frac{t\,dt}{(2t^2+5)t} = \int \frac{dt}{2t^2+5} = \frac{1}{2}\int \frac{dt}{t^2+\frac{5}{2}}$

$= \frac{1}{2}.\sqrt{\frac{2}{5}} \tan^{-1}\left(t.\sqrt{\frac{2}{5}}\right) + c$

$= \frac{1}{\sqrt{10}} \tan^{-1}\left(\frac{\sqrt{2}}{\sqrt{5}}\sqrt{x^2-1}\right) + c$

INTEGRALS OF THE FORM

$$\int \frac{dx}{(ax^2+b)\sqrt{(cx^2+e)}}$$

Here 1st put $x = \frac{1}{t}$ and then take the expression inside the square root as y^2

Student Task

Q. What happens if we put $cx^2 + e = t^2$ in above form?

***Illustration 38*:** Evaluate : $\int \frac{dx}{(x^2+5)\sqrt{2x^2-3}}$

Solution : Put $x = \frac{1}{t}$

$\Rightarrow dx = -\frac{dt}{t^2}$

So $I = \int \frac{-dt}{t^2\left(\frac{1}{t^2}+5\right)\sqrt{\frac{2}{t^2}-3}}$

$= \int \frac{-t\,dt}{(1+5t^2)\sqrt{2-3t^2}}$

INTEGRALS OF THE TYPE

$\int x^m(a+bx^n)^p\,dx (p \neq 0)$. Here we have the following cases.

Case I : If p is a natural number, then expand $(a + bx^n)^p$ by binomial theorem and integrate.

Case II : If p is a negative integer and m and n are rational number, then put $x = t^k$, when k is the LCM of denominators of m and n.

Case III : $\frac{m+1}{n} + p$ is an integer,

put $\frac{a+bx^n}{x^n} = t^k$, where k is the denominator of p.

***Illustration 39*:** Evaluate : $\int x^{\frac{-2}{3}}\left(1+x^{\frac{2}{3}}\right)^{-1}$.

Solution : Here $p = -1$, is the negative integer and m and n are rational numbers.

Put $x = t^3$

$\Rightarrow dx = 3t^2\,dt$

So $I = \int t^{-2}(1+t^2)^{-1}3t^2dt = \int \frac{3dt}{1+t^2} = 3\tan^{-1}t + c = 3\tan^{-1}(x^{1/3}) + c$

***Illustration 40*:** Evaluate : $\int x^{\frac{-1}{3}}\left(1+x^{\frac{1}{3}}\right)^{1/4} dx$

Solution : Here $m = -\frac{1}{3}, n = \frac{1}{3}, p = \frac{1}{4}$

$\frac{m+1}{n} = 2$, which is an integer

So put $(1 + x^{1/3}) = t^4$

$\Rightarrow \frac{dx}{3x^{2/3}} = 4t^3dt$

$I = 12\int (t^4 - 1)t^4dt$

$= \frac{12t^9}{9} - \frac{12t^5}{5} + c = \frac{4}{15}\left[5t^9 - 9t^5\right] + c$

$= \frac{4}{15}t^5\left[55^4 - 9\right] + c$

$= -\frac{4}{15}\left(1+x^{1/3}\right)^{5/4}[4 + 9x^{1/3}] + c$

***Illustration 41*:** Evaluate : $\int x^{-11}(1+x^4)^{-1/2}dx$

Solution : Here $m = -11$

$n = 4$

$p = -\frac{1}{2}$

$\frac{m+1}{n} + p = -\frac{10}{4} - \frac{1}{2} = -3$, which is the integer

So put $\frac{1+x^4}{x^4} = t^2$

$\Rightarrow 1 + \frac{1}{x^4} = t^2 \;\Rightarrow\; \frac{-4}{x^5}dx = 2t\,dt$

So $I = \int \frac{dx}{x^{13}\left(1+\frac{1}{x^4}\right)^{1/2}} = -\frac{1}{4}\int\left(\frac{1}{x^4}\right)^2 \frac{1}{\left(1+\frac{1}{x^4}\right)^{1/2}}\left(\frac{-4dx}{x^5}\right)$

$= -\frac{1}{4}\int (t^2-1)^2.\frac{1}{t}.2t\,dt$

$$= -\frac{1}{2}\int (t^4 - 2t^2 + 1)dt = \frac{t^5}{-10} + \frac{t^3}{3} - \frac{t}{2} + c$$

where $t = \sqrt{1 + \frac{1}{x^4}}$

9. TRIGONOMETRIC INTEGRALS

INTEGRALS OF THE FORM

$$\int \left(\frac{f(\sin x, \cos x)}{g(\sin x, \cos x)} \right) dx = \int R(\sin x, \cos x) dx$$

where f and g both are polynomials in sin x and cos x.

Here we can convert them in algebraic form by putting $\tan\frac{x}{2} = t$ after writing $\sin x = \frac{2\tan\frac{x}{2}}{1 + \tan^2\frac{x}{2}}$ and $\cos x = \frac{1 - \tan^2\frac{x}{2}}{1 + \tan^2\frac{x}{2}}$

Now, put $\tan\frac{x}{2} = t$ and $\sec^2\frac{x}{2}dx = 2dt$.

The new integrand can be easily integrated.

Somtime instead of putting the above substitution we go for below procedure.

(i) If $R(-\sin x, \cos x) = -R(\sin x, \cos x)$, put $\cos x = t$

(ii) If $R(\sin x, -\cos x) = -R(\sin x, \cos x)$, put $\sin x = t$

(iii) If $R(-\sin x, -\cos x) = R(\sin x, \cos x)$ put $\tan x = t$, after dividing numerator and denominator by $\cos^2 x$.

***Illustration 42*:** Evaluate : $\int \frac{dx}{\sin x(2\cos^2 x - 1)}$

Solution : Here

$$R(\sin x, \cos x) = \frac{1}{\sin x(2\cos^2 x - 1)}$$

$$R(-\sin x, \cos x) = \frac{1}{-\sin x(2\cos^2 x - 1)} = -R(\sin x, \cos x)$$

So we put $\cos x = t$

$\Rightarrow$ $-\sin x\, dx = dt$

$$I = \int \frac{-\sin x\, dx}{(1 - \cos^2 x)(2\cos^2 x - 1)} = \int \frac{dt}{(t^2 - 1)(2t^2 - 1)}$$

$$= \int \frac{dt}{t^2 - 1} - 2\int \frac{dt}{2t^2 - 1}$$

$$= \frac{1}{2}\ln\left|\frac{t-1}{t+1}\right| - \frac{1}{\sqrt{2}}\ln\left|\frac{t - \frac{1}{\sqrt{2}}}{t + \frac{1}{\sqrt{2}}}\right| + C = \frac{1}{2}\ln\left|\frac{\cos x - 1}{\cos x + 1}\right| - \frac{1}{\sqrt{2}}\ln\left|\frac{\sqrt{2}\cos x - 1}{\sqrt{2}\cos x + 1}\right| + C$$

***Illustration 43*:** Evaluate : $\int \frac{\cos x\, dx}{\sin^2 x(\sin x + \cos x)}$

Solution : Here $R(\sin x, \cos x) = \frac{\cos x}{\sin^2 x(\sin x + \cos x)}$

$R(-\sin x, -\cos x) = R(\sin x, \cos x)$

So put $\tan x = t \Rightarrow \sec^2 x\, dx = dt$

$$I = \int \frac{\cos x \sec^2 x\, dx}{\sec^2 x \sin^2 x(\sin x + \cos x)} = \int \frac{\sec^2 x\, dx}{\tan^2 x(1 + \tan x)} = \int \frac{dt}{t^2(1+t)}$$

Let $\frac{1}{t^2(1+t)} = \frac{A}{t} + \frac{B}{t^2} + \frac{C}{(1+t)}$

or $1 = At(1 + t) + B(1 + t) + ct^2$

Put $t = 0$, we get $B = 1$

put $t = -1$, we get $C = 1$

compare the coefficients of t^2, we get $0 = A + C \Rightarrow A = -1$

So $I = \int \frac{dt}{t} + \int \frac{dt}{t^2} + \int \frac{dt}{1+t} = \ln|t| - \frac{1}{t} + \ln|1+t| + c = \ln\left|\frac{1+\tan x}{\tan x}\right| - \cot x + c$

***Illustration44*:** Evaluate : $\int \frac{dx}{(2 + \cos x - \sin x)}$

Solution : Writing $\cos x = \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)}$ and $\sin x = \frac{2\tan(x/2)}{1 + \tan^2(x/2)}$

We have $\int \frac{dx}{(2 + \cos x - \sin x)} = \int \frac{dx}{2 + \left\{\frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)}\right\} - \left\{\frac{2\tan(x/2)}{1 + \tan^2(x/2)}\right\}}$

$$= \int \frac{[1 + \tan^2(x/2)]}{[\tan^2(x/2) - 2\tan(x/2) + 3]} dx$$

$$= \int \frac{\sec^2(x/2)}{[\tan^2(x/2) - 2\tan(x/2) + 3]} dx$$

$= 2\int \frac{dt}{(t^2 - 2t + 3)}$, where $\tan\frac{x}{2} = t$

and $\sec^2\frac{x}{2} dx = 2dt$

$$= 2\int \frac{dt}{(t-1)^2 + \left(\sqrt{2}\right)^2} = 2.\frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{t-1}{\sqrt{2}}\right) + c$$

$$= \sqrt{2}\tan^{-1}\left[\frac{\tan(x/2) - 1}{\sqrt{2}}\right] + c$$

***Illustration 45*:** Evaluate : $\int \frac{dx}{(4\sin^2 x + 5\cos^2 x)}$

Solution : Dividing numerator and denominator by $\cos^2 x$, we get

$$\int \frac{dx}{(4\sin^2 x + 5\cos^2 x)} = \int \frac{\sec^2 x}{(4\tan^2 x + 5)} dx$$

$$= \int \frac{dt}{4t^2 + 5} \qquad \text{[putting } \tan x = t\text{]}$$

$$= \frac{1}{4}\int \frac{dt}{\left(t^2 + \frac{5}{4}\right)} = \frac{1}{4}.\int \frac{dt}{\left[t^2 + \left(\frac{\sqrt{5}}{2}\right)^2\right]} = \frac{1}{4}.\frac{1}{\left(\frac{\sqrt{5}}{2}\right)} \tan^{-1} \frac{t}{\left(\frac{\sqrt{5}}{2}\right)} + c$$

$$= \frac{1}{2\sqrt{5}} \tan^{-1}\left(\frac{2t}{\sqrt{5}}\right) + c = \frac{1}{2\sqrt{5}} \tan^{-1}\left(\frac{2\tan x}{\sqrt{5}}\right) + c$$

INTEGRALS OF THE FORM

$$\int \left(\frac{p\sin x + q\cos x + r}{a\sin x + b\cos x + c}\right) dx$$

Here put $p \sin x + q \cos x + r = A(a \sin x + b \cos x + c) + B(a \cos x - b \sin x) + C$

Values of A, B and C can be obtained by comparing the coefficients of $\sin x$, $\cos x$ and constant term by this technique. The given integral becomes sum of 3 integrals in which 1st two are very easy and in 3rd we can put $\tan \frac{x}{2} = t$.

***Illustration 46*:** Evaluate : $\int \frac{(5\sin x + 6)dx}{\sin x + 2\cos x + 3}$

Solution : Let $5 \sin x + 6 = A(\sin x + 2 \cos x + 3) + B(\cos x - 2 \sin x) + C$
Equating the coefficients of $\sin x$, $\cos x$ and constnat term, we get

$$\left.\begin{aligned} A - 2B &= 5 \\ 2A + B &= 0 \\ 3A + C &= 6 \end{aligned}\right\} \Rightarrow A = 1, B = -2, C = 3$$

So $I = \int dx - 2\int \frac{(\cos x - 2\sin x)dx}{\sin x + 2\cos x + 3} + 3\int \frac{dx}{\sin x + 2\cos x + 3}$

$I = x - 2 \ln |\sin x + 2 \cos x + 3| + 3I_1$ ....(i)

Now $I_1 = \int \frac{dx}{\sin x + 2\cos x + 3} = \int \frac{\sec^2 \frac{x}{2}}{2\tan\frac{x}{2} + 2\left(1 - \tan^2 \frac{x}{2}\right) + 3\left(1 + \tan^2 \frac{x}{2}\right)}$

put $\tan\frac{x}{2} = t \Rightarrow \sec^2 \frac{x}{2} dx = 2dt$

So $I_1 = \int \frac{2dt}{t^2 + 2t + 5} = \int \frac{2dt}{(t+1)^2 + 4} = \tan^{-1}\left(\frac{t+1}{2}\right) + C = \tan^{-1}\left(\frac{1 + \tan\frac{x}{2}}{2}\right) + C$

Now substitute the value of I, in eq. (i) and we get the answer.

INTEGRALS OF THE FORM

$\int \sin^p x \cos^q x\, dx$

When any one or both of p and q are odd integers, then put $\cos x = t$ or $\sin x = t$ and when p and q are rational number such that $\frac{p+q-2}{2}$ is a negative integer, then put $\tan x = t$ or $\cot x = t$.

***Illustration 47*:** Evaluate : $\int \sin^3 x \cos^4 x\, dx$

Solution : $\int \sin^3 x \cos^4 x\, dx = \int \sin^2 x \cos^4 x \sin x\, dx$

$= \int (1-\cos^2 x)\cos^4 x \sin x\, dx = -\int (1-t^2)t^4 dt$

[putting $\cos x = t$ and $\sin x\, dx = dt$]

$= \int t^6 dt - \int t^4 dt = \frac{t^7}{7} - \frac{t^5}{5} + c = \frac{1}{7}\cos^7 x - \frac{1}{5}\cos^5 x + c$

***Illustration 48*:** Evaluate : $\int \sin^{-7/5} x \cos^{-3/5} x\, dx$

Solution : Here $p = -\frac{7}{5}, q = -\frac{3}{5} \Rightarrow \frac{p+q-2}{2} = -2$

$I = \int \sin^{-7/5} x \cos^{-3/5} x\, dx = \int \frac{\cos^{-3/5} x}{\sin^{-3/5} x \sin^2 x} dx = \int (\cot x)^{-3/5} \operatorname{cosec}^2 x\, dx$

Put $\cot x = t \Rightarrow \operatorname{cosec}^2 x = -dt$

So $I = -\int t^{-3/5} dt = -\frac{5}{2} t^{2/4} + c = -\frac{5}{2}(\cot x)^{2/5} + c$

10. SUCCESSIVE REDUCTION IN INTEGRATION

Sometimes the integrand is a function of x as well as of $n (n \in N)$, then we use this method.

***Illustration 49*:** Evaluate : If $I_n = \int \tan^n x\, dx$, then prove that $I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$. Hence find $\int \tan^7 x\, dx$.

Solution : $I_n = \int \tan^n x\, dx = \int \tan^{n-2} x . \tan^2 x\, dx$

$= \int \tan^{n-2} x . \sec^2 x\, dx - \int \tan^{n-2} x\, dx = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$

$= \int \tan^{n-2} x . \sec^2 x\, dx - \int \tan^{n-2} x\, dx = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$

Now $I_7 = \frac{\tan^6 x}{6} - I_5$

$= \frac{\tan^6 x}{6} - \frac{\tan^4 x}{4} + \frac{\tan^2 x}{2} - I_1$

$I_1 = \int \tan x\, dx = \ln|\sec x| + c$

so $\int \tan^7 x\, dx = \frac{\tan^6 x}{6} - \frac{\tan^4 x}{4} + \frac{\tan^2 x}{2} - \ln|\sec x| + c$

EXERCISE -3

Integrate the following :

Q.1 $\dfrac{1}{\sqrt{x^2+4x+5}}$

Q.2 $\dfrac{2x-3}{\sqrt{x^2+x+1}}$

Q.3 $\dfrac{1}{x\sqrt{x^2+x+1}}$

Q.4 $\dfrac{1}{1+2\cos x}$

Q.5 $x^{1/3}\,(1+x^{1/3})^{-2}$

Q.6 $\dfrac{1}{x^5(1+x^4)}$

Q.7 $\dfrac{1}{\sin^2 x+\sin 2x}$

Q.8 $\dfrac{\sin x}{\sin x+\cos x}$

Q.9 $\dfrac{\cos^2 x}{\sin x}$

Q.10 $\dfrac{1}{\cos x\sqrt{\sin^2 x-\cos^2 x}}$

Answer Key

1. $\ln\left|(x+2)+\sqrt{x^2+4x+5}\right|+c$

2. $2\sqrt{x^2+x+1}-4\ln\left|\left(x+\dfrac{1}{2}\right)+\sqrt{x^2+x+1}\right|+c$

3. $-\ln\left|\dfrac{1}{x}+\dfrac{1}{2}+\sqrt{\dfrac{1}{x^2}+\dfrac{1}{x}+1}\right|+c$

4. $\dfrac{1}{\sqrt{3}}\ln\left|\dfrac{\sqrt{3}+\tan\frac{x}{2}}{\sqrt{3}-\tan\frac{x}{2}}\right|+c$

5. $3\left[\dfrac{y^2}{2}-3y+3\ln|y|+\dfrac{1}{y}\right]+c$, where $y = 1 + x^{1/3}$

6. $\dfrac{1}{4}\left[\ln\left|1+\dfrac{1}{x^4}\right|-\left(1+\dfrac{1}{x^4}\right)\right]+c$

7. $\dfrac{1}{2}\ln\left|\dfrac{\tan x}{\tan x+2}\right|+c$

8. $\dfrac{1}{2}\left(x-\ln|\sin x+\cos x|\right)+c$

9. $\cos x+\dfrac{1}{2}\ln\left|\dfrac{\cos x-1}{\cos x+1}\right|+c$

10. $\ln\left|\tan x+\sqrt{\tan^2 x-1}\right|+c$

MISCELLANEOUS EXAMPLES

Example 1 : $\displaystyle\int\frac{\sqrt{\tan x}}{\sin x\cos x}dx =$

(a) $2\sqrt{\tan x}$ (b) $2\sqrt{\cot x}$ (c) $\sqrt{\cot x}$ (d) $\sqrt{\tan x}$

Solution : **(a)** $\displaystyle\int\frac{\sqrt{\tan x}}{\sin x\cos x}dx=\int\frac{\sqrt{\tan x}}{\tan x\cos^2 x}dx=\int\frac{1}{\sqrt{\tan x}}\sec^2 x\,dx\ =2\sqrt{\tan x}$

Example 2 : $\displaystyle\int\frac{dx}{\sin x+\sqrt{3}\cos x} =$

(a) $\dfrac{1}{2}\log\tan\left(\dfrac{x}{2}+\dfrac{\pi}{6}\right)$

(b) $\dfrac{1}{2}\log\left\{\operatorname{cosec}\left(x+\dfrac{\pi}{3}\right)-\cot\left(x+\dfrac{\pi}{3}\right)\right\}$

(c) $\dfrac{1}{2}\log\left\{\sec\left(x-\dfrac{\pi}{6}\right)+\tan\left(x-\dfrac{\pi}{6}\right)\right\}$

(d) $-\dfrac{1}{2}\log\left\{\operatorname{cosec}\left(x+\dfrac{\pi}{3}\right)+\cot\left(x+\dfrac{\pi}{3}\right)\right\}$

Solution : **(a, b, c, d)** $\int \frac{dx}{\sin x+\sqrt{3}\cos x}=\frac{1}{2}\int \frac{dx}{\frac{1}{2}\sin x+\frac{\sqrt{3}}{2}\cos x}=\frac{1}{2}\int \frac{dx}{\sin\left(x+\frac{\pi}{3}\right)}$

$$=\frac{1}{2}\int \frac{dx}{2\sin\left(\frac{x}{2}+\frac{\pi}{6}\right)\cos\left(\frac{x}{2}+\frac{\pi}{6}\right)}=\frac{1}{4}\int \frac{\sec^2\left(\frac{x}{2}+\frac{\pi}{6}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}dx$$

$$=\frac{1}{2}\int \frac{dt}{t} \text{ where } t=\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)=\frac{1}{2}\log t$$

But $\log\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)=\log\frac{\sin\left(\frac{x}{2}+\frac{\pi}{6}\right)}{\cos\left(\frac{x}{2}+\frac{\pi}{6}\right)}=\log\frac{2\sin^2\left(\frac{x}{2}+\frac{\pi}{6}\right)}{2\sin\left(\frac{x}{2}+\frac{\pi}{6}\right)\cos\left(\frac{x}{2}+\frac{\pi}{6}\right)}$

$$=\log\frac{1-\cos\left(x+\frac{\pi}{3}\right)}{\sin\left(x+\frac{\pi}{3}\right)}=\log\left\{\operatorname{cosec}\left(x+\frac{\pi}{3}\right)-\cot\left(x+\frac{\pi}{3}\right)\right\}$$

$$=\log\frac{1}{\left\{\operatorname{cosec}\left(x+\frac{\pi}{3}\right)+\cot\left(x+\frac{\pi}{3}\right)\right\}}$$

In the same way we can prove $\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)=\sec\left(x-\frac{\pi}{6}\right)+\tan\left(x-\frac{\pi}{6}\right)$

Example 3 : $\int \frac{\sin 2x\,dx}{a^2\cos^2 x+b^2\sin^2 x}=$

(a) $(b-a)\log(a^2\cos^2 x+b^2\sin^2 x)$ (b) $\frac{1}{b^2-a^2}\log(a^2\cos^2 x+b^2\sin^2 x)$

(c) $\frac{1}{b^2-a^2}\log(a^2\cos^2 x+b^2\sin^2 x)$ (d) $\frac{1}{a^2+b^2}\log(a^2\cos^2 x+b^2\sin^2 x)$

Solution : **(c)** $I=\int \frac{\sin 2x\,dx}{a^2\cos^2 x+b^2\sin^2 x}$, put $t=a^2\cos^2 x+b^2\sin^2 x$

$\Rightarrow$ $dt=(-a^2\sin 2x+b^2\sin 2x)dx=(b^2-a^2)\sin 2x\,dx$

$\therefore$ $I=\frac{1}{b^2-a^2}\int\frac{dt}{t}=\frac{1}{b^2-a^2}\log|t|=\frac{1}{b^2-a^2}\log(a^2\cos^2 x+b^2\sin^2 x)$

Example 4 : $\int \frac{\sqrt{x}}{\sqrt{a^3-x^3}}dx=$

(a) $\frac{2}{3}\left(\frac{x}{a}\right)^{3/2}$ (b) $\frac{2}{3}\sin^{-1}\left(\frac{x}{a}\right)^{3/2}$ (c) $\frac{2}{3}\cos^{-1}\left(\frac{x}{a}\right)^{3/2}$ (d) $\frac{2}{3}\sin^{-1}\left(\frac{a}{x}\right)^{3/2}$

Solution : **(b)** $I=\int \frac{\sqrt{x}\,dx}{\sqrt{a^3-x^3}}$, put $x^3=a^3\sin^2\theta$; $3x^2\,dx=2a^3\sin\theta\cos\theta\,d\theta$

$$=\int\frac{a^{1/2}\sin^{1/3}\theta\,2a^3\sin\theta\cos\theta}{3a^2\sin^{2/3}\theta\,a^{3/2}\cos\theta}d\theta=\frac{2}{3}\int d\theta=\frac{2\theta}{3}=\frac{2}{3}\sin^{-1}\left(\frac{x}{a}\right)^{3/2}$$

Example 5 : $\int \sqrt{1+x^2}\, d(x^2) =$

(a) $\frac{2}{3x}(1+x^2)^{3/2}$ (b) $\frac{2}{3}(1+x^2)^{3/2}$ (c) $\frac{2x}{3}(1+x^2)^{3/2}$ (d) $\frac{2x}{3}(1+x^2)$

Solution : **(b)** $\int \sqrt{1+x^2}\, d(x^2) = \int \sqrt{1+x^2}\, d(1+x^2) = \frac{2}{3}(1+x^2)^{3/2}$

Example 6 : $\int \frac{3^x\, dx}{\sqrt{1-9^x}} =$

(a) $\log_e 3(\sin^{-1}(3^x))$ (b) $\frac{1}{\log_e 3}\sin^{-1}(3^x)$ (c) $\log_e 3(\sin^{-1}(3^{x/2}))$ (d) $\frac{1}{\log_e 3}\sin^{-1}(3^{x/2})$

Solution : **(b)**

$\int \frac{3^x\, dx}{\sqrt{1-9^x}} = \frac{1}{\log 3}\int \frac{dt}{\sqrt{1-t^2}} = \frac{1}{\log 3}\sin^{-1} t = \frac{1}{\log 3}\sin^{-1}(3^x)$ where $t = 3^x$

Example 7 : $\int \frac{1}{x(x^4-1)}\, dx =$

(a) $\log\left(\frac{x^4}{x^4-1}\right)$ (b) $\frac{1}{2}\log\left(\frac{x^2-1}{x^2+1}\right)$ (c) $\frac{1}{4}\log\left(\frac{x^4-1}{x^4}\right)$ (d) $\log\frac{x(x^2-1)}{x^2+1}$

Solution : **(c)** $I = \int \frac{x\, dx}{x^2(x^4-1)} = \frac{1}{2}\int \frac{dt}{t(t^2-1)}$ where $t = x^2$

$= \frac{1}{4}\int\left[\frac{1}{t-1} - \frac{2}{t} + \frac{1}{t+1}\right]dt = \frac{1}{4}\log\left(\frac{t^2-1}{t^2}\right) = \frac{1}{4}\log\left(\frac{x^4-1}{x^4}\right)$

Example 8 : $\int \frac{e^{\log\left(1+\frac{1}{x^2}\right)}dx}{x^2+\frac{1}{x^2}} =$

(a) $\frac{1}{\sqrt{2}}\tan^{-1}\left(x+\frac{1}{x}\right)$ (b) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right)$

(c) $\sqrt{2}\tan^{-1}\left(x+\frac{1}{x}\right)$ (d) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{1}{x}-x\right)$

Solution : **(b)** $\int \frac{e^{\log\left(1+\frac{1}{x^2}\right)}}{x^2+\frac{1}{x^2}}dx = \int \frac{d\left(x-\frac{1}{x}\right)}{x^2+\frac{1}{x^2}} \int \frac{d\left(x-\frac{1}{x}\right)}{\left(x-\frac{1}{x}\right)^2+2} = \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right)$

Example 9 : $\int \frac{dx}{(2+x)\sqrt{1+x}} =$

(a) $2\tan^{-1}\sqrt{1+x}$ (b) $\frac{1}{2}\tan^{-1}\sqrt{1+x}$ (c) $\tan^{-1}\sqrt{1+x}$ (d) $\log\left\{(2+x)\sqrt{1+x}\right\}$

Solution : **(a)** $I = \int \frac{dx}{(2+x)\sqrt{1+x}}$

put $1 + x = t^2$ $\therefore$ $dx = 2t\, dt$

$\therefore \quad I = \int \frac{2t\, dt}{(1+t^2)t} = 2\tan^{-1}\sqrt{1+x}$

Example 10: If $\int \tan^4 x\,dx = a\tan^3 x + b\tan x + cx$, then

(a) $a = \frac{1}{3}$ (b) $b = -1$ (c) $a = 1$ (d) $c = 1$

Solution : **(a, b, d)** Given $\int \tan^4 x\,dx = a\tan^3 x + b\tan x + cx$

$\int \tan^4 x = \int \tan^2 x(\sec^2 x - 1)dx = \frac{1}{3}\tan^3 x - \int(\sec^2 x - 1)dx = \frac{1}{3}\tan^3 x - \tan x + x$

$\therefore \quad a = \frac{1}{3}, b = -1, c = 1$

Example 11: Evaluate : $\int \frac{dx}{x^4(x^3+2)^3}$

Solution : $I = \int \frac{dx}{x^4(x^3+2)^3} = \int \frac{dx}{x^{13}\left(1+\frac{2}{x^3}\right)^3}$

Put $1 + \frac{2}{x^3} = t \;\Rightarrow\; -\frac{6}{x^4}dx = dt$

$I = -\frac{1}{6}\int\left(\frac{1}{x^3}\right)^3 \frac{1}{\left(1+\frac{2}{x^3}\right)^3} - \left(\frac{6dx}{x^4}\right) = -\frac{1}{6}\int\left(\frac{t-1}{2}\right)^3 \frac{1}{t^3}dt$

$= -\frac{1}{48}\int \frac{(t^3 - 3t^2 + 3t - 1)}{t^3}dt = -\frac{1}{48}\int\left(1 - \frac{3}{t} + \frac{3}{t^2} + \frac{1}{t^3}\right)dt$

$= -\frac{1}{48}\int \frac{(t^3 - 3t^2 + 3t - 1)}{t^3}dt = -\frac{1}{48}\int\left(1 - \frac{3}{t} + \frac{3}{t^2} + \frac{1}{t^3}\right)dt$ where $t = 1 + \frac{2}{x^3}$

Example 12: Evaluate : $\int \frac{2dx}{(1-x)(1+x^2)}$

Solution : The integrand $\frac{2}{(1-x)(1+x^2)} = \frac{1}{1-x} + \frac{x+1}{x^2+1} = \frac{1}{1-x} + \frac{1}{2}\cdot\frac{2x}{x^2+1} + \frac{1}{x^2+1}$

Hence $\int \frac{2\,dx}{(1-x)(1+x^2)} = \int \frac{dx}{1-x} + \frac{1}{2}\int \frac{2x\,dx}{x^2+1} + \int \frac{dx}{x^2+1} = -\log(1-x) + \frac{1}{2}\log(x^2+1) + \tan^{-1}x + C$

Example 13: Evaluate : $\int \frac{x}{\sqrt{x^2+x+1}}dx$

Solution : Express the numerator x as $A \times$ (d.c. of $x^2 + x + 1$) + B i.e., $x = \frac{1}{2}(2x+1) - \frac{1}{2}$

$\therefore \quad I = \frac{1}{2}\int \frac{(2x+1)dx}{\sqrt{x^2+x+1}} - \frac{1}{2}\int \frac{dx}{\sqrt{x^2+x+1}}$

$= \frac{1}{2}\int \frac{du}{\sqrt{u}} - \frac{1}{2}\int \frac{dx}{\sqrt{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}}$ where $u = x^2 + x + 1$

$= \sqrt{x^2+x+1} - \frac{1}{2} \times \log\left\{x + \frac{1}{2} + \sqrt{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}\right\}$

$= \sqrt{x^2+x+1} - \frac{1}{2}\log\left\{\left(x+\frac{1}{2}\right) + \sqrt{x^2+x+1}\right\} + C$

Example 14: Evaluate : $\int\left(\frac{1}{\log x}-\frac{1}{(\log x)^2}\right)dx$

Solution : Let $z = \log_e x \quad \therefore \quad dz = \frac{1}{x}dx = \frac{dx}{e^z}$

$= e^z\left(\frac{1}{z}\right)$ since the integrand is the form $\int e^x(f(x)+f'(x))\,dx = \frac{x}{\log_e x} + C$

Example 15: Evaluate : $\int\left(\sqrt{\cot x}+\sqrt{\tan x}\right)dx$

Solution : $I = \int\left(\sqrt{\cot x}+\sqrt{\tan x}\right)dx = \int\frac{\cos x+\sin x}{\sqrt{\sin x\cos x}}dx$

$= \sqrt{2}\int\frac{\cos x+\sin x}{\sqrt{2\sin x\cos x}} = \sqrt{2}\int\frac{(\cos x+\sin x)}{\sqrt{1-(\sin x-\cos x)^2}}dx$

$= \sqrt{2}\int\frac{dz}{\sqrt{1-z^2}}$ where $z = \sin x - \cos x$

$= \sqrt{2}\sin^{-1} z = \sqrt{2}\sin^{-1}(\sin x-\cos x) + C$

Example 16: Find $\int\frac{x^2-3}{x^3-2x^2-x+2}dx$

Solution : Note that $(x^3 - 2x^2 - x + 2) = (x + 1)(x - 1)(x - 2)$

$\therefore \quad \frac{x^2-3}{(x+1)(x-1)(x-2)} = \frac{A}{x+1}+\frac{B}{x-1}+\frac{C}{x-2}$

where $A = \frac{(-1)^2-3}{((-1)-(1))(-1-2)} = \frac{-2}{(-2)(-3)} = -\frac{1}{3}$, $B = \frac{1^2-3}{(1+1)(1-2)} = \frac{-2}{2(-1)} = 1$

$C = \frac{2^2-3}{(2+1)(2-1)} = \frac{1}{3}$

(Note : The constants are obtained respectively by putting
$x = -1$, in all the terms except $x + 1$
$x = 1$, in all the terms except $x - 1$
$x = 2$, in all the terms except $x - 2$)

$\int\frac{(x^2-3)dx}{(x+1)(x-1)(x-2)} = \int\frac{A}{x+1}dx+\int\frac{B}{x-1}dx+\int\frac{C}{x-2}dx$

$= -\frac{1}{3}\ln(x+1)+\ln(x-1)+\frac{1}{3}\ln(x-2)+C = \frac{1}{3}\ln\left\{\frac{A(x-2)(x-1)^3}{(x+1)}\right\}$ where $C = \frac{1}{3}\ln A$

Example 17: Evaluate : $\int\frac{dx}{\cos^6 x+\sin^6 x}$

Solution : Let $I = \int\frac{dx}{(\cos^2 x+\sin^2 x)(\cos^4 x-\cos^2 x+\sin^4 x)} = \int\frac{dx}{1-3\sin^2 x\cos^2 x}$

$= \int\frac{\sec^4 x\,dx}{\sec^4 x-3\tan^2 x}$ Taking $t = \tan x$

$= \int\frac{(1+t^2)dt}{(1+t^2)^2-3t^2} = \int\frac{(1+t^2)dt}{t^4-t^2+1} = \int\frac{\left(1+\frac{1}{t^2}\right)dt}{t^2+\frac{1}{t^2}-1}$

$= \int\frac{du}{u^2+1}$, where $u = t - \frac{1}{t} = \tan^{-1}(u)$

$$= \tan^{-1}\left(\tan x - \frac{1}{\tan x}\right) + C$$

$$= \tan^{-1}(\tan x - \cot x) + C$$

Example 18: Evaluate : $\int \frac{2+3\cos x}{(3+2\cos x)^2}\,dx$

Solution : Here integrand is of the form $\int \frac{f(x)}{(g(x))^2}\,dx$

Let $I = \int \frac{(2+3\cos x)}{(3+2\cos x)^2}\,dx = \int (2\operatorname{cosec} x + 3\cot x).\frac{\sin x}{(3+2\cos x)^2}\,dx$

Integrating by parts

$$I = (2\operatorname{cosec} x + 3\cot x)\left(\frac{1}{2(3+2\cos x)}\right) - \frac{1}{2}\int\left(\frac{-2\operatorname{cosec} x\cot x - 3\operatorname{cosec}^2 x}{(3+2\cos x)}\right)dx$$

$$= \frac{2\operatorname{cosec} x + 3\cot x}{2(3+2\cos x)} + \frac{1}{2}\int \frac{\operatorname{cosec}^2 x(2\cos x+3)}{(3+2\cos x)}\,dx$$

$$= \frac{2\operatorname{cosec} x + 3\cot x}{2(3+2\cos x)} - \frac{1}{2}\cot x + C$$

Example 19: Evaluate : $\int \frac{dx}{2\sin x + 3\sec x}$

Solution : $I = \int \frac{dx}{2\sin x + 3\sec x} = \frac{1}{2}\int \frac{(\cos x + \sin x)dx}{3+\sin 2x} + \frac{1}{2}\int \frac{(\cos x - \sin x)dx}{3+\sin 2x}$

$$= \frac{1}{2}\int \frac{(\cos x + \sin x)dx}{3+(1-(\sin x - \cos x)^2)} + \frac{1}{2}\int \frac{(\cos x - \sin x)dx}{3+(\sin x + \cos x)^2 - 1}$$

in 1st integral put $\sin x - \cos x = t$

in 2nd integral put $\cos x + \sin x = y$

So $I = \frac{1}{2}\int \frac{dt}{4-t^2} + \frac{1}{2}\int \frac{dy}{y^2+2} = \frac{1}{8}\ln\left|\frac{\sin x - \cos x + 2}{\sin x - \cos x - 2}\right| + \frac{1}{2\sqrt{2}}\tan^{-1}\left(\frac{\sin x + \cos x}{\sqrt{2}}\right) + C$

Example 20: Find the values of a and b such that $\int \frac{dx}{1+\sin x} = \tan\left(\frac{x}{2} + a\right) + b$.

Solution : $\int \frac{dx}{1+\sin x} = \int \frac{dx}{1+\cos\left(\frac{\pi}{2} - x\right)} = \int \frac{dx}{2\cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right)}$

$$= \int \frac{1}{2}\sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right)dx = -\tan\left(\frac{\pi}{4} - \frac{x}{2}\right) + c \quad = \tan\left(\frac{x}{2} - \frac{\pi}{4}\right) + c$$

Hence by comparison we get $a = -\frac{\pi}{4}$ and $b = c$ an arbitrary constant

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

1. Integration by Trigonometric Identities

Q.1 $\int \frac{\cos 2x}{(\sin x+\cos x)^2}dx =$

(a) $\frac{-1}{\sin x+\cos x}+c$

(b) $\log|\sin x + \cos x| + c$

(c) $\log(\sin x - \cos x) + c$

(d) $\log(\sin x + \cos x)^2 + c$

Q.2 If $f(x)=\frac{x^2}{1+x^2}$ and $g(x) = \sin x$, then $\int (fog)(x)\cos(x)dx =$

(a) $\sin x - \tan^{-1}(\sin x) + c$

(b) $\cos x - \tan^{-1}(\sin x) + c$

(c) $\cos x + \tan^{-1}(\sin x) + c$

(d) $\sin x - \tan^{-1}(\cos x) + c$

Q.3 If $\int(\sin 2x+\cos 2x)dx=\frac{1}{\sqrt{2}}\sin(2x-c)+a$, then

(a) $c = \pi/2$, a = arbitrary constant

(b) $c = \pi/6$, a = arbitrary constant

(c) $c = \pi/4$, a = arbitrary constant

(d) None of these

Q.4 $\int \frac{\sin x}{\sin x-\cos x}dx$ is equal to

(a) $\frac{1}{2}x-\frac{1}{2}\log|\sin x-\cos x|+c$

(b) $\frac{1}{2}x+\frac{1}{2}\log|\sin x-\cos x|+c$

(c) $x + \log|\sin x - \cos x| + c$

(d) None of these

Q.5 If $f(x) = \int \cot^4 x\,dx+\frac{1}{3}\cot^3 x-\cot x$ and $f\left(\frac{\pi}{2}\right)=\frac{\pi}{2}$, then $f(x) =$

(a) $\frac{\pi}{2}-\pi$ (b) $x - \pi$

(c) $\pi - x$ (d) x

Q.6 $\int \frac{dx}{1+\tan x}$ is

(a) $\log\frac{(1+\tan x)}{\tan x}+C$

(b) $\frac{1}{2}\log|\sin x + \cos x| + \frac{x}{2} + C$

(c) $\log|\sin x + \cos x| + C$

(d) None of these

Q.7 $\int \frac{\sin x-\cos x}{\sqrt{1-\sin 2x}}e^{\sin x}\cos x dx$ is equal to

(a) $e^{\sin x} + C$

(b) $e^{\sin x - \cos x} + C$

(c) $e^{\sin x + \cos x} + C$

(d) $e^{\cos x - \sin x} + C$

Q.8 If $\int \frac{1}{1+\sin x}dx=\tan\left(\frac{x}{2}+a\right)+b$, then

(a) $a = -\frac{\pi}{4}, b \in R$

(b) $a = \frac{\pi}{4}, b \in R$

(c) $a = \frac{5\pi}{4}, b \in R$

(d) None of these

Q.9 $\int(\sin 2x-\cos 2x)dx=\frac{1}{\sqrt{2}}\sin(2x-a)+b$, then

(a) $a = 5\pi/4$, $b \in R$

(b) $a = -5\pi/4$, $b \in R$

(c) $a = \pi/4$, $b \in R$

(d) None of these

Q.10 $\int \frac{\cos 2x}{\cos x}dx$ is equal

(a) $2\sin x + \log(\sec x - \tan x) + C$

(b) $2\sin x - \log(\sec x - \tan x) + C$

(c) $2\sin x + \log(\sec x + \tan x) + C$

(d) $2\sin x - \log(\sec x + \tan x) + C$

Q.11 Integral of $\sqrt{1+2\cot x(\cot x+\operatorname{cosec} x)}$ w.r.t. x is

(a) $2\,\ell n \cos\frac{x}{2}+c$

(b) $2\,\ell n \sin\frac{x}{2}+c$

(c) $\frac{1}{2}\ell n \cos\frac{x}{2}+c$

(d) $\ell n \sin x-\ell n(\operatorname{cosec} x-\cot x)+c$

2. Integration by Substitution

Q.12 $\int x^3 e^{x^2}dx=$

(a) $x^2\left(e^{x^2}-1\right)+c$

(b) $\frac{1}{2}x^2\left(e^{x^2}-1\right)+c$

(c) $\frac{1}{2}e^{x^2}(x^2-1)+c$

(d) $\frac{1}{2}\left(e^{x^2}-1\right)+c$

Q.13 If I^m stands for log log .. repeated m times, then $\int\{xI(x)I^2(x)....I^n(x)\}^{-1}dx$ is

(a) $I^{n+1}(x)+c$

(b) $I^n(x)+c$

(c) $I^{n-1}(x)+c$

(d) $\frac{1}{x}+c$

Q.14 The value of $\int x^3\log x\,dx$ is

(a) $\frac{1}{8}\{x^4\log x-4x^4+c\}$

(b) $\frac{1}{16}\{4x^4\log x-x^4+c\}$

(c) $\frac{x^4\log x}{4}+c$

(d) $\frac{1}{16}\{4x^4\log x-x^4+c\}$

Q.15 If $\int f(x)dx=F(x)$, then $\int x^3 f(x^2)dx$ is equal to

(a) $\frac{1}{2}x^2F(x^2)-\int xF(x^2)dx$

(b) $\frac{1}{2}\left(x^2F(x^2)-\int F(x^2)dx\right)$

(c) $\frac{1}{2}\left(x^2F(x)-\frac{1}{2}\int F(x^2)dx\right)$

(d) $\frac{1}{2}\left(x^2F(x)-\frac{3}{2}\int F(x^2)dx\right)$

Q.16 $\int \sin x\,d(\cos x)$ is equal to

(a) $\frac{1}{4}\sin 2x+\frac{x}{2}+c$

(b) $\frac{1}{4}\sin 2x-\frac{x}{2}+c$

(c) $2\sin 2x+c$

(d) $\sin x+\cos x+c$

Q.17 The value of $\int e^{\tan^{-1}x}\frac{(1+x+x^2)}{1+x^2}dx$ is

(a) $xe^{\tan^{-1}x}+C$

(b) $\tan^{-1}x+C$

(c) $e^{\tan^{-1}x}+2x+C$

(d) None of these

Q.18 $\int e^{3\log x}(x^4+1)^{-1}dx$ is equal to

(a) $\log(x^4+1)+C$

(b) $\frac{1}{4}\log(x^4+1)+C$

(c) $-\log(x^4+1)+C$

(d) None f these

Q.19 $\int\frac{a^{\sqrt{x}}}{\sqrt{x}}dx$ is equal to

(a) $\frac{a^{\sqrt{x}}}{\log a}+C$

(b) $\frac{2a^{\sqrt{x}}}{\log a}+C$

(c) $2a^{\sqrt{x}}.\log a+C$

(d) None of these

Q.20 $\int e^{x\log a}.e^x dx$ is equal to

(a) $(ae)^x+C$

(b) $\frac{(ae)^x}{\log(ae)}+C$

(c) $\frac{e^x}{1+\log a}+C$

(d) None of these

Q.21 $\int\frac{\sin^8 x-\cos^8 x}{1-2\sin^2 x\cos^2 x}dx=$

(a) $\frac{1}{2}\sin 2x+C$

(b) $-\frac{1}{2}\sin 2x+C$

(c) $-\frac{1}{2}\sin x + C$

(d) $-\sin^2 x + C$

Q.22 If $\int \frac{2^x}{\sqrt{1-4^x}}\,dx = K\sin^{-1}(2^x) + C$, then K is equal

(a) log 2 (b) (1/2) log 2
(c) 1/2 (d) 1/log 2

Q.23 If $\int f(x)\sin x\cos x\,dx = \frac{1}{2(b^2-a^2)}\log(f(x)) + C$, then $f(x)$ is equal to

(a) $\frac{1}{a^2\sin^2 x + b^2\cos^2 x}$

(b) $\frac{1}{a^2\sin^2 x - b^2\cos^2 x}$

(c) $\frac{1}{a^2\cos^2 x + b^2\sin^2 x}$

(d) $\frac{1}{a^2\cos^2 x - b^2\sin^2 x}$

Q.24 Integral of $f(x) = \sqrt{(1+x^2)}$ with respect to x^2 is

(a) $\frac{2}{3}\frac{(1+x^2)^{3/2}}{x} + k$

(b) $\frac{2}{3}\left(1+x^2\right)^{3/2} + k$

(c) $\frac{2}{3}x\left(1+x^2\right)^{3/2} + k$

(d) None of these

Q.25 $\int \frac{d(x^2+1)}{\sqrt{x^2+2}}$ is equal to

(a) $2\sqrt{x^2+2} + k$ (b) $\sqrt{x^2+2} + k$
(c) $\frac{1}{(x^2+2)^{3/2}} + k$ (d) None of these

Q.26 $\int 5^{5^{5^x}}.5^{5^x}.5^x\,dx$ is equal to

(a) $\frac{5^{5^x}}{(\log 5)^3} + C$ (b) $5^{5^{5^x}}(\log 5)^3 + C$
(c) $\frac{5^{5^{5^x}}}{(\log 5)^3} + C$ (d) None of these

Q.27 $\int \cos\left\{2\tan^{-1}\sqrt{\frac{1-x}{1+x}}\right\}dx$ is equal to

(a) $\frac{1}{8}(x^2-1) + k$ (b) $\frac{1}{2}x^2 + k$
(c) $\frac{1}{2}x + k$ (d) None of these

Q.28 $\int \frac{\{f(x).\phi'(x) - f'(x).\phi(x)\}}{f(x).\phi(x)}\{\log\phi(x) - \log f(x)\}dx$ is equal to

(a) $\log\frac{\phi(x)}{f(x)} + k$

(b) $\frac{1}{2}\left\{\log\frac{\phi(x)}{f(x)}\right\}^2 + k$

(c) $\frac{\phi(x)}{f(x)}\log\frac{\phi(x)}{f(x)} + k$

(d) None of these

Q.29 $\int \sin 2x.\log\cos x\,dx$ is equal to

(a) $\cos^2 x\left(\frac{1}{2} + \log\cos x\right) + k$

(b) $\cos^2 x$. log cos x + k

(c) $\cos^2 x\left(\frac{1}{2} - \log\cos x\right) + k$

(d) None of these

Q.30 If $\ell^r(x)$ means log log log .... x, the log being repeated r times, then $\int\left[x\ell(x)\ell^2(x)\ell^3(x).....\ell^r(x)\right]^{-1}dx$ is equal to

(a) $\ell^{r+1}(x) + C$ (b) $\frac{\ell^{r+1}(x)}{r+1} + C$
(c) $\ell^r(x) + C$ (d) None of these

Q.31 $\int \frac{\cot^{-1}(e^x)}{e^x}\,dx$ is equal to

(a) $\frac{1}{2}\ell n\left(e^{2x}+1\right) - \frac{\cot^{-1}(e^x)}{e^x} + x + c$

(b) $\frac{1}{2}\ell n\left(e^{2x}+1\right) + \frac{\cot^{-1}(e^x)}{e^x} + x + c$

(c) $\frac{1}{2}\ell n\left(e^{2x}+1\right) - \frac{\cot^{-1}(e^x)}{e^x} - x + c$

(d) $\frac{1}{2}\ell n\left(e^{2x}+1\right) + \frac{\cot^{-1}(e^x)}{e^x} - x + c$

Q.32 $\int \frac{\cos^3 x + \cos^5 x}{\sin^2 x + \sin^4 x}dx$

(a) $\sin x - 6\tan^{-1}(\sin x) + c$
(b) $\sin x - 2\sin^{-1} x + c$
(c) $\sin x - 2(\sin x)^{-1} - 6\tan^{-1}(\sin x) + c$
(d) $\sin x - 2(\sin x)^{-1} + 5\tan^{-1}(\sin x) + c$

Q.33 The evaluation of $\int \frac{px^{p+2q-1}+qx^{q-1}}{x^{2p+2q}+2x^{p+q}+1}dx$ is

(a) $-\frac{x^p}{x^{p+q}+1}+C$

(b) $\frac{x^q}{x^{p+q}+1}+C$

(c) $-\frac{x^q}{x^{p+q}+1}+C$

(d) $\frac{x^p}{x^{p+q}+1}+C$

Q.34 If $I_n = \int (\sin x)^n dx$, $n \in N$. Then $5I_4 - 6I_6$ is equal to

(a) $\sin x \cdot (\cos x)^5 + C$

(b) $\sin 2x \cdot \cos 2x + C$

(c) $\frac{\sin 2x}{8}\left[\cos^2 2x + 1 - 2\cos 2x\right] + C$

(d) $\frac{\sin 2x}{8}\left[\cos^2 2x + 1 + 2\cos 2x\right] + C$

3. Integration by Part

Q.35 $\int \frac{d(x^2+1)}{\sqrt{x^2+2}} =$

(a) $2\sqrt{x^2+2}+c$

(b) $\sqrt{x^2+2}+c$

(c) $\frac{1}{(x^2+2)^{3/2}}+c$

(d) None of these

Q.36 $\int e^{e^{e^x}} . e^{e^x} . e^x dx =$

(a) $e^{e^{e^x}}+c$

(b) $\left(e^{e^{e^x}}\right)^2+c$

(c) $\frac{1}{2}e^{e^x}+c$

(d) None of these

Q.37 $\int \frac{a^x(x\log_a x - 1)}{x(\log_e x)^2}dx$ equals

(a) $\frac{a^x \log_e x}{x} + c$

(b) $\frac{a^x}{\log_e x}+c$

(c) $\frac{a^x}{\log_a x}+c$

(d) None of these

Q.38 $\int \frac{(1+x^2)\tan^{-1}x - x\log x}{x(1+x^2)(\tan^{-1}x)^2}dx$ equals

(a) $\frac{\log x}{x\tan^{-1}x}+c$

(b) $\frac{x\log x}{\tan^{-1}x}+c$

(c) $\frac{\log x}{\tan^{-1}x}+c$

(d) None of these

Q.39 If $\int\left[\left(\frac{\sec^2 x}{\tan^{-1}x}\right)-\left(\frac{\tan x}{(\tan^{-1}x)^2(1+x^2)}\right)\right]dx = \frac{f(x)}{\tan^{-1}x}+c$ then $f(x)$ equals

(a) $\sin x$ (b) $\sec x$

(c) $\tan x$ (d) None of these

Q.40 $\int \frac{x+\sin x}{1+\cos x}dx$ is equal to

(a) $x\tan\frac{x}{2} + C$

(b) $\cot\frac{x}{2} + C$

(c) $\log(1 + \cos x) + C$

(d) $\log(x + \sin x) + C$

Q.41 $\int \frac{xe^x}{(1+x)^2}dx$ is equal to

(a) $\frac{e^x}{x+1}+C$

(b) $e^x(x + 1) + C$

(c) $-\frac{e^x}{(x+1)^2}+C$

(d) $\frac{e^x}{(x+1)^2}+C$

Q.42 If $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x}dx = \ell n(f(x)) + g(x) + C$ where C is the constant of integration and $f(x)$ is positive, then $f(x) + g(x)$ has the value equal to

(a) $e^x + \sin x + 2x$ (b) $e^x + \sin x$

(c) $e^x - \sin x$ (d) $e^x + \sin x + x$

Q.43 $\int\left(\sin(101x).\sin^{99}x\right)dx$ equals

(a) $\frac{\sin(100x)(\sin x)^{100}}{100}+C$

(b) $\frac{\cos(100x)(\sin x)^{100}}{100}+C$

(c) $\frac{\cos(100x)(\cos x)^{100}}{100}+C$

(d) $\frac{\sin(100x)(\cos x)^{100}}{100}+C$

Q.44 $\int\frac{e^{\tan^{-1}x}}{(1+x^2)}\left[\left(\sec^{-1}\sqrt{1+x^2}\right)^2+\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right]dx$ $(x>0)$

(a) $e^{\tan^{-1}x}\cdot\tan^{-1}x+C$]

(b) $\frac{e^{\tan^{-1}x}.\left(\tan^{-1}x\right)^2}{2}+C$

(c) $e^{\tan^{-1}x}.\left(\sec^{-1}\left(\sqrt{1+x^2}\right)\right)^2+C$

(d) $e^{\tan^{-1}x}.\left(\operatorname{cosec}^{-1}\left(\sqrt{1+x^2}\right)\right)^2+C$

4. Integration, Irrational Function, Partial Fraction

Q.45 $\int\frac{x^2+1}{x^4+x^2+1}dx=$

(a) $\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{3}x}\right)+c$

(b) $\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{x^2+1}{\sqrt{3}x}\right)+c$

(c) $\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{x^2-1}{3x}\right)+c$

(d) None of these

Q.46 $\int\frac{dx}{x\sqrt{1-x^3}}=$

(a) $\frac{1}{3}\log\left|\frac{\sqrt{1-x^2}+1}{\sqrt{1-x^2}-1}\right|+c$]

(b) $\frac{1}{3}\log\left|\frac{\sqrt{1-x^3}-1}{\sqrt{1-x^3}+1}\right|+c$

(c) $\frac{2}{3}\log\left|\frac{1}{\sqrt{1-x^3}}\right|+c$

(d) $\frac{1}{3}\log\left|1-x^3\right|+c$

Q.47 $\int\frac{1-x^7}{x(1+x^7)}dx$ equals

(a) $\ln x+\frac{2}{7}\ln(1+x^7)+C$

(b) $\ln x-\frac{2}{7}\ln(1-x^7)+C$

(c) $\ln x-\frac{2}{7}\ln(1+x^7)+C$

(d) $\ln x+\frac{2}{7}\ln(1-x^7)+C$

Q.48 If $xf(x) = 3(f(x))^2 + 2$, then $\int\frac{2x^2-12xf(x)+f(x)}{(6f(x)-x)(x^2-f(x))^2}dx$ equals

(a) $\frac{1}{x^2-f(x)}+c$

(b) $\frac{1}{x^2+f(x)}+c$

(c) $\frac{1}{x-f(x)}$

(d) None of these

Q.49 One of the function represented by the integral $\int\frac{dx}{\sqrt{x}+\sqrt[3]{x^2}}$ is

(a) $6\left[\sqrt[3]{x^2}+\sqrt[3]{x}+\log\left|1+\sqrt[3]{x}\right|\right]$

(b) $3\sqrt[3]{x}-6\sqrt[6]{x}+6\log\left|1+\sqrt[6]{x}\right|$

(c) $3\sqrt[3]{x}+6\sqrt[6]{x}+6\log\left|1+\sqrt[6]{x}\right|$

(d) $6\sqrt[3]{x^2}-3\sqrt[3]{x}+6\log\left|1+\sqrt[3]{x}\right|$

Q.50 $\int\frac{1}{[(x-1)^3(x+2)^5]^{1/4}}dx$ is equal to

(a) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4}+C$

(b) $\frac{4}{3}\left(\frac{x+2}{x-2}\right)^{1/4}+C$

(c) $\frac{1}{3}\left(\frac{x-1}{x+2}\right)^{1/4}+C$

(d) $\frac{1}{3}\left(\frac{x+2}{x-1}\right)^{1/4}+C$

Q.51 $\int \frac{(x^4 - x)^{1/4}}{x^5} dx$ is equal to

(a) $\frac{4}{15}\left(1 - \frac{1}{x^3}\right)^{5/4} + C$

(b) $\frac{4}{5}\left(1 - \frac{1}{x^3}\right)^{5/4} + C$

(c) $\frac{4}{15}\left(1 + \frac{1}{x^3}\right)^{5/4} + C$

(d) None of these

Q.52 $\int \frac{x^3 - 1}{x^3 + x} dx$ is equal to

(a) $x - \log x + \log(x^2 + 1) - \tan^{-1} x + C$

(b) $x - \log x + \frac{1}{2}\log(x^2 + 1) - \tan^{-1} x + C$

(c) $x + \log x + \frac{1}{2}\log(x^2 + 1) + \tan^{-1} x + C$

(d) None of these

Q.53 $\int \frac{2x^2 + 3}{(x^2 - 1)(x^2 + 4)} dx = a\log\left(\frac{x+1}{x-1}\right) + b\tan^{-1}\frac{x}{2}$, then (a, b) is

(a) $(-1/2, -1/2)$

(b) $(1/2, 1/2)$

(c) $(1, -1)$

(d) $(1, -1)$

Q.54 $\int \frac{1}{(1 + x^2)\sqrt{1 - x^2}} dx$ is equal to

(a) $\frac{1}{2}\tan^{-1}\left(\frac{\sqrt{2}x}{\sqrt{1 - x^2}}\right) + c$

(b) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\sqrt{2}x}{\sqrt{1 + x^2}}\right) + c$

(c) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\sqrt{2}x}{\sqrt{1 - x^2}}\right) + c$

(d) None of these

Q.55 $\int \frac{1}{x\sqrt{1 - x^3}} dx = a\log\left|\frac{\sqrt{1 - x^3} - 1}{\sqrt{1 - x^3} + 1}\right| + b$, then a equal to

(a) 1/3 (b) 2/3

(c) –1/3 (d) –2/3

Q.56 If $\int \frac{dx}{\sqrt{x^2 + 2x + 1}} = A\log|x + 1| + C$ for $x < -1$, then A is

(a) 0 (b) 1

(c) –1 (d) None of these

Q.57 $\int \frac{x^{5/2}}{\sqrt{1 + x^7}} dx$ is

(a) $\frac{2}{7}\left(x^{7/2} + \sqrt{x^7 + 1}\right) + c$

(b) $\frac{1}{2}\log\frac{x^7 + 1}{x^7 - 1} + c$

(c) $2\sqrt{1 + x^7} + c$

(d) None of these

Q.58 $\int \frac{1 - x^7}{x(1 + x^7)} dx$ equals

(a) $\ell n x + \frac{2}{7}\ell n\left(1 + x^7\right) + c$

(b) $\ell n x - \frac{2}{7}\ell n\left(1 - x^7\right) + c$

(c) $\ell n x - \frac{2}{7}\ell n\left(1 + x^7\right) + c$

(d) $\ell n x + \frac{2}{7}\ell n\left(1 - x^7\right) + c$

Q.59 $\int \frac{x dx}{\sqrt{1 + x^2 + \sqrt{\left(1 + x^2\right)^3}}}$ is equal to

(a) $\frac{1}{2}\ell n x\left(1 + \sqrt{1 + x^2}\right) + c$

(b) $2\sqrt{1 + \sqrt{1 + x^2}} + c$

(c) $2\left(1 + \sqrt{1 + x^2}\right) + c$

(d) None of these

Q.60 $\int \frac{(2x + 1)}{(x^2 + 4x + 1)} dx$

(a) $\frac{x^3}{(x^2 + 4x + 1)^{1/2}} + C$

(b) $\frac{x}{(x^2 + 4x + 1)^{1/2}} + C$

(c) $\frac{x^2}{(x^2 + 4x + 1)^{1/2}} + C$

(d) $\frac{1}{(x^2 + 4x + 1)^{1/2}} + C$

Section-B
(Previous Years Question)

Q.1 If $g(x)=\int_{\sin x}^{\sin 2x}\sin^{-1}(t)\,dt$, then

(a) $g'\left(\frac{\pi}{2}\right)=0$

(b) $g'\left(\frac{\pi}{2}\right)=-2\pi$

(c) $g'\left(-\frac{\pi}{2}\right)=-2\pi$

(d) $g'\left(-\frac{\pi}{2}\right)=2\pi$

Q.2 If $I=\sum_{k=1}^{98}\int_{k}^{k+1}\frac{k+1}{x(x+1)}dx$, then

(a) $I<\frac{49}{50}$

(b) $I>\log_e 99$

(c) $I<\log_e 99$

(d) $I>\frac{49}{50}$

Q.3 The value of $\int_{-\pi/2}^{\pi/2}\frac{x^2\cos x}{1+e^x}dx$ is equal to

(a) $\frac{\pi^2}{4}-2$

(b) $\frac{\pi^2}{4}+2$

(c) $\pi^2-e^{\pi/2}$

(d) $\pi^2+e^{\pi/2}$

Q.4 Let $f(x)=\lim_{n\to\infty}\left(\frac{n^n(x+n)\left(x+\frac{n}{2}\right)....\left(x+\frac{n}{n}\right)}{n!\left(x^2+n^2\right)\left(x^2+\frac{n^2}{4}\right)...\left(x^2+\frac{n^2}{n^2}\right)}\right)^{\frac{x}{n}}$, for all $x>0$. Then

(a) $f\left(\frac{1}{2}\right)\geq f(1)$

(b) $f\left(\frac{1}{2}\right)\leq f\left(\frac{2}{3}\right)$

(c) $f'(2)\leq 0$

(d) $\frac{f'(3)}{f(3)}\geq\frac{f'(2)}{f(2)}$

Q.5 If $\alpha=\int_0^1\left(e^{9x+3\tan^{-1}x}\right)\left(\frac{12+9x^2}{1+x^2}\right)dx$ where $\tan^{-1}x$ takes only principal values, then the value of $\left(\log_e|1+\alpha|-\frac{3\pi}{4}\right)$ is

Q.6 Let $f:R\to R$ be a continuous odd function, which vanishes exactly at one point and $f(1)=\frac{1}{2}$.

Suppose that $F(x)=\int_{-1}^{x}f(t)dx$ for all $x\in[-1,2]$ and $G(x)=\int_{-1}^{x}t\left|f(f(t))\right|dt$ for all $x\in[-1,2]$. If $\lim_{x\to1}\frac{F(x)}{G(x)}=\frac{1}{14}$, then the value of $f\left(\frac{1}{2}\right)$ is

Q.7 The option(s) with the values of a and L that satidfy the following equation is (are)

$$\frac{\int_0^{4\pi}e^t\left(\sin^6 at+\cos^4 at\right)dt}{\int_0^{\pi}e^t\left(\sin^6 at+\cos^4 at\right)dt}=L\,?$$

(a) $a=2, L=\frac{e^{4\pi}-1}{e^{\pi}-1}$

(b) $a=2, L=\frac{e^{4\pi}+1}{e^{\pi}+1}$

(c) $a=4, L=\frac{e^{4\pi}-1}{e^{\pi}-1}$

(d) $a=4, L=\frac{e^{4\pi}+1}{e^{\pi}+1}$

Q.8 Let $f(x)=7\tan^8 x+7\tan^6 x-3\tan^4 x-3\tan^2 x$ for all $x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, Then the correct expression (s) is (are)

(a) $\int_0^{\pi/4}xf(x)dx=\frac{1}{12}$

(b) $\int_0^{\pi/4}f(x)dx=0$

(c) $\int_0^{\pi/4}xf(x)dx=\frac{1}{6}$

(d) $\int_0^{\pi/4}f(x)dx=1$

Q.9 Let $f'(x)=\frac{192x^3}{2+\sin^4\pi x}$ for all $x\in R$ with $f\left(\frac{1}{2}\right)=0$. If $m\le\int_{1/2}^{1}(x)dx\le M$, then the possible values of m and M are

(a) $m=13, M=24$

(b) $m=\frac{1}{4}, M=\frac{1}{2}$

(c) $m=-11, M=0$

(d) $m=1, M=12$

Paragraph

Let F : R $\to$ R be a thirce differentiable function. Suppose that F(1) = 0, F(3)=–4 and F'(x)<0 for all $x\in(1/2, 3)$. Let f(x) = xF(x)for all $x\in R$.

Q.10 The correct statement(s) is (are)

(a) f′(1) < 0

(b) f(2)< 0

(c) f′(x) ≠ 0 for any $x\in(1, 3)$

(d) f′(x) = 0 for some $x\in(1, 3)$

Q.11 If $\int_1^3 x^2F'(x)dx = -12$ and $\int_1^3 x^3F''(x)dx = 40$, then the correct expression (s) is (are)

(a) $9f'(3)+f'(1)-32=0$

(b) $\int_1^3 f(x)dx = 12$

(c) $9f'(3)-f'(1)+32=0$

(d) $\int_1^3 f(x)dx = -12$

Section-C
(Board pattern)

Q.1 Integrate $\left(\frac{2a}{\sqrt{x}}-\frac{b}{x^2}+3c\sqrt[3]{x^2}\right)$ w.r.t. *x*.

Q.2 Evaluate : $\int\frac{3ax}{b^2+c^2x^2}dx$.

Q.3 Verify the following using the concept of integration as an antiderivative.

$$\int\frac{x^3dx}{x+1}=x-\frac{x^2}{2}+\frac{x^3}{3}-\log|x+1|+c$$

Q.4 Evaluate : $\int\frac{e^{6\log x}-e^{5\log x}}{e^{4\log x}-e^{3\log x}}dx$.

Q.5 Evaluate : $\int\frac{dx}{1+\cos x}$ .s

Q.6 Evaluate : $\int\sqrt{1+\sin x}\,dx$.

Q.7 Evaluate : $\int\frac{x^{1/2}}{1+x^{3/4}}dx$.

Q.8 Evaluate $\int\frac{dt}{\sqrt{3t-2t^2}}$

Q.9 Evaluate : $\int e^{\tan^{-1}x}\left(\frac{1+x+x^2}{1+x^2}\right)dx$.

Q.10 Find $\int\sin^{-1}\sqrt{\frac{x}{a+x}}dx$.

Q.11 Find $\int e^{-3x}\cos^3 x\,dx$.

Q.12 Find $\int\frac{x^9}{(4x^2+1)^6}dx$.

Q.13 Find $\int\sqrt{\tan x}\,dx$.

Q.14 Evaluate : $\int\left(\log(\log x)+\frac{1}{(\log x)^2}\right)dx$.

Q.15 Evaluate : $\int\frac{3x+1}{2x^2-2x+3}dx$.

ANSWER KEY

CHAPTER-01 (APPLICATION OF DERIVATIVES)

SECTION-A (Straight objective type questions)

1. (d)	2. (a)	3. (b)	4. (b)	5. (c)	6. (c)	7. (a)
8. (a)	9. (a)	10. (a)	11. (c)	12. (a)	13. (a)	14. (b)
15. (a)	16. (c)	17. (d)	18. (d)	19. (c)	20. (b)	21. (c)
22. (a)	23. (d)	24. (b)	25. (b)	26. (b)	27. (d)	28. (c)
29. (b)	30. (b)	31. (d)	32. (c)	33. (c)	34. (a)	35. (a)

SECTION-B(Previous Years questions)

1. (2,4)	2. (2,3)	3. (4)	4. (3,4)	5. (1,3)	6. (3)	7. (3)
8. (3)	9. (4)	10. (3)	11. (3)	12. (4)	13. (1)	14. (1)
15. (4)	16. (2)	17. (1)	18. (2)	19. (1,2)	20. (1,3)	21. (1,2)
22. (2,3)	23. (1,2,3,4)	24. (2,3)	25. (3,4)	26. (2)	27. (1, 4)	28. (4)
29. (1)	30. (2)	31. (3)				

SECTION-C (School/Board level type questions)

1. $\tan^{-1}\frac{1}{\sqrt{2}}$ 2. $\frac{1000}{27\pi}m^3$ 4. $2\sqrt{2}$ units

6. Base radius = $8\sqrt{2}$ cm, height = 32 cm 7. $\pi R^2(\sqrt{5}+1)$ sq. units

8. $6 : (6+\pi)$ 9. $2\left(\frac{9C}{16}\right)^{1/3}, \left(\frac{9C}{16}\right)^{1/3}, \left(\frac{32C}{81}\right)^{1/3}$

10. $(c - d\cot\alpha)$ from his house 12. $\frac{2}{3}r^3(27+2\pi)$, $x = 3r$

13. $4r/3$ 14. $\frac{8}{27}.\frac{4}{3}\pi r^3$

16. Breadth of the stiffest beam = $a/2$ 17. $\sqrt{5}$, (1, 3)

18. 40 km/hr 19. t = 2 hr, Shortest distance = $15\sqrt{13}$ km.

20. $\pi : 4$

21. Radius of semi-circle = $\frac{20}{4+\pi}$ meters, Side of the rectangle = $\frac{20}{4+\pi}$ m

22. 25 cm^2 25. $\frac{1000}{3\sqrt{3\pi}}$ cm^3 26. 9/2 cm, 9 cm, 6 cm

28. Sides of the triangle are $2\sqrt{s}, \sqrt{2s}$ and $\sqrt{2s}$ 29. 1 : 1 31. $\left(\frac{1}{4},\frac{1}{2}\right), \frac{3\sqrt{2}}{8}$ units

32. $\frac{3\sqrt{3}}{4}ab$ sq. units 36. 6 37. 5 unit 38. (2,1)

39. Maxima at $x = \frac{\left(b-\sqrt{b^2-1}\right)}{4}$ and minima at $x = \frac{1}{4}\left(b+\sqrt{b^2-1}\right)$

40. 2hk 41. $\frac{4\sqrt{3}}{9}$ 42. $\sqrt{2}x-2y+2=0, \sqrt{2}x+2y-2=0$

43. $\lambda \in \left(-\frac{3}{2},\frac{3}{2}\right)$ 44. 5 45. 9 46. 1 47. 2 48. 7

CHAPTER-02 (INDEFINITE INTEGRALS)

SECTION-A (Straight objective type questions)

1. (b)	2. (a)	3. (c)	4. (b)	5. (d)	6. (b)	7. (a)
8. (a)	9. (b)	10. (d)	11. (b)	12. (c)	13. (a)	14. (b)
15. (a)	16. (a)	17. (a)	18. (b)	19. (b)	20. (b)	21. (b)
22. (d)	23. (a)	24. (b)	25. (a)	26. (c)	27. (b)	28. (b)
29. (c)	30. (a)	31. (c)	32. (c)	33. (c)	34. (c)	35. (a)
36. (a)	37. (b)	38. (c)	39. (c)	40. (a)	41. (a)	42. (b)
43. (a)	44. (c)	45. (a)	46. (b)	47. (c)	48. (a)	49. (b)
50. (a)	51. (a)	52. (b)	53. (b)	54. (c)	55. (a)	56. (c)
57. (a)	58. (c)	59. (b)	60. (b)			

SECTION-B (Previous Years questions)

1. (a)	2. (c,d)	3. (a)	4. (b,c)	5. (9)	6. (7)	7. (a)
8. (a,b)	9. (d)	10. (a,b,c)	11. (c,d)			

SECTION-C(School/Board level type questions)

1. $4a\sqrt{x}+\frac{b}{x}+\frac{9cx^{5/3}}{5}+C$

2. $\frac{3a}{2c^2}\log|b^2+c^2x^2|+C$

4. $\frac{x^3}{3}+c$

5. $\tan\frac{x}{2}+c$

6. $2\sin\frac{x}{2}-2\cos\frac{x}{2}+c$

7. $\frac{4}{3}\left(x^{3/4}-\frac{1}{3}\log|1+x^{3/4}|\right)+c$

8. $\frac{1}{\sqrt{2}}\sin^{-1}\frac{4t-3}{3}+c$

9. $xe^{\tan^{-1}x}+c$

10. $a\left[\frac{x}{a}\tan^{-1}\sqrt{\frac{x}{a}}-\sqrt{\frac{x}{a}}+\tan^{-1}\sqrt{\frac{x}{a}}\right]+c$

11. $\frac{1}{24}e^{-3x}[\sin3x-\cos3x]+\frac{3}{40}e^{-3x}[\sin x-3\cos x]$

12. $-\frac{1}{10}\frac{1}{\left(4-\frac{1}{x^2}\right)^5}+c$

13. $\frac{1}{\sqrt{x}}\tan^{-1}\frac{(\tan x-1)}{\sqrt{2\tan x}}+\frac{1}{2\sqrt{2}}\log\left|\frac{\tan x-\sqrt{2\tan x}+1}{\tan x+\sqrt{2\tan x}+1}\right|+c$

14. $x\log(\log x)-\frac{x}{\log x}+c$

15. $\frac{3}{4}\log|2x^2-2x+3|+\frac{\sqrt{5}}{2}\tan^{-1}\frac{2x-1}{\sqrt{5}}+c$